Self-reinforced geometric absorption:
a memory-dependent kinetic framework for vacuum, dark matter and localized structure

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Abstract

We develop a single kinetic law for the dark sector and follow it to its consequences. Regions of an effective substrate convert irreversibly into a pressureless localized product and a growing vacuum energy, at a rate that increases with the fraction already converted and with a causal memory of that history. We show that the population-level memory law is exactly the average of a per-unit relaxation law with a common relaxation time, so it is equivalent to a local assumption — one of equal or greater strength, not a weaker one — rather than to an extra non-local one, that the finite-population process has a controlled deterministic limit, with an exactly binomial $N^{-1/2}$ law where the rate does not depend on the state and numerical evidence for the same scaling where it does, and that memory makes the kinetics path dependent: two histories that reach the same converted fraction with different memory states convert at different epochs and, integrated along a prescribed expansion history, accumulate different vacuum energies. An interaction written as a function of the densities and the expansion rate closes on those densities, so they fix its entire future whatever its functional form and however its sign behaves with redshift; here they do not, because the memory is an independent state variable.

Coupled to general relativity, the model has exact identities relating the converted fraction to the surviving densities, global positivity with bounded expansion, and a de Sitter attractor reached without the Hubble rate ever increasing. Positivity is structural rather than tuned, because every outflow is proportional to the density of the sector it leaves; this is the property lacking in the recipient-proportional couplings among those recently catalogued as admitting negative densities. We give the exact map onto the standard parametrizations, prove that in the one-way model the effective dark matter can only lose energy, and derive a parameter-free cosmographic inequality, $j \geq 1 + 2\Omega_r$, whose sign is opposite to that of running vacuum and of the currently favoured evolving dark-energy quadrant. The memory parameters cancel from the jerk and reappear in the snap, though not in isolation: the snap constrains their lag together with the activation profile. In linear theory the branching ratio between the two products cancels identically, and under the geodesic realization of the transfer — which we show is forced rather than chosen, given the hypothesis that the transfer carries no momentum: a pressureless donor whose only source then points along its own flow cannot be accelerated — the model predicts enhanced density growth together with suppressed velocity growth, a signature of opposite signs which we prove a coupling proportional to the local density cannot produce, the cancellation there being exact and independent of the transfer's size. The same framework's reverse channel carries the opposite sign of $j - 1$, and it is the best-fitting extension reported here, so the sign predictions belong to the one-way restriction and not to the framework.

Two defects of an earlier version of this work are corrected rather than dropped: a rate with a positive floor admits no viable early universe, which we prove and repair with an activation law; and the growth equation previously written for the precursor contrast is wrong whenever both products are present. Three obstructions are recorded rather than passed over. The conjecture that conversion regularizes the caustics of the clock congruence is false

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for the model as published, because a rate bounded on the approach cannot change the exponent of the divergence. The branching ratio required for regularity there vanishes, while a relic abundance requires it not to. And the compression-driven rate that would repair the first runs into a covariant obstruction we leave unresolved. The obstructions sort into two families with two different roots, and naming them is what a successor needs. One root is the recipient: an exact vacuum receiving energy with no momentum transfer is what forbids vorticity where the transfer is on, forces the rate to be a function of the clock rather than of the local compression, and — with a pressureless donor — forces the geodesic realization of the transfer outright. The other root is the localized product, which is dust and is compressed by the same expansion as everything else, so it carries the divergence at a caustic by itself however it was made. Each family lifts under one modification and neither lifts under the other's. A cosmographic bound relating the matter deficit to the jerk is also shown not to apply to this model, its hypothesis on the transfer profile failing for every activation redshift and every rate we tried. Covariant realizations are exhibited for the vacuum sector, including a four-form whose membrane nucleation events map onto absorption events and yield the sign and size of the reinforcement coefficient — rigorously, but within that auxiliary system and under hypotheses, detailed balance among them, that the model itself does not adopt, so the displayed results there carry a level of their own, and the constrained-clock potential is fixed by the kinetics to a five-parameter, monotone, bounded family rather than a free function, which narrows but does not remove the standard objection to mimetic constructions. A finite-energy unit-charge defect is solved numerically, and an open quantum realization reproduces the classical chain with a mean-field closure error measured to scale as N^{-1}, subleading to the intrinsic $N^{-1/2}$ fluctuations. The model is confronted with Pantheon+, DESI DR2 baryon acoustic oscillations, cosmic chronometers and the Planck distance priors, and the outcome is a null result reported as one: ΛCDM is preferred, the convertible fraction is not measurable because the rate can always vanish, and what the data do bound is the matter deficit, to below 0.0100 of the critical density. Profiling the predicted jerk excess directly bounds it more than an order of magnitude below the resolution of present late-time data, so the model's sign predictions are refutable in principle and not yet in practice.

Every displayed key result carries an evidential level, from postulate through identity and theorem to conjecture, printed beside it and checked for presence and placement as part of the build. We do not derive quantum mechanics, electromagnetism, or any Standard-Model content, and we do not claim that the defects are the dark matter. What we claim is narrower and testable: an effective model of self-reinforced absorption with memory, with exact cosmological consequences under stated hypotheses and three sign predictions that data can refute.

Keywords: interacting vacuum; dark sector; absorbing-state kinetics; memory and hysteresis; cosmography; mimetic gravity; topological defects.

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1 Problem, hypotheses, and what is claimed

1.1 The question

Suppose a substrate whose regions explore configurations, some of which are persistent; suppose the accumulated conversion changes the probability of further conversion; and suppose the products gravitate while remaining electromagnetically quiet. This paper asks what that idea implies once it is written as equations, and where it stops implying anything.

The word "space" carries two meanings in such a proposal. Microscopically it names a substrate that is not specified here. Macroscopically it names a Lorentzian manifold. We do not identify the two by definition: the coarse-graining map that would connect them is part of the open problem, not part of the construction. Throughout, *absorption* means the absence of return on the timescale and in the set of observables considered. An absorbing state need not be a global energy minimum, a relativistic vacuum, or a particle; each of those is a separate calculation.

1.2 Hypotheses

Five assumptions carry the construction, and it is worth listing them plainly because the paper's results are conditional on them.

1. Identifiable mesoscopic units exist, each active or absorbed.
2. The conversion rate is non-negative and may depend on a memory of the conversion already accumulated.
3. The macroscopic description admits a pressureless precursor, a pressureless localized product, and a product of vacuum type.
4. A fixed fraction of the transferred energy feeds each product.
5. General relativity and covariant conservation of the total tensor hold in the regime studied.

The third assumption does not follow from the first two, which is why Sec. 8 studies explicit realizations of a vacuum tensor and of a localized configuration rather than asserting them. The rate is not derived from gravity either: an effective law is specified and its consequences are computed.

1.3 Relation to existing work

Most of the machinery used here is not new, and saying so precisely is necessary before the contribution can be identified.

Interacting vacuum cosmology, in which a component with $p = -\rho$ exchanges energy with matter and therefore has no rest frame of its own, is older than the reference we lean on most: Wands et al. [1] gave it the modern covariant treatment we use, and its covariant perturbation theory is set out by Sebastianutti et al. [2]. The particular branch this paper works in — a vacuum exchanging energy with *geodesic* cold dark matter — is a named scenario with its own observational literature, and the constraints on it are the right point of comparison for Sec. 10: see Martinelli et al. [3] and Hogg et al. [4]. What Proposition 9.4 adds to that scenario is not the scenario but an argument that its defining closure is forced rather than chosen. The constrained-clock action of Sec. 8 is the "dust of dark energy" of Lim et al. [5], equivalently mimetic matter with a potential [6], whose known pathologies – vanishing sound speed, caustics, and gradient instabilities in extended versions – are reviewed by Sebastiani et al. [7] and Malaeb [8]. A silent sector combining these ingredients with an interacting vacuum, fitted to recent data, has been constructed by Cruz Pérez et al. [9]. Four-form realizations of a vacuum sector go back to Aurilia et al. [10] and Hawking [11], with the membrane-nucleation mechanism that Sec. 8.5 builds on due to Brown and Teitelboim [12, 13]; Bousso and Polchinski [14] is the multi-flux landscape refinement of it. Open-system descriptions of dark energy have been developed by Salcedo et al. [15]. The link between the third derivative of the scale factor and energy transfer to the vacuum was made by Szydlowski [16] and revisited by Worsley et al. [17].

The kinetics is not new either, and the claim is worth stating only after that is said plainly. A rate that grows with the fraction already converted is the autocatalytic law of Prout and Tompkins [18]; a conversion fraction governed by nucleation and growth is the theory of Avrami [19], of which Eq. (3.8) is the elementary case; irreversible conversion into an absorbing state is the setting reviewed by Hinrichsen [20]; and path dependence arising from self-reinforcement is the defining property of reinforced stochastic processes, going back to the urn of Eggenberger and Pólya [21] and surveyed by Pemantle [22]. None of those ingredients is claimed here. Nor is Eq. (3.2) an unusual object in isolation: it is a one-pole relaxation, and Proposition 3.2 identifies it as exactly that.

What we have not found elsewhere is the combination, used as a dark-sector dynamical law: a self-reinforced absorbing-state kinetics whose reinforcement responds to a lagged memory of the conversion, coupled to general relativity so that the converted fraction is the vacuum energy. That is where this paper plants its flag, and it is a narrow flag. Theorem 3.8 states what the memory buys that no coupling written directly between densities can provide — a state variable those couplings do not have — and Sec. 10 reports that the data do not currently measure it. The reader should weigh the claim against both.

1.4 Contributions

1. A kinetic law with a controlled deterministic limit, whose memory variable is re-expressed as the population mean of a per-unit relaxation law with a *common* relaxation time — a local assumption of equal or greater strength rather than a weaker one (Proposition 3.2 and Theorem 3.4) — and which makes the kinetics path dependent: at equal converted fraction the memory state still carries information about the future, so along a prescribed expansion history two such histories accumulate different vacuum (Theorem 3.8).

2. A positivity theorem that is structural rather than tuned (Theorem 5.1), which is what distinguishes this coupling from the recipient-proportional models among those catalogued as admitting negative densities [23–25]; together with an exact map onto the standard parametrizations (Proposition 5.3) and a sign prediction that no choice of parameters can evade (Theorem 5.4).

3. A proof that the rate law cannot be extended into the early universe as originally written, and the minimal repair (Theorem 6.1 and Assumption 6.2).

4. A parameter-free cosmographic inequality and a snap relation (Theorems 7.1 and 7.4), whose signs separate this model from running vacuum and from the currently favoured dark-energy parametrizations, and in which the memory parameters appear in the snap while cancelling from the jerk (Theorem 7.5).

5. A closed linear perturbation theory in which the branching ratio cancels identically (Theorem 9.5), correcting a result of the earlier version of this work, and which predicts enhanced density growth together with suppressed velocity growth (Eq. (9.10)).

6. A confrontation with Pantheon+, DESI DR2, cosmic chronometers and the Planck distance priors (Sec. 10), with the code and data checksums archived. Its outcome is negative and is reported as such: ΛCDM is preferred on the evidence over every extension fitted, and the convertible fraction is not measurable at all, for a reason the section identifies.

7. A solved, finite-energy, unit-charge defect (Eq. (11.6)), and an open quantum realization whose mean-field closure error is measured to scale as N^{-1} (Proposition 12.2).

8. Three obstructions, stated rather than passed over (Sec. 11.5): that a bounded rate cannot regularize a caustic, which refutes a conjecture of the earlier version of this work; that the branching ratio required for regularity there is the one a relic abundance forbids; and that the rate which would repair the first has no covariant realization we could construct. A fourth, in Sec. 7.1: the cosmographic bound relating the matter deficit to the jerk excess is not available to this model, its hypothesis on the transfer profile failing for every activation redshift.

1.5 What is claimed, and at what level

This paper is a framework, not a unification. To keep that distinction from eroding as the argument proceeds, every displayed *key* result carries an explicit evidential level, printed with it. "Key" is not left to taste: a displayed equation is levelled if it is referred to anywhere else in the paper, which is the operational test the build applies — the remaining numbered displays are intermediate steps of the argument around them, and a level on those would classify the algebra rather than the claims. A build check refuses an unclassified boxed result, and refuses a conjecture outside Sec. 13. What the check enforces is the presence and the placement of a level, not its correctness: the assignment of a particular result to a particular level is ours, and a reader who disagrees with one is disagreeing with us and not with the software.

Level	Meaning
A	Postulate. Defines the model; it follows from nothing prior in this paper, and everything at level C follows from it.
I	Identity, or a standard result restated. Follows from stated equations and definitions alone.
C	Theorem of the model. Follows from the postulates together with the system specified; the hypotheses are named in the statement.
X	Result of an auxiliary model. Follows rigorously, but from a *different* system than the one this paper postulates, under hypotheses stated where it appears. It supports an identification and does not establish one.
N	Numerical. A solution of the stated equations, obtained to a stated tolerance by the archived code, with no data involved.
P	Phenomenological. A number obtained by fitting to data, valid within the stated likelihood and priors.
S	Conjecture. Requires a microscopic theory or a calculation not performed here; confined to Sec. 13.

Three claims that this paper does *not* make are worth stating in advance, because a reader could otherwise supply them. It does not derive quantum mechanics: Sec. 12 begins from a specified open quantum model and recovers the classical chain from it. It does not derive electromagnetism or any Standard-Model content. And it does not claim that the defects of Sec. 11 are the dark matter; that identification is a conjecture, listed as such.

1.6 Plan

Sec. 2 fixes conventions. Sec. 3 develops the kinetic law. Sec. 4 couples it to gravity and proves the global results. Sec. 5 establishes positivity and the map to interacting dark energy. Sec. 6 gives the solvable family and repairs the early-universe branch. Sec. 7 derives the cosmographic tests. Sec. 8 exhibits covariant realizations. Sec. 9 develops linear perturbation theory. Sec. 10 confronts the model with data. Sec. 11 solves the localized configuration numerically — a minimum within the hedgehog ansatz, which is not a proof of stability against all deformations, as that section says — and Sec. 12 treats the open-system description. Sec. 13 lists what remains conjectural, and Sec. 14 states what has been shown.

2 Covariant setup

This section fixes conventions and records the gravitational identities the rest of the paper uses. Nothing here is new; it is written out because the signs matter for the transfer terms of Sec. 4 and because several statements later depend on distinguishing what Einstein's equations impose from what a choice of matter model adds.

2.1 Conventions

We use $c = \hbar = k_B = 1$, signature $(-+++)$, and

$$M_{\rm Pl}^2 = (8\pi G)^{-1}, \qquad [x^\mu] = -1, \quad [\rho] = [p] = 4, \quad [H] = [\Gamma] = 1, \tag{2.1}$$

with brackets denoting mass dimension. Curvature conventions are

$$[\nabla_\mu, \nabla_\nu] X^\rho = R^\rho{}_{\sigma\mu\nu} X^\sigma, \qquad R_{\mu\nu} = R^\rho{}_{\mu\rho\nu}, \tag{2.2}$$

so that a sphere has positive scalar curvature; these agree with Carroll [26].

The symbol V always denotes a vacuum energy *density*, never a volume; a physical volume is written $\mathcal{V}$. The conversion rate is Γ and the memory variable is M, never to be confused with the Christoffel symbols $\Gamma^{\rho}{}_{\mu\nu}$, which appear only in this section, or with $M_{\rm Pl}$. A complete symbol list is given in Appendix A.

2.2 Field equations and the meaning of local conservation

From

$$S = \int \mathrm{d}^4x\,\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R + \mathcal{L}_{\rm m}\right] + S_\partial, \qquad T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\rm m}}{\delta g^{\mu\nu}}, \tag{2.3}$$

with the boundary term chosen so that metric variations of compact support leave no surface contribution, the standard variational identities $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\,g_{\mu\nu}\delta g^{\mu\nu}$ and $\delta R = R_{\mu\nu}\delta g^{\mu\nu} + (g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu)\delta g^{\mu\nu}$ give

I — Identity

$$M_{\rm Pl}^2 G_{\mu\nu} = T_{\mu\nu}, \tag{2.4}$$

and, through the contracted Bianchi identity together with metric compatibility,

I — Identity

$$\nabla_\mu T^{\mu\nu} = 0. \tag{2.5}$$

Eq. (2.5) is *local* covariant conservation. It does not say that the energy in a comoving volume is constant, and in a cosmology without a timelike Killing vector it does not define a global energy at all. This distinction is what makes a growing vacuum energy possible without any violation of Eq. (2.5): what the equation constrains is the sum over sectors, not each sector separately.

2.3 Transfer between sectors

Write the total tensor as a sum over the five sectors used throughout, $T = T_I + T_D + T_V + T_{\rm b} + T_{\rm r}$. Their individual divergences may be non-zero provided they cancel. For an exact vacuum,

I — Identity

$$T_V^{\mu\nu} = -V g^{\mu\nu}, \qquad \nabla_\mu T_V^{\mu\nu} = -\nabla^\nu V. \tag{2.6}$$

A vacuum that varies in space or time therefore exchanges energy and momentum, and its transfer four-vector is not an independent choice: it is fixed by V itself. This is the constraint that returns in Sec. 9 as an integrability condition, and it is the reason an exact vacuum cannot be assigned a sound speed and a density perturbation independently.

A split of the total tensor into sectors is a definition, not a theorem. In particular the bookkeeping adopted in Sec. 4 is one choice among several that reproduce the same $T_{\mu\nu}$. Sec. 8 exhibits an action whose field equations produce the split between dust and vacuum, which is the split that carries the physics here; it does not produce the further division of the dust into precursor and defects, and Sec. 8.4 explains why that division is bookkeeping at every level this paper computes.

2.4 Friedmann equations

For the spatially flat line element $\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(t)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j$ with $H = \dot{a}/a$, the non-vanishing Christoffel symbols are $\Gamma^0{}_{ij} = H g_{ij}$ and $\Gamma^i{}_{0j} = H\delta^i{}_j$, giving

$$R_{00} = -3(\dot{H} + H^2), \qquad R_{ij} = (\dot{H} + 3H^2)g_{ij}, \qquad R = 6(\dot{H} + 2H^2), \tag{2.7}$$

hence $G_{00} = 3H^2$ and $G_{ij} = -(2\dot{H} + 3H^2)g_{ij}$. With $T_{00} = \rho$ and $T_{ij} = p\,g_{ij}$, Eq. (2.4) becomes

I — Identity

$$3M_{\rm Pl}^2H^2 = \rho, \qquad \dot{H} = -\frac{\rho + p}{2M_{\rm Pl}^2}, \qquad \frac{\ddot{a}}{a} = -\frac{\rho + 3p}{6M_{\rm Pl}^2}, \tag{2.8}$$

and the time component of Eq. (2.5) gives $\dot\rho + 3H(\rho + p) = 0$ for the total. Two consequences are used repeatedly. First, $\dot H \le 0$ whenever the total satisfies $\rho + p \ge 0$, which will hold in every model considered here even while the vacuum grows. Second, acceleration requires $\rho + 3p < 0$, that is, a genuine contribution to the Ricci tensor: being invisible electromagnetically is not by itself relevant to Eq. (2.8).

2.5 Consistency check via Raychaudhuri

For a normalised timelike congruence with expansion Θ, shear $\sigma_{\mu\nu}$, vorticity $\omega_{\mu\nu}$ and acceleration $A^\mu = u^\nu \nabla_\nu u^\mu$,

I — Identity

$$u^\mu \nabla_\mu \Theta = -\frac{\Theta^2}{3} - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}u^\mu u^\nu + \nabla_\mu A^\mu. \tag{2.9}$$

In FLRW the shear, vorticity and acceleration vanish and $\Theta = 3H$, so Eq. (2.9) reduces to $3(\dot H + H^2) = -R_{\mu\nu}u^\mu u^\nu$; the trace of Eq. (2.4) gives $R_{\mu\nu}u^\mu u^\nu = (\rho + 3p)/(2M_{\rm Pl}^2)$, returning the third of Eq. (2.8). We will use Eq. (2.9) again in Sec. 9, where the shear and vorticity terms are second order and the acceleration vanishes because the transfer is geodesic.

What this section establishes. The gravitational side of the problem, with signs fixed. What it does not establish is any property of the matter sector: the decomposition into precursor, defects and vacuum, and the rate at which one becomes another, are additional physical input, introduced in Sec. 3 and constrained in Sec. 8.

3 Absorbing process with reinforcement and memory

The dynamical content of this paper is a single kinetic law. This section states it as a finite-population jump process, derives the deterministic limit under explicit hypotheses, and establishes the two properties that distinguish it from a coupling written directly at the level of densities: the memory variable is not an extra postulate but the mean of an eliminated per-unit degree of freedom, and it makes the dynamics path dependent.

3.1 State, rate and memory

Consider a mesoscopic region containing N labelled units, each either active or absorbed, and let K_t be the number absorbed, $f_N = K_t/N$. An event converts one unit and never returns it. Processes with absorbing states are a well-studied class [20]; we borrow the formalism, and note at the outset that neither the presence of absorption nor a state-dependent rate is by itself evidence of a critical phase, so no universality class is claimed here. A related tradition builds spacetime itself by irreversible stochastic growth [27]; the process below differs in carrying a memory, which that construction does not. The total intensity is

A — Postulate

$$\lambda_k(M,t) = (N-k)\,\Gamma(k/N, M, t), \qquad \lambda_N = 0, \tag{3.1}$$

so that in a short interval the probability of a jump is $\lambda_k \Delta t + o(\Delta t)$. The rate is a conditional probability per unit time, not a probability: it is not bounded by one.

Definition 3.1 (Memory variable)**.** Along a flow line with proper time τ,

A — Postulate

$$\tau_M \frac{\mathrm{d}M}{\mathrm{d}\tau} = f - M, \qquad \tau_M > 0, \tag{3.2}$$

whose solution from τ_i is

I — Identity

$$M(\tau) = M_i e^{-(\tau-\tau_i)/\tau_M} + \frac{1}{\tau_M}\int_{\tau_i}^{\tau} e^{-(\tau-s)/\tau_M} f(s)\,\mathrm{d}s. \tag{3.3}$$

Only the past enters Eq. (3.3), and the weights of M_i and of the history sum to one, so M stays in $[0,1]$ whenever f and M_i do. The covariant form is $\tau_M\, u^\mu \nabla_\mu M = f - M$; the choice of u^μ is physical input, fixed in Sec. 8. A homogeneous reduction is a symmetry assumption, not a mechanism for superluminal communication: the rate at a point depends only on that point's causal past. That reading is what Secs. 3.1 and 3.5 use and it is consistent for the homogeneous mode; promoting it to a local law off the background is not automatic, for a reason given in Sec. 9.1 and listed as open in Sec. 13.2.

The rate used throughout is

A — Postulate

$$\Gamma(f, M) = \Gamma_0\, e^{\alpha f + \beta M}\, S_{\mathrm{act}}(a), \qquad \alpha, \beta \geq 0, \tag{3.4}$$

with the activation factor $S_{\mathrm{act}}(a)$ introduced in Sec. 6; on $[0,1]^2$ it is bounded, $\Gamma_0 S_{\mathrm{act}} \leq \Gamma \leq \Gamma_0 e^{\alpha+\beta} S_{\mathrm{act}}$.

3.2 Where the memory law comes from

Eq. (3.2) looks like an extra assumption at the population level. It is one; what the following shows is that it is equivalent to a microscopically local assumption, which is a different and more useful statement than saying it has been derived.

Proposition 3.2 (Memory as an eliminated degree of freedom)**.** *Attach to each unit i an internal variable $m_i \in [0,1]$ relaxing towards that unit's own state $s_i \in \{0,1\}$, $\tau_M \dot{m}_i = s_i - m_i$. Then the population mean $M = N^{-1}\sum_i m_i$ obeys Eq. (3.2) exactly, for every N and every realisation.*

Proof. The per-unit relaxation is linear in (m_i, s_i), so averaging commutes with it: $\tau_M\, \dot{\overline{m}} = \overline{s} - \overline{m}$, and $\overline{s} = f$ by definition of the converted fraction. □

Averaging a linear equation does not derive that equation, and Proposition 3.2 should not be read as doing so: it assumes each unit carries an internal variable relaxing towards its own state with a *common* time τ_M, which is if anything a stronger assumption than the population law. What it buys is that the population law is not an extra non-local postulate on top of a local dynamics; the two are the same assumption written at different levels.

Two further readings of the same equation are worth recording, because they say what τ_M would be in a microscopic completion. First, an exponential kernel is the one-mode Markovian embedding of a relaxation process: the pair (f, M) is Markov even though f alone is not. This distinguishes the construction from fractional-derivative models of dark energy, which are genuinely non-local and have no such embedding. Second, if conversion proceeds by nucleation and the influence of an event reaches other units only once a front has swept them, the swept fraction lags the event count, and τ_M is the light-crossing time of the correlated patch. Sec. 8 makes this concrete in the four-form realization.

3.3 Deterministic limit

Assumption 3.3. Γ is non-negative, bounded above by $G < \infty$, and uniformly Lipschitz in (f, M) on $[0,1]^2$; the initial conditions converge in probability.

Theorem 3.4 (Law of large numbers)**.** *Under Assumption 3.3, on every finite time interval the process of Eq. (3.1) converges in probability to*

C — Theorem of the model

$$\dot{f} = (1-f)\Gamma(f, M), \qquad \tau_M \dot{M} = f - M. \tag{3.5}$$

Proof. Write $f_N(t) = f_N(0) + \int_0^t F(f_N, M_N)\,\mathrm{d}s + \mathcal{M}_N(t)$ with $F = (1-f)\Gamma$. The compensated jump term $\mathcal{M}_N$ is a martingale whose predictable quadratic variation obeys $\mathbb{E}\langle \mathcal{M}_N \rangle_T \leq GT/N$, so Doob's maximal inequality gives $\mathbb{E}\sup_{t\leq T} |\mathcal{M}_N|^2 \leq 4GT/N$. Subtracting the integral equations

for (f_N, M_N) and (f, M) and using the Lipschitz property, $\sup_{s\le t}|X_N - X| \le |X_N(0) - X(0)| + \sup_{s\le t}|\mathcal{M}_N| + L\int_0^t \sup_{r\le s}|X_N - X|\,\mathrm{d}s$; Gronwall's inequality bounds the left side by the first two terms times e^{Lt}, and both vanish in probability. □

Remark 3.5 (The boundedness hypothesis can be traded for a time change). Assumption 3.3 requires a bounded rate, which fails wherever the rate is driven by a diverging compression. For the converted fraction nothing is lost, though Remark 3.6 shows that the same is not true of the residual density. Let $\Lambda(t) = \int_{t_i}^t \Gamma\,\mathrm{d}s$ be the cumulative hazard and change time to Λ. If Γ does not depend on the state, every unit then carries an independent unit-rate exponential clock, so the converted count is *exactly* binomial,

C — Theorem of the model

$$K \sim \mathrm{Binomial}\Big(N,\, 1 - e^{-\Lambda}\Big), \qquad \mathbb{E}[f_N] = 1 - e^{-\Lambda}, \qquad \sqrt{N}\,\sigma(f_N) = \sqrt{p(1-p)}, \tag{3.6}$$

with $p = 1 - e^{-\Lambda}$. The mean is the deterministic solution exactly, at every finite N, and the $N^{-1/2}$ law holds uniformly with no boundary layer where the rate diverges. For $\Gamma = g/(T-t)$ one has $\Lambda = g\log[(T-t_i)/(T-t)]$, finite for every $t < T$ and divergent only at T, where both the chain and its limit have $f = 1$. The hypothesis that matters here is that Γ does not depend on the state; the reinforced law Eq. (3.4) does, so the statement applies to a compression-driven channel and not to the reinforcement.

Remark 3.6 (But the residual density has a boundary layer). The deterministic limit describes f; it does not describe $1 - f$, and the precursor density is proportional to the latter. From Eq. (3.6) the relative fluctuation of $1 - f$ is $\sqrt{p/[N(1-p)]}$, which *diverges* as $p \to 1$. The continuum description of ρ_I therefore fails once the expected number of unconverted units falls below one, at

C — Theorem of the model

$$Ne^{-\Lambda} = 1 \qquad \Longleftrightarrow \qquad s_* = N^{-1/g} \quad \text{for } \Gamma = g/(T-t). \tag{3.7}$$

Inside that layer the precursor is exhausted discretely rather than continuously: the mean field returns an average over realisations in which the density is typically exactly zero. Any statement about the regularity of ρ_I at a caustic is a statement about the continuum model, and holds only outside s_*.

The hypotheses that remain are material. If collective jumps of macroscopic size occur, or if correlations do not decay with N, the limit can retain noise and Eq. (3.5) fails. The fluctuations that do survive are Gaussian of relative size $N^{-1/2}$, with the diffusion amplitude $\sqrt{(1-f)\Gamma/N}$ read off the second infinitesimal moment; this is not the same as a noise term chosen for convenience, and near $f = 0$ a Gaussian approximation would produce negative excursions the jump process forbids. Appendix F.7 verifies both the limit and the scaling by direct simulation (Fig. 1).

3.4 Survival, acceleration, and what reinforcement does

Proposition 3.7 (Survival). *For $\Gamma \ge 0$ locally integrable along the solution and $f_i \in [0, 1)$,*

I — Identity

$$1 - f(t) = (1 - f_i)\exp\left[-\int_{t_i}^t \Gamma(s)\,\mathrm{d}s\right], \tag{3.8}$$

so $f_i \le f(t) < 1$ at every finite time, and $f \to 1$ if and only if $\int^\infty \Gamma\,\mathrm{d}s$ diverges.

Proof. With $y = 1 - f$, $\dot{y} = -\Gamma y$; integrate. Uniqueness prevents any solution from crossing the stationary boundary $y = 0$. □

Differentiating Eq. (3.5) gives the acceleration criterion in the form used later,

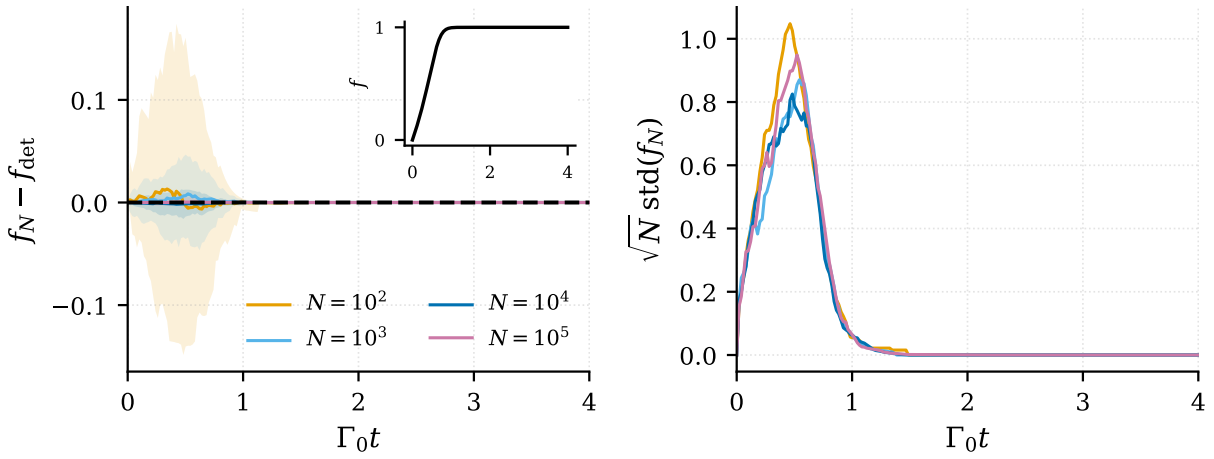


Figure 1. Convergence of the finite-population process to its deterministic limit. *Left:* deviation of the ensemble mean from the deterministic solution, with 5–95% bands, for four population sizes; the inset shows the deterministic $f(t)$ that has been subtracted. *Right:* the same ensembles' standard deviations multiplied by $\sqrt{N}$, which approximately collapse, with a residual upward drift of about five per cent across the range that Appendix F.7 documents and does not resolve. This is numerical evidence for the $N^{-1/2}$ scaling under the reinforced rate, which is not what Theorem 3.4 proves: that theorem gives convergence in probability, and the exact binomial law of Remark 3.5 needs a state-independent rate. Rate parameters as in Appendix F.7.

I — Identity

$$\ddot{f} = (1-f)\Gamma\left[\frac{\mathrm{d}\log\Gamma}{\mathrm{d}t} - \Gamma\right], \tag{3.9}$$

so conversion accelerates precisely while $\mathrm{d}\log\Gamma/\mathrm{d}t > \Gamma$. For the memoryless case $\Gamma = \Gamma_0 e^{\alpha f}$ this reads $\alpha(1-f) > 1$: reinforcement wins early and depletion wins late, and the exact solution is $t - t_i = \Gamma_0^{-1} e^{-\alpha}[\mathrm{Ei}(\alpha(1-f_i)) - \mathrm{Ei}(\alpha(1-f))]$. With memory the criterion becomes $\alpha(1-f)\Gamma + \beta\tau_M^{-1}(f-M) > \Gamma$, which depends on the initial memory state: $\beta > 0$ alone does not guarantee earlier acceleration.

3.5 Hysteresis

The operational content of the memory variable is that the present state of the densities does not determine the future.

Theorem 3.8 (Hysteresis, along a prescribed expansion history)**.** *Fix a scale factor $a(t)$ and let $\beta > 0$. Let two solutions of Eq. (3.5) driven by that $a(t)$ reach the same converted fraction f_* at the same time with different memory states $M^{(1)} > M^{(2)}$. Then $f^{(1)}(t) \geq f^{(2)}(t)$ at every later t, with strict inequality on a set of positive measure, and the vacuum accumulated through Eq. (4.4) along that same $a(t)$ differs between them.*

Remark 3.9 (Why the expansion history is prescribed). The hypothesis is not cosmetic and cannot be removed by rewording. The comparison argument below compares two solutions of one system, so both must be driven by the same $a(t)$; and the conclusion is that they accumulate different vacuum, which through the Friedmann constraint would give them different $a(t)$. Stated for the coupled system the theorem would therefore contradict its own hypothesis. What it establishes is a property of the kinetics: at equal converted fraction the memory state still carries information about the future, so the pair (ρ_I, ρ_D, V) at one instant does not determine the solution. Fig. 2 is computed exactly in this decoupled setting.

Proof. Both solutions are driven by the same $a(t)$, so they solve one system, and Eq. (3.5) is cooperative: $\partial\dot{f}/\partial M = (1-f)\beta\Gamma \geq 0$ and $\partial\dot{M}/\partial f = \tau_M^{-1} > 0$. The Kamke–Müller comparison

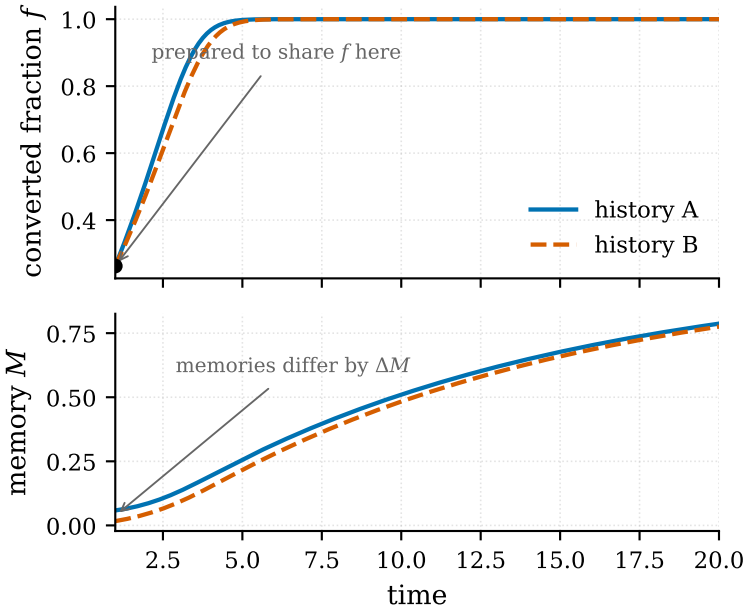


Figure 2. Hysteresis. Two histories are prepared to reach the same converted fraction at the first plotted time while carrying different memory states. Evolved thereafter under identical conditions on a matter-dominated background they separate, both running to $f = 1$ — the converted fractions agree to -5.72×10^{-13} — while their asymptotic vacuum densities differ by 0.0841 in relative terms. In the memoryless limit the two curves coincide to solver tolerance.

theorem for cooperative systems gives componentwise ordering of the two solutions from ordered initial data, and strictness follows because the vector fields differ wherever $M^{(1)} \neq M^{(2)}$ and $f < 1$. The vacuum statement follows by integrating Eq. (4.4), whose integrand depends on the epoch at which each increment of f occurs: the earlier history deposits its conversion when a is smaller and therefore accumulates more. □

It is worth being precise about what does *not* differ. Under Assumption 6.2 the integrated rate diverges, so by Proposition 3.7 both histories have $f \to 1$: the asymptotic converted *fractions* coincide. Hysteresis shows up in the accumulated vacuum, not in the endpoint of the kinetics, because Eq. (4.4) weights each increment of f by a^{-3} and the two histories convert at different epochs. The simulation of Fig. 2 reports exactly this: at the end of the integration window the converted fractions agree to -5.72×10^{-13} — both have run to completion — while the vacuum densities differ by 0.0841 in relative terms. The expansion is what separates the two: with a held constant the vacuum increment *is* the converted fraction, by Eq. (4.4), so the two differences coincide identically and no such run can exhibit the theorem. A test asserts that degeneracy, so the choice of an expanding background here is checked rather than asserted.

It is worth being exact about what this distinguishes the model from, because the slogan version of the statement compares the wrong things. Theorem 3.8 compares two histories at equal *converted fraction*, and by Eq. (4.4) their vacuum densities then differ, so it is not a comparison at equal densities and cannot be read as one. What it does establish is a statement about the state space. An interaction written as $Q(\rho_{\rm dm}, \rho_{\rm de}, H)$ closes on the densities: together with the Friedmann constraint they form an autonomous system, and the present densities fix the entire future, whatever the functional form and however its sign behaves with redshift. Here they do not, because M is an independent state variable that no function of the densities reproduces: the pair (f, M) is Markov and f alone is not (Proposition 3.2). That is the difference, and it is a difference in the dimension of the state space rather than in the shape of a coupling. Sec. 7 turns it into an observable by showing that β and τ_M enter the snap parameter while cancelling out of the jerk.

What this section establishes. A well-posed kinetic law with a controlled deterministic limit, a memory variable that is the population mean of a per-unit relaxation law with a common relaxation time, rather than a weaker assumption, and path dependence. What it does not establish is that any physical substrate realises this law: Γ_0, α, β and τ_M are parameters of an effective description, and Sec. 8 exhibits candidate mechanisms without deriving their values.

4 Homogeneous cosmology with two products

We now couple the kinetic law of Sec. 3 to gravity. The results of this section are exact consequences of the closure stated below: identities relating the converted fraction to the surviving densities, global positivity and boundedness, and a de Sitter attractor reached without the Hubble rate ever growing.

4.1 The closure

Assumption 4.1 (Homogeneous closure)**.** Space is flat and FLRW. The precursor ρ_I and the localized product ρ_D are pressureless, the third product is an exact vacuum with $p_V = -V$, and baryons and radiation are separately conserved. A fraction $b \in [0,1]$ of the converted energy feeds the localized product and the rest feeds the vacuum. No other sector of negative pressure is present.

A — Postulate

$$\begin{aligned} &\dot{\rho_I} + 3H\rho_I = -\Gamma\rho_I, \quad \dot{\rho_D} + 3H\rho_D = b\,\Gamma\rho_I, \qquad \dot{V} = (1-b)\Gamma\rho_I \coloneqq Q_V, \\ &\dot{\rho_{\rm b}} + 3H\rho_{\rm b} = 0, \qquad \dot{\rho_{\rm r}} + 4H\rho_{\rm r} = 0, \qquad 3M_{\rm Pl}^2H^2 = \rho_I + \rho_D + \rho_{\rm b} + \rho_{\rm r} + V, \end{aligned} \tag{4.1}$$

together with Eq. (3.5) for (f, M) and $\dot{a} = Ha$. The total pressure is $p = \rho_{\rm r}/3 - V$, and summing the five density equations returns the total continuity equation exactly: the transfer terms $-\Gamma\rho_I + b\Gamma\rho_I + (1-b)\Gamma\rho_I$ cancel, so Eq. (2.5) holds identically. In a given realization V already carries the entire effective tensor of the homogeneous product; there is no separate energy of the field f to add again.

4.2 What the kinetics is not responsible for

Before deriving what the transfer does, it is worth recording what it cannot do, because the point is easy to lose and it fixes the division of labour in everything that follows.

Proposition 4.2 (Conversion cannot manufacture an equation of state)**.** *Let the dark sector consist of components exchanging energy with no momentum transfer, each a perfect fluid with its own density ρ_i and pressure p_i* and a common four-velocity*. Then the total pressure is $\sum_i p_i$, independent of every transfer rate. In particular, if every component is pressureless the total is pressureless, and, provided the total density is non-negative,*

C — Theorem of the model

$$\ddot{a} \le 0 \quad \textit{for all } t, \tag{4.2}$$

whatever the rate, the reinforcement, or the memory.

Proof. Energy exchange moves ρ between components and enters each continuity equation as a source; it does not appear in the stress tensor's spatial trace, which is $\sum_i p_i$ by construction. The acceleration equation depends on the total through $\rho + 3p$ alone, and the sources cancel in the sum by total conservation. □

Both added hypotheses earn their place. A common four-velocity is needed because perfect fluids in relative motion generate an isotropic pressure from that motion, so "each a perfect fluid" alone does not give $p_{\rm tot} = \sum_i p_i$; in this model the dark components share the congruence

of Assumption 4.1 and the hypothesis holds. Non-negativity of the total density is needed for the second conclusion and not the first: with $p_{\rm tot} = 0$ the acceleration equation gives $\ddot{a}/a = -\rho_{\rm tot}/(6M_{\rm Pl}^2)$, so $\ddot{a} \le 0$ requires $\rho_{\rm tot} \ge 0$ — which Theorem 5.1 supplies towards the future, and which Sec. 6 shows is exactly what can fail towards the past.

Read as an effective bulk pressure the two halves are individually non-trivial and individually misleading. Solving $\dot{\rho}_i + 3H(\rho_i + \Pi_i) = 0$ for the Π_i that reproduces the model's equations gives $\Pi_I = +\Gamma\rho_I/3H$ for the donor and $\Pi_D = -b\Gamma\rho_I/3H$ for the recipient: the recipient does acquire a negative effective pressure, which is the familiar result from particle-creation cosmology, and the donor acquires an equal and opposite one. Their sum is $(1-b)\Gamma\rho_I/3H$, which vanishes identically at $b = 1$. Read as a viscosity the donor's coefficient is $\zeta_I = -\Gamma\rho_I/9H^2 < 0$, which the second law forbids — the tell that the precursor is not dissipating but handing energy to something else.

Integrating the two-sector system with all conversion going to the defects, over random draws spanning two decades in rate and the full ranges of reinforcement and memory, gives $(\rho + 3p)/\rho = 1$ to the solver tolerance and $\mathrm{d}\rho/\mathrm{d}N = -3\rho$ to 6×10^{-13}.

The consequence is the division of labour this paper works under. The acceleration comes from the vacuum's $p = -V$, which is postulated in Assumption 4.1 and not derived; the kinetics, the reinforcement and the memory decide only how much vacuum exists by today. Any programme that hopes to produce acceleration from conversion alone, from vortical structure, or from averaging, has to break one of the hypotheses above — and the only one this paper's framework leaves available is that the recipient be pressureless.

4.3 Exact identities

Proposition 4.3 (Survival and product identities)**.** *With $a_i = 1$, constant b and $f_i < 1$, the system Eq. (4.1) satisfies*

I — Identity

$$\rho_I a^3 = \rho_{I,i}\,\frac{1-f}{1-f_i}, \qquad \rho_D a^3 = \rho_{D,i} + b\,\rho_{I,i}\,\frac{f-f_i}{1-f_i}, \tag{4.3}$$

and, wherever $\dot{f} > 0$,

I — Identity

$$\frac{\mathrm{d}V}{\mathrm{d}f} = \frac{(1-b)\rho_{I,i}}{a^3(1-f_i)}. \tag{4.4}$$

Proof. The first equation of Eq. (4.1) gives $\mathrm{d}\log(\rho_I a^3)/\mathrm{d}t = -\Gamma$, while $\mathrm{d}\log(1-f)/\mathrm{d}t = -\Gamma$ by Eq. (3.8); their difference is constant. Substituting into $\mathrm{d}(\rho_D a^3)/\mathrm{d}t = b\Gamma\rho_I a^3$ and using $\dot{f} = (1-f)\Gamma$ integrates directly. Dividing $\dot{V}$ by $\dot{f}$ gives Eq. (4.4). □

Eq. (4.4) is the reason the converted fraction does not by itself determine the vacuum density: the same amount of conversion occurring at different epochs deposits different amounts of energy, because the donor has been diluted by different factors. It is also the mechanism behind Theorem 3.8 at the cosmological level, since two histories reaching a given f along different paths integrate Eq. (4.4) against different $a(f)$.

A related warning: the statistical fraction is not the energy fraction. In the two-sector subcase $b = 0$ with $x = V/(\rho_I + V)$ one finds $\dot{x} = (1-x)(\Gamma + 3Hx)$, whose second term is absent from the equation for f. Even with no conversion at all a pre-existing vacuum increases its energy share as matter dilutes, so $f = \Omega_V$ is not an admissible identity.

4.4 Positivity, conservation and global bounds

Theorem 4.4 (Regular future)**.** *Assume Assumption 4.1; non-negative initial densities with positive sum; $a_i = 1$, $f_i \in [0,1)$, $M_i \in [0,1]$, $b \in [0,1]$, $\tau_M > 0$; and Γ locally Lipschitz with*

$0 \le \Gamma \le G < \infty$ *on the physical domain. Then a unique solution exists for all finite future times, all densities remain non-negative,* $f, M \in [0,1]$*,* $H > 0$*, and*

C — Theorem of the model

$$\dot{H} = -\frac{\rho_I + \rho_D + \rho_{\rm b} + \frac{4}{3}\rho_{\rm r}}{2M_{\rm Pl}^2} \le 0, \qquad 0 < H(t) \le H_i, \qquad V_i \le V(t) \le V_i + (1-b)\rho_{I,i}. \tag{4.5}$$

Proof. Each source-free equation integrates to a positive exponential times its initial value; the equations for ρ_D and V have non-negative sources, and their integrating-factor solutions preserve sign. Formally, the vector field of Eq. (4.1) is inward-pointing on every face of the non-negative orthant, because every outflow is proportional to the density of the sector it leaves; the Nagumo subtangent condition then gives forward invariance. Proposition 3.7 and Eq. (3.3) confine (f, M) to the unit square. Summing the density equations gives the total continuity equation, and differentiating the Friedmann constraint gives the first of Eq. (4.5), hence $H \le H_i$. For the vacuum bound, use $a \ge 1$ and Eq. (4.3):

$$V(t) - V_i = (1-b)\int_{t_i}^{t} \Gamma \rho_I \, \mathrm{d}s \le (1-b)\rho_{I,i}\left[1 - e^{-\int_{t_i}^{t} \Gamma \, \mathrm{d}s}\right] \le (1-b)\rho_{I,i}. \tag{4.6}$$

Finally $\dot{\rho_{\rm tot}} \ge -4H_i\rho_{\rm tot}$, so the total density cannot reach zero in finite time and the positive Friedmann root stays bounded away from zero; local existence therefore extends globally to the future. □

This is a statement about a system of ordinary differential equations. It is not a global existence result for the Einstein equations with inhomogeneities, caustics or gravitational collapse.

4.5 Late-time attractor

Theorem 4.5 (de Sitter attractor without superacceleration)**.** *Add to Theorem 4.4 either* $V_i > 0$*, or* $b < 1$ *together with* $\rho_{I,i} > 0$ *and* $\int^{\infty} \Gamma \, \mathrm{d}t = \infty$*. Then* $V \to V_\infty > 0$*, all other densities tend to zero,*

C — Theorem of the model

$$H(t) \longrightarrow H_\infty = \sqrt{\frac{V_\infty}{3M_{\rm Pl}^2}} > 0, \qquad a(t)e^{-H_\infty(t-t_i)} \longrightarrow C \in (0,\infty), \tag{4.7}$$

and the expansion accelerates from some finite time onwards.

Proof. Under the second alternative $\int^{\infty} \Gamma \, \mathrm{d}t = \infty$, so Γ cannot vanish identically and there is t_1 with $\Gamma(t_1) > 0$; since $\rho_I(t_1) > 0$ by Eq. (4.3) and $b < 1$, the source Q_V is positive there and $V(t) > 0$ for $t > t_1$. Note that Q_V need *not* be positive initially — under Assumption 6.2 it is not, since $\Gamma \to 0$ as $a \to 0$ — so $H > 0$ before t_1 comes from Theorem 4.4 and not from this argument. For $t > t_1$, by monotonicity $H \ge \sqrt{V(t_1)/3M_{\rm Pl}^2} > 0$ thereafter, hence a grows at least exponentially. Eq. (4.3) bounds the comoving dust density, so dust and radiation decay exponentially in t, while V is monotone and bounded by Eq. (4.5) and therefore converges to a positive limit; the Friedmann constraint gives the first of Eq. (4.7). Moreover $V_\infty - V(t)$ decays at least exponentially, so $H - H_\infty$ is integrable and integrating $\mathrm{d}\log a/\mathrm{d}t - H_\infty$ gives the second. Finally $\ddot{a}/a \to H_\infty^2 > 0$, so acceleration sets in at some finite time. □

Corollary 4.6 (At most one transition)**.** *With* $\rho_{\rm m} = \rho_I + \rho_D + \rho_{\rm b}$ *and* $\mathcal{A} = \ddot{a}/a$*,*

I — Identity

$$\dot{\mathcal{A}} = \frac{3Q_V + 3H\rho_{\rm m} + 8H\rho_{\rm r}}{6M_{\rm Pl}^2} \ge 0, \tag{4.8}$$

strictly when matter or radiation is present. A solution that begins decelerating and satisfies Theorem 4.5 therefore crosses $\mathcal{A} = 0$ *exactly once. A history requiring several late alternations between acceleration and deceleration does not belong to this closure.*

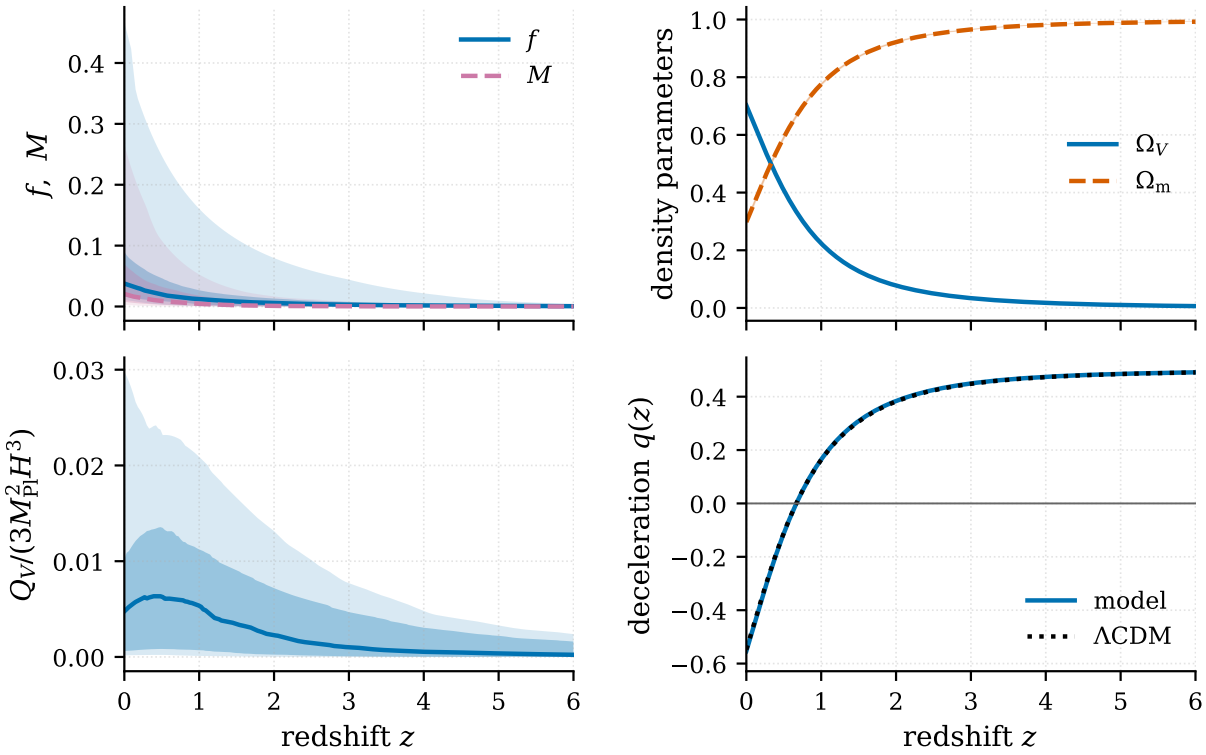


Figure 3. The background solution for the full model, drawn from the nested posterior of Table 3. *Top left:* the density parameters, with the precursor draining into the vacuum after activation. *Top right:* the converted fraction and its memory, the lag between them being what Theorem 3.8 makes operational. *Bottom left:* the deceleration parameter, crossing zero once, as Theorem 4.4 requires. *Bottom right:* the transfer Q_V, non-negative throughout, which is the content of Theorem 5.1. Bands are 68 and 95 per cent of the posterior; the dotted curve is ΛCDM at its own best fit.

Proof. Differentiate $\mathcal{A} = (2V - \rho_{\rm m} - 2\rho_{\rm r})/(6M_{\rm Pl}^2)$ and substitute $\dot{\rho_{\rm m}} = -3H\rho_{\rm m} - Q_V$ and $\dot{\rho}_{\rm r} = -4H\rho_{\rm r}$. □

4.6 Four things that can be true at once

It is worth stating plainly, because the draft version of this work conflated them: the converted fraction grows, the vacuum density grows, the acceleration $\ddot{a}/a$ increases, and the Hubble rate decreases. These are four different statements about four different quantities, related by the equations above and in no tension with one another. In particular, if one insists on writing $\dot{V} + 3H(1 + w_{V,\rm eff})V = 0$ then $w_{V,\rm eff} = -1 - Q_V/(3HV) \leq -1$, yet the physical vacuum stress tensor still has $p_V = -V$ exactly, the total satisfies $\rho + p \geq 0$, and Eq. (4.5) forbids $\dot{H} > 0$. There is no superacceleration and no future singularity within these hypotheses. What $w_{V,\rm eff}$ is *not* is the $w(z)$ that an observational fit assuming conserved matter would report; Sec. 5 replaces this bookkeeping quantity with statements that do not depend on how a fit is anchored.

What this section establishes. Exact identities and global bounds for the homogeneous system, under Assumption 4.1. What it does not establish is that the closure describes our universe, nor that the past extension of these solutions is physical: with a rate that does not switch off, the vacuum density becomes negative at early times, which is the problem Sec. 6 repairs, by assumption rather than by mechanism.

5 Positivity, and the map to interacting dark energy

A recent survey of interacting dark energy models found that in seven of eight standard couplings the energy densities become negative somewhere in the physical domain, and it is important to quote its conclusion in full rather than in the half that suits the present model: the negative densities arise *if energy flows from dark matter to dark energy*, which the authors read as a strong theoretical preference for the opposite direction [23–25]. The model of this paper flows the disfavoured way. Its precursor is pressureless and its product is vacuum, so if that preference is a statement about interacting dark energy in general, this model is excluded on theoretical grounds before any data are considered.

It is not such a statement, and the survey's own results show why. Its eight couplings all have the form $Q = 3H\delta\, F(\rho_{\rm dm}, \rho_{\rm de})$, and exactly one of them keeps every density positive: $F = \rho_{\rm dm}\rho_{\rm de}/(\rho_{\rm dm} + \rho_{\rm de})$. That is the one whose outflow vanishes on each face it could leave through, which is the Nagumo criterion stated below and the property the present model has by construction. The direction of flow is the same for all eight; what separates the one from the seven is whether the transfer switches off when the donor runs out. A transfer whose size is set by the sector *receiving* energy does not, and keeps draining a reservoir that is already empty. The survey's conclusion is therefore a property of the couplings it surveys and not of the direction of flow — a claim this paper owes the reader, having placed itself on the wrong side of it, and one its own table supports. The section then maps the model onto the standard parametrizations and states what it predicts that cannot be fitted away.

5.1 Positivity is structural

Theorem 5.1 (Quasi-positivity)**.** *Assume Assumption 4.1 with non-negative initial data, and let $\Gamma \geq 0$ be merely locally integrable along the solution: no lower bound, no upper bound, and no continuity are required. Then the non-negative orthant is forward invariant: every density in Eq. (4.1) remains non-negative for as long as the solution exists.*

Proof. Not by the Nagumo subtangent criterion, which is a statement about continuous vector fields and would need a continuity hypothesis the theorem declines to make. Integrating factors suffice and need nothing beyond local integrability. The precursor equation is linear in ρ_I, so

$$\rho_I(t) = \rho_I(t_i)\exp\left[-\int_{t_i}^{t}(3H+\Gamma)\,\mathrm{d}s\right] \geq 0,$$

the integral being finite on any interval where the solution exists. With $\rho_I \geq 0$ the sources $b\Gamma\rho_I$ and $(1-b)\Gamma\rho_I$ are non-negative, and the same integrating factor applied to ρ_D and to V writes each as a non-negative initial value times a positive exponential plus a non-negative integral of those sources. Baryons and radiation are unsourced and the same argument applies with no source term. □

Two remarks on what the theorem does and does not assert. It is conditional on existence — it says the densities cannot change sign while a solution lasts, not that one lasts forever — and existence on a maximal interval follows from Theorem 4.4 under the extra hypotheses stated there, not from this argument. And "forward" is essential; see the first of the two consequences recorded below.

The mechanism is visible in the proof and is worth naming, because it is what the pathological models lack: *every outflow is proportional to the density of the sector it leaves.* The converse half of the dichotomy is what makes the comparison informative, and it is equally elementary.

Proposition 5.2 (Recipient-proportional couplings are not positivity-safe)**.** *Consider two sectors with $\dot\rho_1 + 3H\rho_1 = -Q$ and $\dot\rho_2 + 3H(1+w)\rho_2 = +Q$, and let $Q = \xi H\rho_2$ with $\xi > 0$.*

1. *On the face* $\{\rho_1 = 0\}$ *the field is* $\dot{\rho}_1 = -\xi H \rho_2 < 0$ *whenever* $\rho_2 > 0$*, so the subtangent condition fails on a set of codimension one and the non-negative orthant is* not *forward invariant.*

2. *Whether a given trajectory reaches that face is a separate question with a separate answer. Writing* $u = \rho_1 a^3$ *gives* $\mathrm{d}u/\mathrm{d}N = -\xi \rho_{2i} e^{-3wN}$*, so* ρ_1 *reaches zero at finite* a *for every* $\xi > 0$ *if and only if* $w \leq 0$*, and for* $w > 0$ *only if* $\xi > 3w\rho_{1i}/\rho_{2i}$*.*

The same argument with the roles exchanged applies to $Q = \xi H \rho_1$ *draining* ρ_2*.*

The second part is not a technicality, and the first is what the comparison rests on. A stiff recipient, $w = 1$, makes the integral converge: the outflow dies away faster than the donor dilutes, and for small enough ξ the donor survives. The failure is therefore guaranteed exactly in the regime the catalogued models occupy — a recipient that behaves like dark energy, w close to -1 — and the unqualified statement that ρ_1 "is driven through zero" is true there and not in general.

It is worth stating the sharp criterion, because *donor versus recipient* is a useful label and not the underlying condition. By Nagumo the orthant is forward invariant if and only if the outflow vanishes on the face it would leave through, $Q|_{\rho_1=0} \leq 0$. Donor-proportionality, $Q \propto \rho_1$, is the simplest sufficient case and the one this model realises — and it additionally gives uniqueness, which a coupling like $Q = \xi H \sqrt{\rho_1 \rho_2}$ satisfies the invariance condition without. What the recipient-proportional couplings fail is the condition itself: $Q|_{\rho_1=0} = \xi H \rho_2 > 0$, an outflow from a sector that has nothing left. Here the precursor is a reservoir that empties, and an empty reservoir supplies nothing.

Two consequences should be recorded immediately. First, Theorem 5.1 is a statement about the future. It is not time-symmetric: run backwards, V decreases and can reach zero at a finite scale factor. That is not a pathology of the theorem but a genuine feature of the model, and Sec. 6 shows what must be assumed to avoid it. Second, positivity here is structural rather than a consequence of tuning: no choice of Γ_0, α, β or τ_M can violate it.

5.2 Which interacting dark energy model is this?

Proposition 5.3 (Exact map)**.** *With* $\rho_{\rm dm}^{\rm eff} = \rho_I + \rho_D$ *and* $s = \rho_I/\rho_{\rm dm}^{\rm eff}$*,*

I — Identity

$$Q_V = \xi_{\rm eff}\, H\, \rho_{\rm dm}^{\rm eff}, \qquad \xi_{\rm eff} = \frac{(1-b)\Gamma s}{H} \geq 0. \tag{5.1}$$

The model reduces to the standard coupling $Q = \xi H \rho_{\rm dm}$ *with constant* ξ *if and only if* $b = 0$ *and* $\Gamma = \xi H$*. It can never take the form* $Q \propto \rho_{\rm de}$*.*

So the model is not a relabelling of a familiar parametrization: it occupies a restricted corner of one family, with $\xi_{\rm eff}$ a specific decreasing function of scale factor determined by the kinetics, and it is disjoint from the other. This matters when comparing with published constraints, because those are quoted for the general family.

5.3 The prediction that cannot be fitted away

Theorem 5.4 (No sign flip)**.** *Define* $\delta_{\rm eff} = \mathrm{d}\ln(\rho_{\rm dm}^{\rm eff} a^3)/\mathrm{d}\ln a$*. Under Assumption 4.1 with* $\rho_{\rm dm}^{\rm eff} > 0$*,*

C — Theorem of the model

$$\delta_{\rm eff}(a) = -\frac{Q_V}{H\rho_{\rm dm}^{\rm eff}} = -\xi_{\rm eff}(a) \leq 0 \qquad \textit{for every } a, \tag{5.2}$$

for every admissible rate law, with equality only where $\Gamma \rho_I = 0$ *or* $b = 1$*. The same holds for the total dust.*

Proof. Summing the first two equations of Eq. (4.1) gives $\mathrm{d}(\rho_{\mathrm{dm}}^{\mathrm{eff}} a^3)/\mathrm{d}t = -Q_V a^3$. Divide by $\rho_{\mathrm{dm}}^{\mathrm{eff}} a^3 > 0$ and by $H > 0$; then $Q_V = (1-b)\Gamma\rho_I \geq 0$ by Theorem 5.1. Baryons are conserved and only enter the denominator. □

Corollary 5.5 (Memory does not enter the sign)**.** *The memory parameters (β, τ_M) affect Eq. (5.2) only through the scalar $\Gamma \geq 0$. They can change the magnitude of δ_{eff} by a bounded factor and shift the epoch at which it peaks, but they cannot change its sign.*

We state the confrontation with data plainly, because it runs against the model. Fits in the DESI DR2 era weakly prefer the opposite sign, that is, energy flowing from dark energy into dark matter [28–30]. By Theorem 5.4 this is not something the one-way model can accommodate by adjusting parameters: a robust detection of $\delta_{\mathrm{eff}} > 0$ falsifies it, independently of b, α, β, τ_M and the activation profile. Two qualifications belong alongside that statement, and neither is a defence of the model. The current preference is weak and is entangled with the choice of coupling; and by Proposition 5.2 some of the parametrizations carrying that preference are recipient-proportional and therefore admit negative densities somewhere in their own parameter space, which argues for repeating such analyses with a donor-proportional coupling rather than for dismissing the result. If the preference strengthens, the disciplined response is the reverse channel of Sec. 5.5, which is a different model, not a reparametrization of this one.

5.4 The expansion history is bracketed

The draft version of this work argued that the model would appear phantom to an observer. That argument does not survive scrutiny: what a fit reports depends on what it holds fixed. The following replaces it with a statement that is independent of that choice.

Theorem 5.6 (Sandwich)**.** *Let the solution be defined and non-negative on $[a_e, 1]$ with radiation exactly conserved, and define the matter deficit*

I — Identity

$$\Delta := \frac{\rho_{\mathrm{m}}^{\mathrm{early}} - \rho_{\mathrm{m0}}}{3M_{\mathrm{Pl}}^2 H_0^2} = \frac{1}{3M_{\mathrm{Pl}}^2 H_0^2}\int_{t_e}^{t_0} Q_V a^3\,\mathrm{d}t \;\geq 0, \qquad \rho_{\mathrm{m}}^{\mathrm{early}} := \rho_{\mathrm{m}}(t_e)\, a_e^3, \tag{5.3}$$

both densities being comoving, that is, referred to $a = 1$ today. Then for all $a \in [a_e, 1]$

C — Theorem of the model

$$E^2_{\Lambda\mathrm{CDM}}(\Omega_{\mathrm{m0}}, \Omega_{V0}) \leq E^2(a) \leq E^2_{\Lambda\mathrm{CDM}}(\Omega_{\mathrm{m0}} + \Delta,\ \Omega_{V0} - \Delta)\,, \tag{5.4}$$

where both comparison models are spatially flat, carry the same radiation density, and share the observed H_0, differing only by moving the amount Δ from the cosmological constant to matter. Equality holds at $a = 1$ on both sides. If $\Delta > \Omega_{V0}$ the upper comparison model has a negative cosmological constant; it remains a valid bound on E^2, but ceases to be a physical ΛCDM *cosmology.*

Remark 5.7 (The naive bracketing is wrong)*.* The version of this statement one writes down first is false, so it is worth recording. Bracketing instead by the pair $(\rho_{\mathrm{m}}^{\mathrm{early}}, V_{\mathrm{early}})$ does *not* give an upper bound: that model is a lower bound, and it does not share H_0.

Proof. Write $3M_{\mathrm{Pl}}^2 H^2 = m(t)a^{-3} + \rho_{\mathrm{r0}} a^{-4} + V(t)$ with $m = \rho_{\mathrm{m}} a^3$, exact by the Friedmann constraint; both comparison models carry the same radiation density ρ_{r0}, which therefore cancels from every difference below. From Eq. (4.1), $\dot{m} = -Q_V a^3$ and $\dot{V} = Q_V$.

For the lower bound, integrate both backwards from today: $m(t) - \rho_{\mathrm{m0}} = \int_t^{t_0} Q_V a'^3\,\mathrm{d}t'$ and $V(t) - V_0 = -\int_t^{t_0} Q_V\,\mathrm{d}t'$, so

$$3M_{\mathrm{Pl}}^2\left[H^2 - H_{\mathrm{low}}^2\right] = \int_t^{t_0} Q_V\left[\left(\frac{a'}{a}\right)^3 - 1\right]\mathrm{d}t' \;\geq\; 0, \tag{5.5}$$

since $a' \geq a$ and $Q_V \geq 0$.

The upper bound is not the same integral with the endpoints exchanged; it has a different structure, because the high comparison model is anchored at t_e while H_0 is fixed today. Splitting the range at t,

$$3M_{\rm Pl}^2\left[H_{\rm high}^2 - H^2\right] = \int_{t_e}^{t} Q_V a'^3\left(a^{-3}-1\right)\mathrm{d}t' + \int_{t}^{t_0} Q_V\left(1-a'^3\right)\mathrm{d}t', \tag{5.6}$$

and both terms are non-negative for $a \leq 1$, since $a' \leq 1$ throughout. Equality holds at $a = 1$ on both sides because both integrals then have vanishing range or vanishing integrand. □

The practical content is that the model cannot mimic an arbitrary expansion history: it lives in a band whose width is one number, the matter deficit Δ, which is the difference between the matter density inferred from the early universe and the one inferred locally. That is the honest observable, and it is what Sec. 10 reports, together with $H(z)$ and $j(z)$, in place of an anchoring-dependent effective w.

5.5 The reverse channel

Adding a channel that returns vacuum energy to the localized sector,

$$\dot{V} = (1-b)\Gamma\rho_I - \Gamma_{\downarrow}V, \qquad \dot{\rho_D} + 3H\rho_D = b\Gamma\rho_I + \Gamma_{\downarrow}V, \tag{5.7}$$

preserves Theorem 5.1 verbatim, because the new outflow is again proportional to the density of the sector it leaves. What it does change is Theorem 5.4: the net transfer $Q_{\rm net} = (1-b)\Gamma\rho_I - \Gamma_{\downarrow}V$ becomes negative once the vacuum dominates, generically at late times, so $\delta_{\rm eff}$ can flip sign. In the four-form picture of Sec. 8.5 this channel is nothing exotic: it is ordinary downward membrane nucleation, the process the upward one competes against.

The two tracks are therefore distinguished by the sign of a single cosmographic quantity, which is the subject of Sec. 7. Both appear in the model comparison of Sec. 10.

What this section establishes. That positivity holds structurally rather than by tuning; the exact correspondence with the standard interacting parametrizations; and a falsifiable sign prediction that no choice of parameters can evade. What it does not establish is that the sign predicted is the one nature chose; current data lean the other way, weakly, and the paper reports that rather than working around it.

6 A solvable family and the early-universe branch

This section does two things. It exhibits a one-parameter family for which the whole background is elementary, used throughout as a numerical benchmark; and it repairs a defect of the rate law as originally written, which is that a positive rate floor is incompatible with a standard early universe.

6.1 Conversion proportional to the expansion

Take $\Gamma = \xi H$ with $\xi \geq 0$ constant, activation switched hard at $a_{\rm act}$. Then $\mathrm{d}f/\mathrm{d}N = (1-f)\xi$ is independent of the expansion rate, and with $n = 3+\xi$ the system integrates in closed form:

I — Identity

$$\begin{aligned} f &= 1-(a/a_{\rm act})^{-\xi}, & \rho_I &= \rho_{I,\rm act}\,(a/a_{\rm act})^{-n}, \\ \rho_D &= (a/a_{\rm act})^{-3}\left[\rho_{D,\rm act} + b\,\rho_{I,\rm act}\left(1-(a/a_{\rm act})^{-\xi}\right)\right], & V &= V_{\rm act} + \frac{(1-b)\xi}{3+\xi}\rho_{I,\rm act}\left(1-(a/a_{\rm act})^{-n}\right). \end{aligned} \tag{6.1}$$

Summing gives the expansion history explicitly,

$$3M_{\rm Pl}^2H^2 = V_\infty + (\rho_{\rm b,act} + \rho_{D,\rm act} + b\rho_{I,\rm act})\,\hat a^{-3} + \frac{3(1-b)}{3+\xi}\rho_{I,\rm act}\,\hat a^{-(3+\xi)} + \rho_{\rm r,act}\,\hat a^{-4}, \qquad \hat a := a/a_{\rm act}, \tag{6.2}$$

with $V_\infty = V_{\rm act} + (1-b)\xi\rho_{I,\rm act}/(3+\xi)$. Every density subscripted "act" is the physical density at $a = a_{\rm act}$, not a comoving one referred to $a = 1$, which is why each power is written in $a/a_{\rm act}$; the two conventions differ by factors of $a_{\rm act}$ and agree only for $a_{\rm act} = 1$. In the two-sector subcase one may go further and solve for the time dependence, $H = H_\infty \coth y$ with $y = \frac{n}{2}H_\infty(t - t_B)$ and $a^n \propto \sinh^2 y$; substituting into Eq. (8.3) gives a closed-form potential $V(T) = M_{\rm Pl}^2H_\infty^2[3 - \xi\,\mathrm{csch}^2(\frac{n}{2}H_\infty(T - t_B))]$ that realises this trajectory exactly. This is an existence statement about a local covariant action, nothing more: t_B is a parameter of the mathematical extension, not necessarily a physical big bang.

Eq. (6.1) is the benchmark against which the numerical integrator is verified in Appendix F; agreement is at the level of 10^{-9} in the converted fraction.

6.2 A positive rate floor has no early universe

Extending Eq. (6.1) backwards makes the problem visible: V crosses zero at a finite scale factor. The draft treated this as a restriction on the domain. It is worse than that, and the following makes the obstruction quantitative for the rate law as originally written.

Theorem 6.1 (Rate-floor no-go)**.** *Suppose the solution exists with non-negative densities on $[a_i, 1]$, that $\Gamma \geq \Gamma_0 > 0$ throughout, that $b < 1$ and $\rho_{I0} > 0$, and that H is non-increasing. Then*

C — Theorem of the model

$$V_0 - V(a_i) \;\geq\; c\,(1-b)M_{\rm Pl}^2\,\Gamma_0 H_i, \qquad \Gamma_0 \;\lesssim\; \frac{H_0^2}{H_i}, \tag{6.3}$$

with $c = \mathcal{O}(1)$ set by the precursor fraction at a_i. Consequently the fraction converted over the entire late universe is $\Delta f \lesssim H_0/H_i$.

Proof. From $V_0 - V(a_i) = (1-b)\int \Gamma\rho_I\,\mathrm{d}t \geq (1-b)\Gamma_0\int \rho_I\,\mathrm{d}t$, change variables with $\mathrm{d}t = \mathrm{d}a/(aH)$ and use $\rho_I \geq \rho_{I0}a^{-3}$ together with $H \leq H_i$ to get $\int \rho_I\,\mathrm{d}t \geq \rho_{I0}(a_i^{-3} - 1)/(3H_i)$. The second inequality follows from $V(a_i) \geq 0$, which is Theorem 5.1, and $V_0 \leq 3M_{\rm Pl}^2H_0^2$. The statement about Δf follows from $\dot f \leq \Gamma \leq \Gamma_0 e^{\alpha+\beta}$. □

Anchoring a_i anywhere in the radiation era makes H_i/H_0 astronomically large, so the converted fraction is astronomically small: a rate with a positive floor either violates positivity in the past or converts nothing at all. The rate law of the draft therefore has no viable early universe, and the hypothesis $\Gamma \geq g > 0$ used in its convergence theorems cannot be assumed back to early times. This is why Theorem 4.4 was stated above with the weak hypothesis $0 \leq \Gamma \leq G$ rather than the draft's $0 < g \leq \Gamma \leq G$: the activation law satisfies the former everywhere and the latter nowhere in the past. Theorem 4.5 needs one condition that the floor supplied for free, namely $\int^\infty \Gamma\,\mathrm{d}t = \infty$, and that is why it appears there as an explicit hypothesis rather than as a consequence. Neither theorem needs a restriction of its time domain; what changes is which hypothesis does the work.

6.3 The activation law

Assumption 6.2 (Activation)**.** Replace the floor by a switch,

A — Postulate

$$\Gamma(a, f, M) = \Gamma_0\,S_{\rm act}(a)\,e^{\alpha f + \beta M}, \qquad S_{\rm act}(a) = \frac{a^p}{a^p + a_{\rm act}^p}, \tag{6.4}$$

with the exponent p constrained by Proposition 6.4 below.

Remark 6.3 (The switch is local, and it closes under compression). As written, $S_{\rm act}$ refers to the scale factor, which is not a local scalar, so Eq. (6.4) is a statement about the homogeneous mode. It need not stay one. The survival identity $\rho_I a^3 = \rho_{I,\rm act}(1-f)$ is exact, so a can be eliminated with no approximation: writing $\rho_* = \rho_{I,\rm act}/a_{\rm act}^3$ for the comoving precursor density at activation,

I — Identity

$$S_{\rm act} = \left[1 + \left(\frac{\rho_I}{(1-f)\rho_*}\right)^{p/3}\right]^{-1}, \tag{6.5}$$

which involves only ρ_I and f and agrees with $S_{\rm act}(a)$ identically on FLRW. It also reads more physically than the original: conversion switches on once the precursor has diluted below a threshold density, rather than at a particular epoch.

The rewriting has a consequence that matters for Sec. 11.5. Under compression at fixed f the switch *closes*, $S_{\rm act} \to ((1-f)\rho_*/\rho_I)^{p/3}$, and near a fold caustic $\rho_I \sim |J|^{-1}$, so $\Gamma \sim |J|^{p/3} \to 0$ — with the exponent carrying the caveat recorded below. The rate does not merely stay bounded on the approach, as Theorem 11.2 assumes; it vanishes, which is strictly worse for any hope of regularization.

Two caveats, and neither is small.

First, this is *a* local completion, not *the* local completion: it is distinguished by using only an exact identity of the model and by reproducing $S_{\rm act}(a)$ exactly rather than approximately, but a switch written in the expansion Θ also reduces to a function of a on FLRW and behaves in the opposite way at a caustic, where $\Theta \to -\infty$ and such a switch would saturate rather than close. The completions disagree precisely where the question is interesting, so we record Eq. (6.5) as what the model's own identity gives and not as a derivation of what happens at a caustic.

Second, and more sharply: Eq. (6.5) is not consistent with Assumption 9.2. Linearising at fixed f gives $\delta \ln \Gamma = \delta \ln S_{\rm act} = -\frac{p}{3}(1 - S_{\rm act})\,\delta_I$, which equals the $-\delta_I$ that Assumption 9.2 requires — and that Proposition 9.4 shows is forced — only where $S_{\rm act} = 1 - 3/p$, a single value of the scale factor, $S_{\rm act} = \frac{1}{2}$ for the $p = 6$ used here. Away from that instant this completion describes a theory in which the transfer is not comoving-uniform, which is a different theory from the one the rest of this paper analyses.

The consequence for Sec. 11.5 is that the exponent quoted above belongs to that other theory. In the theory as defined, $\delta Q_V = 0$ forces $\Gamma \propto 1/\rho_I$, so near a fold caustic $\Gamma \sim |J|$ — exponent one rather than $p/3$. The qualitative conclusion is unchanged and is in fact reached by both routes: the rate vanishes on the approach rather than merely staying bounded, so Theorem 11.2 applies a fortiori. It is the exponent, and the route to it, that depend on which completion one is willing to adopt.

The reinforcement and memory structure is untouched; only the floor is replaced. All results of Secs. 5, 7 and 9 survive verbatim, because each needs only $\Gamma \geq 0$.

Two rate laws are in play in this paper and the required exponent differs between them, so it is worth separating them explicitly. Eq. (6.4) is the law used for the fits of Sec. 10: Γ_0 is a constant rate. The solvable family of Sec. 6.1 instead has the rate track the expansion, $\Gamma = \xi_0 H S_{\rm act}(a) e^{\alpha f + \beta M}$, and is fitted separately as a benchmark; ξ_0 denotes its dimensionless amplitude.

Proposition 6.4 (Convergence and positivity). *Under Assumption 6.2 with $b < 1$, the limit $V_{\rm early} = \lim_{a\to 0^+} V(a)$ exists and is finite, V is non-decreasing, and therefore $V \geq 0$ on $(0,1]$ if and only if $V_{\rm early} \geq 0$. The vacuum increment is $V_0 - V_{\rm early} = (1-b)\int_0^1 \Gamma \rho_I \, {\rm d}a/(aH)$, so with $\rho_I \propto a^{-3}$ and a radiation-dominated deep past the integrand behaves as $S_{\rm act}(a)a^{-2}$ for a constant rate and as $S_{\rm act}(a)a^{-4}$ for the expansion-tracking rate. Convergence therefore requires*

C — Theorem of the model

$$p > 1 \quad \textit{for } \Gamma = \Gamma_0 S_{\rm act} e^{\alpha f + \beta M}, \qquad p > 3 \quad \textit{for } \Gamma = \xi_0 H S_{\rm act} e^{\alpha f + \beta M}. \tag{6.6}$$

In the second case a sufficient condition explicit in the parameters is $\xi_0 \mathcal{I}_{S_{\rm act}} \leq \Omega_{V0}(1-f_0)/[(1-b)e^{\alpha+\beta}\Omega_{I0}]$ with $\mathcal{I}_{S_{\rm act}} = \int_0^1 S_{\rm act}(a)a^{-4}{\rm d}a \simeq \left[(p-3)^{-1} + \frac{1}{3}\right] a_{\rm act}^{-3}$.

Eq. (6.6) is the analytic content of the repair: the switch must close faster than the donor dilutes. Note that it is a weaker requirement for the constant-rate law, which is the one fitted here; we nevertheless adopt $p = 6$ throughout, which satisfies both and puts the conversion rate at recombination far below anything that could matter. Two sub-models follow. In the *fully generated* case $V_{\rm early} = 0$, the matter deficit of Eq. (5.3) would have to equal the entire present vacuum density, which is excluded unless activation is pushed to redshifts where it is unobservable. In the *partially generated* case $V_{\rm early} > 0$ the deficit is small and the model is testable; this is the regime studied in Sec. 10.

Numerically, with $p = 6$ and activation at $z_{\rm act} = 1$ the conversion rate at recombination is below 10^{-21} of the expansion rate, so nucleosynthesis, recombination and the sound horizon are untouched, and the vacuum density generated after activation is a minority of its present value. The exponent is not cosmetic: with $p = 2$ the expansion-tracking law leaves a rate at recombination nine orders of magnitude larger, and by Eq. (6.6) its vacuum integral does not converge at all.

What this section establishes. A closed-form benchmark, a proof that the original rate law cannot be extended into the early universe, and a minimal repair with an explicit positivity condition. What it does not establish is a mechanism that switches conversion on at a particular epoch: $a_{\rm act}$ and p are parameters, and Sec. 13 lists deriving them as an open problem.

7 Cosmographic bounds: jerk and snap

The results of the previous sections are statements about a model. This section extracts from them two statements about the universe that do not depend on the rate law: an inequality on the third derivative of the scale factor, whose sign separates this model from its competitors, and a relation on the fourth derivative that is the only place where the memory parameters leave a background signature.

7.1 The jerk

Theorem 7.1 (Jerk identity and bound)**.** *For a spatially flat universe containing pressureless matter, conserved radiation and an exact vacuum that receives energy at rate Q_V, and nothing else,*

I — Identity

$$j = 1 + 2\Omega_{\rm r} + \varepsilon, \qquad \varepsilon \coloneqq \frac{Q_V}{2M_{\rm Pl}^2 H^3}, \tag{7.1}$$

and therefore, whenever the transfer is towards the vacuum,

C — Theorem of the model

$$Q_V \geq 0 \implies j \geq 1 + 2\Omega_{\rm r}. \tag{7.2}$$

Adding any further conserved species whose pressure is non-increasing in time — a massive neutrino, for instance — leaves Eq. (7.1) an inequality, $j \geq 1 + 2\Omega_{\rm r} + \varepsilon$, and therefore strengthens Eq. (7.2) rather than weakening it. The identity and the bound are stated separately because only the second survives the addition; Remark 7.2 quantifies the extra term for the fits of Sec. 10.

Remark 7.2 (Massive neutrinos strengthen the bound). The last hypothesis is there because the fits of Sec. 10 carry one massive neutrino eigenstate, which is neither pressureless nor conserved radiation, so a bound advertised as parameter-free must say what it does with it. Such a species contributes $-\dot{p}/(2M_{\rm Pl}^2 H^3)$ to j, by the same identity as every other pressure, and its pressure falls monotonically as it becomes non-relativistic, so the contribution is non-negative and Eq. (7.2) survives — indeed with room to spare. Measured on the ΛCDM best fit for a single 0.06 eV eigenstate, the term is 4.9×10^{-5} today and rises to 0.2528 by $z = 1500$, against $2\Omega_{\rm r} = 1.06 \times 10^{-4}$

today. It is therefore not negligible at the precision the fits reach, which is why it is carried in the code and stated here rather than dropped. In the reduced variables the code works in — densities in units of the critical density and derivatives with respect to N — the same term reads $-\frac{3}{2}(\mathrm{d}p/\mathrm{d}N)/E^2$, since $2M_{\rm Pl}^2H^2 = \frac{2}{3}\rho_{\rm tot}$; the two forms have been confused in an earlier version of this remark, and only the first belongs in Eq. (7.1).

Proof. The kinematic identity $j = 1 - \dot{p}_{\rm tot}/(2M_{\rm Pl}^2H^3)$ follows from $j = 1 + 3\dot{H}/H^2 + \ddot{H}/H^3$ together with Eq. (2.8) and total conservation. With $p_{\rm tot} = \rho_{\rm r}/3 - V$ and $\dot{V} = Q_V$, the radiation term gives $2\Omega_{\rm r}$ and the vacuum term gives ε. □

Two features make Eq. (7.2) worth stating. It is parameter-free: it holds for every admissible rate, so no choice of Γ_0, α, β, τ_M, $a_{\rm act}$ or p can evade it. And it is not new in kind: that a matter–vacuum interaction shows up in the third and higher derivatives of the scale factor was established by Szydlowski [16] for decaying-vacuum cosmologies, and the jerk as a diagnostic of dark-sector interaction has been revisited recently [17]. That precedent is two-sided and should not be read as support for this paper's sign: the same work's data analysis reported a transfer in the *opposite* direction to the one assumed here. What is added here is the inequality form and the sign table below.

In the model, $\varepsilon = \frac{3}{2}(1-b)(\Gamma/H)\Omega_I$. The sandwich theorem alone gives only a trivial bound on j_0, because the two bracketing ΛCDM models are tangent to first order at $a = 1$. A non-trivial bound needs one further hypothesis, and it matters which quantity that hypothesis is placed on. Write $\mathsf{q}(N) = Q_V/(3M_{\rm Pl}^2H_0^2H)$, so that $\Delta = \int_{-\infty}^{0}\mathsf{q}\,e^{3N}\,\mathrm{d}N$ and $\varepsilon_0 = \frac{3}{2}\mathsf{q}_0$. If q is non-decreasing up to today, which says that the transfer measured against the expansion rate is still switching on, then $\mathsf{q} \le \mathsf{q}_0$ throughout and

C — Theorem of the model

$$\mathsf{q}\ \text{non-decreasing} \implies j_0 - 1 \ \ge\ 2\Omega_{r0} + \frac{9}{2}\,\Delta, \tag{7.3}$$

The hypothesis is displayed inside the box because the level marked on it — theorem of the model — would otherwise overstate it: the implication is proved, and the paragraphs below show that this model's activation law does not satisfy the antecedent for any activation redshift tried, so the consequent is never actually available here. With the relation exact, $\varepsilon_0 = \frac{3}{2}(p_{\rm eff}+3)\Delta$, when $\mathsf{q} \propto a^{p_{\rm eff}}$ near today. The hypothesis is on Q_V/H, not on Q_Va^3/H. The latter is strictly weaker and gives no floor: since Δ already weights q by a^3, a q that falls towards today can leave Q_Va^3/H increasing while $\frac{9}{2}\Delta$ exceeds ε_0. Subject to the correct hypothesis the two anchor-free observables of Sec. 5 would not be independent: a measured matter deficit would predict a minimum jerk excess.

The hypothesis fails for the activation law of this paper. We checked it rather than assuming it, and the result is negative. The activation switch turns the transfer on at $z_{\rm act}$, after which the rate falls relative to the expansion; q therefore *peaks at the activation redshift* and declines from there, for every activation redshift and every rate we tried. The floor is correspondingly violated: over 399 uniform prior draws with a non-zero deficit the hypothesis holds in 0 of them, and the floor fails in 399. The ratio ε_0/Δ has median 0.45 where the bound requires 4.5, and — more to the point — it is not a fixed number at all: over the same draws its central ninety per cent spans 3.8×10^{-8} to 1.39, and it reaches down to zero. The reason is visible in the kinetics. An early activation finishes its conversion long before today, leaving a large deficit and a transfer that has already switched off, so ε_0 is small; a late one has the opposite. The deficit and the jerk excess therefore measure different things, and neither bounds the other.

What survives of Eq. (7.3) is therefore a statement about a class of transfer histories that this model does not realise, and nothing in this paper may be inferred from it. In particular a bound on the deficit does not bound the jerk excess: the relation runs the wrong way for that

even when it holds. Where the jerk excess is needed it is computed directly, as in Sec. 10.3. We keep the bound because the hypothesis is a statement about the transfer profile rather than about the kinetics, so a different activation law could satisfy it; but it is not available here.

7.2 The sign separates the alternatives

Theorem 7.3 (Sign table)**.** *Neglecting radiation, the competing late-time descriptions give*

Model	$j-1$	*Sign*
ΛCDM	0	*zero*
This model, one-way	$\frac{3}{2}(1-b)(\Gamma/H)\Omega_I$	>0
This model, reverse channel dominant	$\varepsilon - \Gamma_{\downarrow}V/(2M_{\rm Pl}^2H^3)$	<0
Running vacuum, $\nu>0$ [31]	$-3\nu(1+q)$	<0
CPL [32, 33] in the DESI-preferred quadrant	$\Omega_{\rm DE}\left[\frac{9}{2}w_0(1+w_0)+\frac{3}{2}w_a\right]$	<0

A robust measurement of $j_0<1$ therefore falsifies the one-way model and selects either the reverse channel of Sec. 5.5 or a genuinely different dark-energy sector. It is worth emphasising how unusual this is: the one-way model and the two families currently favoured by the data predict opposite signs for the same kinematic quantity.

How sensitive is this test? Not yet sensitive enough, and the paper says so. Late-time cosmographic reconstructions currently give $\sigma(j_0)\approx 0.4$, against which the excess this model permits is measured directly in Sec. 10.3 rather than inferred from the deficit. Values of j_0 obtained by combining the CMB with low-redshift data are tighter but model-dependent, derived within ΛCDM or CPL and then quoted as if kinematic; they cannot be used to test Eq. (7.2). The practical route is not a direct measurement of a third derivative but a consistency test: fit $H(z)$, then check that the reconstructed $Q_V = 2M_{\rm Pl}^2(\ddot{H}+3H\dot{H})-\frac{4}{3}H\rho_{\rm r}$ is non-negative within errors. That is what Sec. 10 does.

7.3 The snap, where memory shows

Theorem 7.4 (Snap)**.** *With $s=\dot{j}/H-j(2+3q)$ and $s_{\Lambda\rm CDM}(q,\Omega_{\rm r})=2\Omega_{\rm r}'-(1+2\Omega_{\rm r})(2+3q)$ its value at $Q_V=0$, evaluated at the same q and $\Omega_{\rm r}$,*

C — Theorem of the model

$$s-s_{\Lambda\rm CDM}(q,\Omega_{\rm r})=\varepsilon\left[\frac{\mathrm{d}\ln\Gamma}{\mathrm{d}N}-\frac{\Gamma}{H}-2\right], \tag{7.4}$$

which for the rate law Eq. (3.4) reads

C — Theorem of the model

$$s-s_{\Lambda\rm CDM}(q,\Omega_{\rm r})=-\varepsilon\left[2+(1-\alpha(1-f))\,\frac{\Gamma}{H}-\mathcal{B}-p\,(1-S_{\rm act})\right], \qquad \mathcal{B}\coloneqq\frac{\beta(f-M)}{\tau_M H}. \tag{7.5}$$

Proof. From $s=j'-j(2+3q)$ with $'=\mathrm{d}/\mathrm{d}N$ and Eq. (7.1), the radiation terms assemble into $s_{\Lambda\rm CDM}(q,\Omega_{\rm r})$ and the remainder is $\varepsilon[\varepsilon'/\varepsilon-(2+3q)]$. Differentiating $\ln\varepsilon=\text{const}+\ln\Gamma+\ln\rho_I-3\ln H$ and using $\mathrm{d}\ln\rho_I/\mathrm{d}N=-(3+\Gamma/H)$ and $\mathrm{d}\ln H/\mathrm{d}N=-(1+q)$ gives $\varepsilon'/\varepsilon=\mathrm{d}\ln\Gamma/\mathrm{d}N-\Gamma/H+3q$, which is Eq. (7.4). For Eq. (3.4) the logarithmic derivative has three terms, not two: $\mathrm{d}\ln\Gamma/\mathrm{d}N=\alpha(1-f)\Gamma/H+\mathcal{B}+p(1-S_{\rm act})$, the last because $\mathrm{d}\ln S_{\rm act}/\mathrm{d}N=p\,(1-S_{\rm act})$ for the switch of Eq. (6.4). That term is easy to drop — an earlier version of Eq. (7.5) did, while the text around it and the implementation both carried it — and dropping it changes the sign of the bracket wherever the switch is still opening, which is where the effect is largest. □

Theorem 7.5 (Memory fingerprint, and what limits it)**.** *At fixed* (H_0, q_0, j_0) *the memory parameters are unconstrained by the jerk, and do enter the snap:*

C — Theorem of the model

$$\frac{\partial j_0}{\partial \mathcal{B}} = 0, \qquad \frac{\partial s_0}{\partial \mathcal{B}} = +\varepsilon_0 \;\neq 0. \tag{7.6}$$

The activation term of Eq. (7.5), however, enters the same bracket with the same *coefficient as* $\mathcal{B}$*, so what the snap measures is the sum* $\mathcal{B} + p(1 - S_{\rm act})$*. Memory is identified by cosmography alone only when the activation profile is known independently, or when* $p(1 - S_{\rm act})$ *is negligible today.*

Proof. The jerk depends on the memory sector only through the instantaneous value of Γ, and there β and M enter only through the factor $e^{\beta M}$, which is exactly degenerate with the normalisation Γ_0. The lag $\mathcal{B}$ is not so degenerate: in the bracket of Eq. (7.5) the coefficient of Γ/H is $1 - \alpha(1 - f)$ while that of $\mathcal{B}$ is -1, and these are independent directions unless f is held fixed. □

This is the observational counterpart of Theorem 3.8. The jerk measures how much energy is being transferred now; the snap measures whether that transfer is being driven by a lagged history. Two limits check the statement. Without memory, $\mathcal{B} = 0$, and the snap excess is strictly negative for $\alpha(1 - f) \leq 1$ *once the switch has closed*; while it is still opening the activation term $p(1 - S_{\rm act})$ can exceed $2 + (1 - \alpha(1 - f))\Gamma/H$ and reverse the sign, which is the same caveat as the one below and not a separate one. At saturation, $f, M \to 1$, both ε and $\mathcal{B}$ vanish and the de Sitter attractor of Theorem 4.5 becomes cosmographically indistinguishable from ΛCDM, as it must. The honest limitation is that only the single combination $\mathcal{B}_0$ is measurable at $z = 0$, so β and τ_M are not separately determined by cosmography alone.

What this section establishes. A parameter-free inequality on the jerk whose sign distinguishes this model from running vacuum and from the DESI-preferred CPL quadrant, and a snap relation that carries the only background signature of memory. What it does not establish is that either is currently measurable: the predicted jerk excess is about a factor of ten below present cosmographic errors, and the snap is further out of reach. Nor does it establish that the snap isolates memory: the activation profile contributes to the same combination, so a measurement of s_0 constrains $\mathcal{B} + p(1 - S_{\rm act})$ and not $\mathcal{B}$ alone.

8 Covariant realizations of a silent vacuum

Assumption 4.1 postulates a sector with $p = -\rho$ that receives energy. This section asks what can actually produce such a sector. Three constructions are examined, and it is important to say at the outset that they establish different things: a four-form gives an exact vacuum tensor with no local propagating modes; a constrained clock gives a covariant action whose field equations reproduce the background transfer; and neither, on its own, produces the reinforced conversion law. The last subsection proposes the bridge between them, at the level of a conjecture with a stated test.

8.1 Absence of modes is not absence of a tensor

Proposition 8.1 (Four-form vacuum)**.** *Let* $C_{\mu\nu\rho}$ *be a three-form potential with* $\mathcal{H}_{\mu\nu\rho\sigma} = 4\nabla_{[\mu} C_{\nu\rho\sigma]}$ *and*

$$S_4 = -\frac{1}{2 \cdot 4!} \int \mathrm{d}^4x \, \sqrt{-g}\, \mathcal{H}_{\alpha\beta\gamma\delta} \mathcal{H}^{\alpha\beta\gamma\delta}. \tag{8.1}$$

In a connected source-free region the field equations imply

I — Identity

$$\mathcal{H}_{\mu\nu\rho\sigma} = h\,\epsilon_{\mu\nu\rho\sigma}, \qquad \partial_\mu h = 0, \qquad T^{(4)}_{\mu\nu} = -\frac{h^2}{2} g_{\mu\nu}. \tag{8.2}$$

Proof. Varying C and integrating by parts gives $\nabla_\mu \mathcal{H}^{\mu\nu\rho\sigma} = 0$. In four dimensions every four-form is proportional to the volume form, and $\nabla_\mu \epsilon^{\mu\nu\rho\sigma} = 0$, so h is constant — a statement about the non-dynamical four-form; for the version in which the flux itself is promoted to a degree of freedom see Albertini et al. [34]. Varying the metric before imposing the solution gives $T^{(4)}_{\mu\nu} = \frac{1}{3!}\mathcal{H}_{\mu\alpha\beta\gamma}\mathcal{H}_\nu{}^{\alpha\beta\gamma} - \frac{1}{2\cdot 4!} g_{\mu\nu}\mathcal{H}^2$; with signature $(-+++)$ the contractions are $\epsilon_{\mu\alpha\beta\gamma}\epsilon_\nu{}^{\alpha\beta\gamma} = -3!\, g_{\mu\nu}$ and $\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\alpha\beta\gamma\delta} = -4!$, giving $T_{\mu\nu} = -h^2 g_{\mu\nu} + \frac{1}{2}h^2 g_{\mu\nu}$, which is Eq. (8.2) with $\rho_4 = h^2/2$ and $p_4 = -h^2/2$. □

A massless three-form gauge field has $\binom{D-2}{3}$ local polarizations, which vanishes for $D = 4$: its only amplitude is the integration constant h. This settles a question that recurs in discussions of a dark sector: a component can be electromagnetically silent, carry no local waves, and still determine the expansion. It does not settle the magnitude of h, and it does not produce conversion, because in the free theory h cannot change.

8.2 An obstruction for a canonical order parameter

It is tempting to identify the converted fraction with a rolling scalar. That identification is not available.

Proposition 8.2 (No rolling exact vacuum)**.** *For a canonical real scalar with $\mathcal{L} = -\frac{1}{2}(\nabla\phi)^2 - W(\phi)$ one has $\rho_\phi + p_\phi = \dot\phi^2$. Hence $w_\phi = -1$ exactly requires $\dot\phi = 0$.*

So one cannot simultaneously impose $\phi \propto f$, $\dot f > 0$ and $p_\phi = -\rho_\phi$ with a non-degenerate kinetic term. This is why f is kept throughout as a statistical fraction and V as a separate variable of the effective description, rather than the two being identified. A double-well potential $W(s) = W_0 + \lambda_W s^2(1-s)^2 - \Delta W(3s^2 - 2s^3)$ with $\lambda_W > 3\Delta W > 0$ has minima at $s = 0, 1$ with $W(1) - W(0) = -\Delta W$, so the transition releases energy; but releasing energy is not the same as increasing a vacuum density, and a barrier picture alone yields no rate. A semiclassical nucleation rate would require a critical configuration, its action and a prefactor, and a rate per four-volume must be converted to a rate per region before it can be compared with Γ, which has dimensions of inverse time.

8.3 A constrained clock

Introduce a scalar T of dimension -1 and a multiplier λ:

A — Postulate

$$S_{\rm clock} = \int \mathrm{d}^4x\,\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R - \frac{\lambda}{2}\left(g^{\mu\nu}\partial_\mu T\partial_\nu T + 1\right) - V(T)\right]. \tag{8.3}$$

Proposition 8.3 (Clock field equations)**.** *On the branch with future-directed timelike gradient and $\lambda \geq 0$, Eq. (8.3) gives*

I — Identity

$$u_\mu = -\partial_\mu T, \quad u^\mu u_\mu = -1, \quad T_{\mu\nu} = \lambda\, u_\mu u_\nu - V g_{\mu\nu}, \tag{8.4}$$

together with $u^\nu\nabla_\nu u^\mu = 0$, $\nabla_\mu(\lambda u^\mu) = -V_{,T}$ and $\nabla_\mu T_V^{\mu\nu} = V_{,T}\,u^\nu$.

Proof. Varying λ imposes $(\partial T)^2 = -1$. The metric variation produces $\lambda\partial_\mu T\partial_\nu T$, and on the constraint the matter Lagrangian equals $-V$. Varying T gives $\nabla_\mu(\lambda\nabla^\mu T) = V_{,T}$, and since $\nabla^\mu T = -u^\mu$ this is the stated continuity equation. The symmetry of $\nabla_\mu\nabla_\nu T$ gives $u^\nu\nabla_\nu u_\mu = \frac{1}{2}\nabla_\mu(u^\nu u_\nu) = 0$, so the flow is geodesic. □

In FLRW one may take $T = t$, and then $\dot{\lambda} + 3H\lambda = -\dot{V}$ with $3M_{\rm Pl}^2 H^2 = \lambda + V$: an exact realization of dust transferring energy to the vacuum whenever $V_{,T} \geq 0$. This construction is not new. It is the "dust of dark energy" of Lim et al. [5], and with a potential it is mimetic matter in the sense of Chamseddine et al. [6]; reviews are given by Sebastiani et al. [7] and Malaeb [8]. A recent construction combines a mimetic-like silent sector with an interacting vacuum and fits it to DESI data [9]. We use it here as an existence proof for the background of Sec. 4, not as a claim of novelty.

8.4 The potential is determined, not chosen

The standard objection to Eq. (8.3) is that $V(T)$ can be reverse-engineered to reproduce almost any expansion history, so the construction explains nothing. The reconstruction freedom is established by Chamseddine et al. [6], who present it as a feature of mimetic matter rather than as an objection; the objection is ours, and we do not claim to answer it. What can be said is weaker and worth stating precisely.

Proposition 8.4 (Kinetically determined potential)**.** *Fix* $(\Gamma_0, \alpha, \beta, \tau_M, b)$*, the activation law, and initial data. By Theorem 4.4 the background is unique, so with* $T = t$*,*

I — Identity

$$V(T) = V_0 + (1-b)\int_{t_0}^{T} \Gamma\rho_I \, \mathrm{d}t, \qquad \lambda = \rho_I + \rho_D, \tag{8.5}$$

is a five-parameter family, monotone non-decreasing and bounded above by Eq. (4.5).

Eq. (8.5) is simply the integral of the third line of Eq. (4.1); the content is in what it is being integrated against. The kinetic law is an additional postulate imposed from outside the action, so the action still does not constrain V, and the mimetic objection stands. What has been gained is that the ansatz for V has five parameters rather than the freedom of a function space, is monotone non-decreasing and bounded above by Eq. (4.5), carries a specified early-time cutoff, and was not chosen by looking at any dataset. That is a statement about naturalness and about the number of free functions, not a derivation, and it is what makes the fits of Sec. 10 interpretable and the prediction of Sec. 7 falsifiable.

One further consequence of the same limitation should be recorded here rather than left for the reader to notice. The matter tensor of Eq. (8.4) is $\lambda u_\mu u_\nu - V g_{\mu\nu}$ with $\lambda = \rho_I + \rho_D$: the action distinguishes the dust sector from the vacuum, but nothing in it distinguishes the precursor from the defects. Since b is also degenerate with ϵ at background level (Proposition 5.3) and cancels from linear growth (Theorem 9.5), the split of the dark sector into ρ_I and ρ_D is bookkeeping at every level this paper computes. It acquires content only through the defect sector of Sec. 11, and only if those objects are real.

Two known problems of this class of theories remain, and both must be stated without softening.

The scalar sound speed vanishes identically. This single fact has two faces that appear in different places in this paper and should not be counted twice: it means there is no scalar sound horizon, and it is also the reason the linear system of Sec. 9 has no k-dependent coefficient. The latter is what makes Proposition 9.8 available, but it is a degeneracy of the scalar sector rather than a restoring force, so it establishes the absence of one specific instability and nothing more.

Pressureless geodesic flow develops caustics, which no linear analysis can exclude [35]. Eq. (8.3) contains no higher-derivative term of the form $(\Box T)^2$, so the gradient instabilities identified by Ijjas et al. [36] in extended mimetic models are absent *from this action*. That scoping matters: the action alone leaves $V(T)$ arbitrary and contains no rate law, so it is not by itself the theory whose perturbations are computed in Sec. 9 and fitted in Sec. 10. The additional prescription those sections use, Assumption 9.2, is not a consequence of Eq. (8.3), and its scalar sector has not been shown free of gradient instabilities by the argument just given. Caustics are taken up where they actually bite, in Sec. 9.1.

8.5 Nucleation events as absorption events

The three constructions above are related by more than analogy, and this subsection states the connection precisely enough to be attacked.

In a four-form sector coupled to membranes, nucleation of a membrane of charge q shifts the flux across its worldvolume, $h \to h \pm \mathsf{q}$, so the vacuum density jumps by $\Delta V = \pm h\mathsf{q} + \mathsf{q}^2/2$ [12–14, 37]. Identify the upward event with one absorption event: a precursor unit of energy E_p per patch volume is consumed, with the accounting $E_p = \Delta V\, \mathcal{V}_p + E_{\rm wall}$, where $E_{\rm wall}$ is the energy that does not end as vacuum. The symbol is E_p and not ϵ because ϵ is the convertible fraction everywhere else in this paper and the two are unrelated. The branching ratio is then not a free parameter but a ratio of energies, $b = E_{\rm wall}/E_p$, which is what the cosmological sections use.

Remark 8.5 (The remnant is not the defect of Sec. 11, and is not stable). It is tempting to read $E_{\rm wall}$ as the rest energy of a localized object and to identify that object with the texture of Sec. 11. Neither step is available here, and saying so costs the bridge half its reach.

First, the thin-wall membrane leaves no object. For a closed wall of radius R enclosing a region lower in vacuum energy by ΔV, the energy $E = 4\pi R^2\varsigma - \frac{4}{3}\pi R^3 \Delta V$ has its only non-zero stationary point at $R_c = 2\varsigma/\Delta V$, where $E'' = -8\pi\varsigma < 0$. That is the critical bubble, a maximum: the wall either collapses or runs away, and $E(R_c) = 16\pi\varsigma^3/3(\Delta V)^2$ is a barrier, not a rest mass. A stable remnant needs an ingredient this action does not contain — angular momentum, a conserved charge on the wall, a repulsive core — and adding one changes the theory rather than interpreting it. Particle-like remnants have been constructed in three-form sectors [38], but with additional structure, not from the action written here.

Second, even granting a remnant, it would not be the object of Sec. 11. That object is a map $S^3 \to S^3$ built from an SU(2)-valued scalar, carrying a degree in $\pi_3(S^3)$ and given its size by the quartic term of Eq. (11.1). The membrane is a domain wall carrying a gauge charge under a three-form potential, which has no propagating modes in four dimensions. The fields differ, the charge groups differ, and nothing sets the size of the second. They are independent constructions, and their coexistence in this paper is not a mechanism.

What the bridge does connect is the *kinetics*: the event sequence, the reinforcement coefficient of Eq. (8.9), and the interpretation of b as the fraction of transferred energy that does not reach the vacuum. Where that fraction goes — radiation, gravitational radiation, or collapse — this paper does not determine.

Remark 8.6 (The event sequence is a reinforced birth process). If each of N precursor units per patch can nucleate once, and the $(k+1)$-th event has rate $\Gamma_k = \Gamma(h_0 + k\mathsf{q})$, then K_t is exactly the chain of Eq. (3.1) with $\Gamma(f) = \Gamma(h_0 + fN\mathsf{q})$. The exponential law Eq. (3.4) corresponds to $\log\Gamma$ linear in h over the relevant range, with $\alpha = N\mathsf{q}\,\mathrm{d}\log\Gamma/\mathrm{d}h$. Reinforcement, $\alpha > 0$, holds if and only if the upward tunnelling exponent decreases with accumulated flux.

The identification is immediate once each unit may nucleate at most once: the number of units still able to nucleate after k events is $N-k$, and each carries the common rate $\Gamma(h_0+k\mathsf{q})$, so the total intensity is $\lambda_k = (N-k)\Gamma(h_0+k\mathsf{q})$, which is Eq. (3.1) with $f = k/N$. The exponential form Eq. (3.4) is then the statement that $\log\Gamma$ is linear in h over the range swept, with the stated identification of α. We label this a remark rather than a proposition because the premises – one nucleation per unit, a common rate, a flux that shifts by a fixed quantum – are modelling choices, not consequences of the four-form dynamics.

Whether this is a constraint or a picture turns on the sign, and the sign needs care. The de Sitter entropy is $S = 8\pi^2 M_{\rm Pl}^2/H^2 = 24\pi^2 M_{\rm Pl}^4/V$, which *decreases* with V. An upward nucleation therefore lowers the horizon entropy and is exponentially suppressed relative to the downward one, $\Gamma_\uparrow/\Gamma_\downarrow = e^{-\Delta S}$ with $\Delta S > 0$: the naive reading, that accumulating flux is entropically favoured, is backwards. What survives is one order more delicate. The suppression of a single upward step,

X — Result of an auxiliary model

$$\Delta S = S(V) - S(V + \mathsf{q}h) \simeq 24\pi^2 M_{\rm Pl}^4 \, \frac{\mathsf{q}h}{V^2}, \tag{8.6}$$

itself decreases as V grows. That is not yet the statement wanted, because ΔS controls the *ratio*: writing $\Gamma_\uparrow = \Gamma_\downarrow e^{-\Delta S}$,

$$\frac{\mathrm{d}\log\Gamma_\uparrow}{\mathrm{d}h} = \frac{\mathrm{d}\log\Gamma_\downarrow}{\mathrm{d}h} - \frac{\mathrm{d}\Delta S}{\mathrm{d}h}, \tag{8.7}$$

and a downward rate falling faster than the suppression relaxes would carry the upward rate down with it. The second term is the one Eq. (8.6) supplies; the first has to be computed.

Take the downward nucleation in the thin-wall limit, $B_\downarrow = 27\pi^2\varsigma^4/2(\mathsf{q}h)^3$ with ς the membrane tension, valid while the critical bubble is small against the horizon, which requires $\varsigma \lesssim (\sqrt{6}/3)M_{\rm Pl}\mathsf{q}$. Both terms then scale the same way in the flux,

X — Result of an auxiliary model

$$\log\Gamma_\uparrow = -\frac{\mathcal{K}}{h^3}, \qquad \mathcal{K} = \frac{\pi^2}{\mathsf{q}^3}\left(96 M_{\rm Pl}^4\mathsf{q}^4 + \tfrac{27}{2}\varsigma^4\right) > 0, \tag{8.8}$$

so $\mathrm{d}\log\Gamma_\uparrow/\mathrm{d}h = 3\mathcal{K}/h^4 > 0$ and the upward rate does rise with accumulated flux — now for a reason, and a stronger one than the original argument gave, since the two contributions push the same way rather than competing. In the regime where the thin-wall form is valid the entropy term dominates, $\mathcal{K}_{\rm bounce}/\mathcal{K}_{\rm entropy} = 9\varsigma^4/64M_{\rm Pl}^4\mathsf{q}^4 < 1/16$: what makes each successive nucleation easier is that the horizon entropy cost of a step falls as the vacuum grows.

Eq. (8.8) also says what the exponential law of Eq. (3.4) costs. Linearity of $\log\Gamma$ in h across a swept range $\Delta h = N\mathsf{q}$ holds with relative curvature $4N\mathsf{q}/h$, and truncating at the linear term leaves a relative error $2N\mathsf{q}/h$ in the exponent at the end of conversion: the form is an approximation valid when the flux converted is small against the flux already there. The coefficient is then

X — Result of an auxiliary model

$$\alpha = 3\,\frac{N\mathsf{q}}{h}\,\Delta S\left[1 + \mathcal{O}\Big(\big(\varsigma/M_{\rm Pl}\mathsf{q}\big)^4\Big)\right], \qquad \text{so} \qquad \alpha < \tfrac{3}{4}\Delta S, \tag{8.9}$$

the bound following from the linearity condition. That is a relation rather than a fit: reinforcement of order unity requires an entropy change per step of at least order unity, and the two cannot be chosen independently.

These two results carry level X, and the reason is worth stating rather than leaving to the table. They are proved, and the algebra is checked symbolically, but they are proved *about a different system*: a four-form with charged membranes, in the thin-wall limit, with detailed balance between the two nucleation directions, one nucleation per unit, a common rate, and a flux shifting by a fixed quantum. Several of those are modelling choices rather than consequences of the four-form dynamics, as Remark 8.5 says, and detailed balance is an equilibrium relation that the driven nucleation assumed here does not satisfy. A result of that kind is not a theorem of the model, and neither is it a conjecture: the calculation is done, and it is the identification that is open. Level C would claim the first, level S would deny the second.

Two honest caveats accompany it. At vacuum densities of order the observed one these rates are astronomically small, so the mechanism can only be relevant if the nucleation is not thermal but driven by the precursor, which is precisely what Assumption 4.1 assumes and does not derive. And the bridge covers α and b but not the memory sector: nothing in this subsection produces β or τ_M. We therefore record Remark 8.6 as a conditional statement about the structure of the process, and the claim that this mechanism operates in nature as a conjecture, listed as such in Sec. 13.

The same bridge identifies what the reverse channel of Sec. 5 would be: an ordinary downward nucleation, returning vacuum energy to the localized sector.

What this section establishes. That a silent vacuum tensor exists (Proposition 8.1), that a covariant action reproduces the background transfer (Proposition 8.3), and that its potential is fixed by the kinetics rather than chosen (Proposition 8.4). What it does not establish is a microscopic derivation of the rate law, the value of α, or that caustics of the clock congruence are benign.

9 Linear perturbations

A background model with an interacting vacuum is not testable until its perturbations are specified, and the specification is not free: an exact vacuum has no rest frame of its own, so the way it exchanges energy determines how it perturbs. This section fixes that choice, shows why the obvious alternative is unstable, derives the growth equation, and corrects a result of the draft version that is wrong when both products are present.

9.1 The integrability condition and the geodesic realization

Suppose the vacuum receives energy with no spatial force in the frame u^μ, so that $-\nabla^\mu V = Q_V u^\mu$. Lowering the index, the left side is a gradient, hence

I — Identity

$$\nabla_{[\mu}\left(Q_V u_{\nu]}\right) = 0. \tag{9.1}$$

This is a genuine constraint, not a definition: a scalar expression for Q_V does not by itself define a consistent transfer. Its spatial projection is sharper than the Frobenius statement below, and is worth extracting because it settles a question the geometric readings of Sec. 13 keep running into.

Proposition 9.1 (An exact vacuum fed without momentum transfer forbids vorticity)**.** *Let the vacuum be exact, $T_V^{\mu\nu} = -V g^{\mu\nu}$, let it receive energy with no spatial momentum transfer in the frame u^μ, so that $\nabla_\mu T_V^{\mu\nu} = Q_V u^\nu$, and let u^μ be a unit timelike field. Then*

I — Identity

$$Q_V\, \omega_{\mu\nu} = 0, \tag{9.2}$$

where $\omega_{\mu\nu}$ is the vorticity of u^μ: the congruence is irrotational wherever the transfer is active.

Proof. The first hypothesis gives $\nabla_\mu T_V^{\mu\nu} = -\nabla^\nu V$, so the second reads $\nabla_\mu V = -Q_V u_\mu$. The left side is a gradient, whose curl vanishes identically, so $\nabla_{[\mu}(Q_V u_{\nu]}) = 0$. Decomposing $\nabla_\mu u_\nu = -u_\mu a_\nu + \frac{1}{3}\Theta h_{\mu\nu} + \sigma_{\mu\nu} + \omega_{\mu\nu}$ and antisymmetrising removes h and σ, leaving $(\nabla_{[\mu}Q_V)u_{\nu]} + Q_V(\omega_{\mu\nu} - u_{[\mu}a_{\nu]}) = 0$. Projecting both indices with $h^\mu{}_\alpha h^\nu{}_\beta$ annihilates every term carrying a u^μ and leaves Eq. (9.2). □

This is more general than the clock argument of Sec. 8.3, which obtains $\omega_{\mu\nu} = 0$ from $u_\mu = -\partial_\mu T$ and therefore covers only that realization; it is not stronger, and the difference matters. Eq. (9.2) permits $\omega_{\mu\nu} \neq 0$ wherever $Q_V = 0$, which the clock realization does not, so what is bought by dropping the realization is generality and what is paid is the set where the transfer is off. Proposition 9.1 uses no realization at all. The consequence is worth stating plainly: a rotating flow cannot be the one that feeds an exact vacuum, so any picture in which vortical structures drive the vacuum production must abandon one of the three hypotheses. The minimal abandonment is the second. Allowing a spatial momentum transfer q_μ, so that the source is $Q_V u^\mu + q_\mu$, the same projection gives $Q_V \omega_{\mu\nu} = -h^\alpha{}_\mu h^\beta{}_\nu \nabla_{[\alpha} q_{\beta]}$: vorticity is permitted exactly to the extent that the momentum transfer carries a spatial curl. That is a requirement on a mechanism, not a freedom. In the clock realization $u_\mu = -\partial_\mu T$, and Eq. (9.1) is equivalent to $\mathrm{d}Q_V \wedge \mathrm{d}T = 0$, so that Q_V and V are functions of T alone wherever the clock is regular. That proviso is not decorative, and it is worth saying exactly what it excludes. The constraint

$(\partial T)^2 = -1$ guarantees $\mathrm{d}T \neq 0$, so the only way the Frobenius step can fail is at a caustic of the congruence, where the level sets of T self-intersect and T ceases to be a coordinate. Caustics are therefore not a remote nonlinear question about the theory in general: they are precisely the locus on which Eq. (9.3), and with it every result of this section, is undefined. That assumption should not be softened, so we state it in the form that is true. The clock congruence is the dark-sector dust congruence, and in any universe with the observed matter power spectrum dark-matter shell crossing begins on the smallest collapsing scales by $z \sim 20$ and is widespread by $z \sim 1$. Caustics are therefore not absent over the redshift range the data of Sec. 10 cover; they are present throughout it. What we use is a separation of scales: the observables fitted here are background distances, $H(z)$, and $f\sigma_8$ at wavenumbers still in the linear regime, where the congruence is single-stream, and Eq. (9.3) holds on that support. We supply no controlled estimate of the error this incurs, and we note that the failure at smaller scales is of a different kind from the one ΛCDM has there: nonlinearity is a hard problem in ΛCDM and an absence of definition here, because the clock ceases to be a coordinate rather than merely ceasing to be perturbative. A theory of the dark sector that is undefined wherever structure has formed is not a complete theory, and this is the sharpest form of that limitation in this paper. On comoving hypersurfaces of constant T this gives

C — Theorem of the model

$$\delta V_{\mathrm{com}} = 0, \qquad \delta Q_{V,\mathrm{com}} = 0. \tag{9.3}$$

Assumption 9.2 (Geodesic realization)**.** The theory is *defined* by imposing Eq. (9.1) with a geodesic clock, so that the volumetric transfer is spatially uniform on comoving hypersurfaces, Eq. (9.3). Equivalently, the local conversion rate anti-correlates exactly with the local precursor density, $\delta\Gamma/\bar{\Gamma} = -\delta_I$. The kinetic variables (f, M) are then *diagnostic* at linear order: their perturbations are reconstructed from that relation together with whichever of their evolution equations one retains.

The draft listed four ways to extend the transfer off the background and chose none of them. It is stated here as an assumption because that is the role it plays in what follows — every later result is proved *under* it — but it is not an independent one: Proposition 9.4 below shows that the hypotheses already in force leave no alternative to it, and the natural alternative is in any case shown to be unstable.

The structural cost, stated plainly. If instead the rate law Eq. (3.4) is read as a *local* law, so that $\delta\Gamma/\bar{\Gamma} = \alpha\,\delta f + \beta\,\delta M$ holds pointwise, then together with the two linearised kinetic evolution equations one has three conditions on the two unknowns $(\delta f, \delta M)$. The system is overdetermined by one, and differentiating the algebraic relation and substituting the evolution equations yields a compatibility condition,

I — Identity

$$\alpha\left[-\bar{\Gamma}\,\delta f - (1-\bar{f})\bar{\Gamma}\,\delta_I\right] + \frac{\beta}{\tau_M}\,(\delta f - \delta M) = -\dot{\delta}_I, \tag{9.4}$$

which the dust dynamics do not satisfy identically. Assumption 9.2 resolves this by taking the transfer condition as primary and demoting the kinetic perturbations to diagnostics, which costs nothing in linear observables: by Theorem 9.5 the entire linear dust sector closes on δ_{m} and ϑ and sees the kinetic sector only through the background quantity $\bar{Q}_V$. It does, however, mean that at linear order f, M, α, β and τ_M are background bookkeeping rather than local fields, so the locality of the rate law asserted in Sec. 3.1 is a statement about the homogeneous mode. Whether a genuinely local rate law admits a consistent inhomogeneous completion, and what that completion does to Eq. (9.3), is an open problem; it is listed as such in Sec. 13.

Proposition 9.3 (A clumping vacuum is gradient-unstable)**.** *If instead* $\delta V_{\mathrm{com}} = \zeta\,\delta_I$ *with* $\zeta > 0$*, the dust experiences a force* $\rho_I A^i = \partial^i V$*, and the growth equation acquires a term* $-k^2\zeta\delta/(a^2\bar{\rho}_I)$*.*

The effective sound speed squared is then negative and short-wavelength modes grow without bound.

Proposition 9.3 excludes one branch and not all alternatives, and it is worth being exact about the scope, because two different questions are easy to run together. *Taken by itself*, as a statement about the growth equation, it says nothing about $\zeta < 0$, which gives a positive effective sound speed; nor about the other completions the draft listed, namely allowing momentum transfer with a compensating acceleration of the donor, or giving the receiving sector its own dynamics and pressure. Those are different theories, not refuted ones. *Given the two stress hypotheses used everywhere else in this paper*, the situation is different, and Proposition 9.4 below is the statement of it: ζ is then not a parameter at all but zero, and the compensating-acceleration branch is excluded rather than untried. The two readings do not conflict — the second assumes more — but the first is the one that survives if the recipient is allowed to stop being an exact vacuum. What can be said is that the geodesic branch is not an invention of this paper: it is the interacting-vacuum–geodesic-CDM scenario, set out in the covariant formalism of Wands et al. [1] and Sebastianutti et al. [2] and constrained against data by Martinelli et al. [3] and Hogg et al. [4]; it is also the branch for which the linear sector closes. More can be said, and should be, because presenting four completions as a menu understates how little freedom there actually is.

Proposition 9.4 (The geodesic realization is forced, not chosen)**.** *Let the dark sector consist of pressureless components with a common four-velocity u^μ and no heat flux, $T^{\mu\nu}_{I,D} = \rho_{I,D} u^\mu u^\nu$, together with an exact vacuum, $T^{\mu\nu}_V = -V g^{\mu\nu}$; let the remaining species be separately conserved, so that the dark sector conserves in total; let the transfer within it carry no momentum in the frame u^μ, so that $\nabla_\mu T^{\mu\nu}_V = Q_V u^\nu$; and let $\rho_I + \rho_D > 0$. Then, on any simply connected region where $Q_V \neq 0$ and the congruence is regular,*

1. *$a^\mu \equiv u^\nu \nabla_\nu u^\mu = 0$: the congruence is geodesic;*
2. *$\omega_{\mu\nu} = 0$, so u^μ is hypersurface-orthogonal and $u_\mu = -\partial_\mu T$ with $(\partial T)^2 = -1$;*
3. *V and Q_V are functions of T alone, hence $\delta V = \delta Q_V = 0$ on the surfaces $T = \text{const}$.*

Eq. (9.3) therefore holds identically, and Assumption 9.2 states a consequence of hypotheses already in force rather than a choice among completions.

Proof. (1) By hypothesis the dark sector conserves in total, $\nabla_\mu (T_I + T_D + T_V)^{\mu\nu} = 0$. Using $\nabla_\mu(\rho u^\mu u^\nu) = u^\nu \nabla_\mu(\rho u^\mu) + \rho\, a^\nu$ and projecting the whole equation with $h^\lambda{}_\nu$, every term carrying a free u^ν is annihilated — including the vacuum source, which by hypothesis has no other part — and what remains is $(\rho_I + \rho_D)\, a^\lambda = 0$.

(2) The first hypothesis gives $\nabla_\mu V = -Q_V u_\mu$, so $Q_V u$ is exact and hence closed, which is Proposition 9.1. With $a^\mu = 0$ from (1), $\nabla_{[\mu} u_{\nu]} = \omega_{\mu\nu} - u_{[\mu} a_{\nu]}$ vanishes as well, so u is itself closed and locally exact; the normalization $u \cdot u = -1$ is the clock constraint.

(3) $\nabla_\mu V = -Q_V u_\mu = Q_V\, \partial_\mu T$ says that ∇V is everywhere parallel to ∇T, so $V = F(T)$ locally, and comparing the two expressions gives $Q_V = F'(T)$. □

One objection should be met before the consequences are drawn, because it is the first a reader will raise: is the hypothesis not already the conclusion? It is not, but the distance is short and worth measuring. The hypothesis is a statement about the *transfer* — that its momentum part vanishes in the frame u^μ — and the conclusion is a statement about the *congruence*. What connects them is the donor's equation of motion, and the content of the proposition is exactly that connection: two of the four completions the draft listed, the geodesic one and the one with a compensating acceleration of the donor, are not two options but one dichotomy, and choosing a transfer with no momentum is choosing a geodesic clock whether or not one notices. A reader who finds that obvious once stated has understood it; it was not stated in the draft, and Assumption 9.2 was written as though the two were independent.

The reason to state this separately from Proposition 9.1 is that the two use different equations. The vorticity result uses only the vacuum's own divergence and holds whatever the donor is. Geodesy comes from the *donor's* equation of motion: a pressureless fluid whose only source points along its own four-velocity feels no transverse force, so it cannot be accelerated, and the exact vacuum by construction exerts none. Once both are in hand the third completion listed above, a compensating acceleration of the donor, is excluded rather than merely untried, and the first, $\zeta \neq 0$, is excluded by (3). What survives is the fourth: give the receiving sector its own dynamics and pressure. That is the one modification this paper's obstructions keep pointing at, and Sec. 13.8 collects the others that point the same way.

9.2 The branching ratio is invisible

Theorem 9.5 (b-independence of linear growth)**.** *Under Assumption 9.2, let $\delta_{\rm m} = \sum_k \bar\rho_k \delta_k / \bar\rho_{\rm m}$ run over all pressureless species $k \in \{I, D, \mathrm{b}\}$ and let ϑ be the perturbed expansion. Then for every $b \in [0,1]$*

C — Theorem of the model

$$\dot\delta_{\rm m} = -\vartheta + \gamma_c \delta_{\rm m}, \qquad \dot\vartheta + 2H\vartheta = -\frac{\bar\rho_{\rm m}}{2M_{\rm Pl}^2}\delta_{\rm m}, \qquad \gamma_c = \frac{\bar Q_V}{\bar\rho_{\rm m}}, \tag{9.5}$$

and hence

C — Theorem of the model

$$\ddot\delta_{\rm m} + (2H - \gamma_c)\dot\delta_{\rm m} - \left(\frac{\bar\rho_{\rm m}}{2M_{\rm Pl}^2} + \dot\gamma_c + 2H\gamma_c\right)\delta_{\rm m} = 0. \tag{9.6}$$

Proof. Perturb the continuity equation of each pressureless species and add. Write the perturbed transfer out of the precursor as $\delta(\Gamma\rho_I)$ without evaluating it: the defects then receive $+b\,\delta(\Gamma\rho_I)$ and the vacuum $+(1-b)\delta(\Gamma\rho_I)$, so in the dust sum the terms proportional to b cancel identically and the residual is exactly $-\delta Q_V$, which vanishes by Eq. (9.3). Leaving $\delta(\Gamma\rho_I)$ unevaluated is not fastidiousness: writing it as $\bar\Gamma\bar\rho_I\delta_I$ would set $\delta\Gamma = 0$, which contradicts Assumption 9.2's $\delta\Gamma/\bar\Gamma = -\delta_I$, under which the two contributions vanish separately. The cancellation needs neither. The $+\gamma_c\delta_{\rm m}$ of Eq. (9.5) therefore does not come from the transfer perturbation at all: it comes from the denominator, because $\delta_{\rm m}$ is defined relative to a background density obeying $\dot{\bar\rho}_{\rm m} = -3H\bar\rho_{\rm m} - \bar Q_V$, which is being drained. The linearised Raychaudhuri equation with geodesic motion, and with shear and vorticity entering quadratically, gives the second, its source being the total dust. Eliminating ϑ yields Eq. (9.6). □

Corollary 9.6. *Every linear observable built from $\delta_{\rm m}$, ϑ and the metric potentials depends on the model only through $H(a)$ and $\gamma_c(a)$. In particular b, and the split of the dark sector into precursor and defects, cannot be measured by any linear probe.*

Correction to the draft. Proposition 7 of the draft version wrote Eq. (9.6) for the precursor contrast δ_I, with $\gamma_c = \bar Q_V/\bar\rho_I$ and a gravitational source $\bar\rho_I\delta_I$. That is correct only at $b = 0$, and the restriction was not stated. For $b > 0$ it fails twice: the friction coefficient in the δ_I equation is $\bar\Gamma$, not $(1-b)\bar\Gamma$; and the Poisson source is the total dust, not the precursor alone, which by Eq. (4.3) underestimates it by a factor tending to zero as $f \to 1$. Theorem 9.5 is the repaired statement, and it has the same form as what the draft wrote, but for the right variable.

9.3 Growth, redshift-space distortions and the potential

In $N = \ln a$, with $E = H/H_{100}$ and $g_c = \gamma_c/H$, Eq. (9.6) is

C — Theorem of the model

$$\delta_{\rm m}'' + \left(2 + \frac{E'}{E} - g_c\right)\delta_{\rm m}' - \left[\frac{3}{2}\Omega_{\rm m} + g_c' + \left(2 + \frac{E'}{E}\right)g_c\right]\delta_{\rm m} = 0, \tag{9.7}$$

where $g_c = \omega_V'/\omega_\mathrm{m}$ in the reduced units of Appendix F, and the growing mode has $\delta_\mathrm{m} \propto a$ deep in matter domination. Setting $g_c = 0$ recovers the standard equation.

The observationally important point is that redshift-space distortions do not measure $\mathrm{d}\ln\delta_\mathrm{m}/\mathrm{d}\ln a$. They measure the velocity divergence, and here the two differ precisely because the dust is being sourced:

C — Theorem of the model

$$f_\mathrm{RSD} = -\frac{\vartheta}{H\delta_\mathrm{m}} = \frac{\mathrm{d}\ln\delta_\mathrm{m}}{\mathrm{d}\ln a} - \frac{Q_V}{H\rho_\mathrm{m}}, \qquad [f\sigma_8]_\mathrm{RSD} = f_\mathrm{RSD}\,\sigma_8(z). \tag{9.8}$$

The correction is not small, and its structure is the sharpest linear signature the model has. Work to first order in g_c during matter domination and write $g_c \propto a^{p_g}$ near the onset of conversion. The exponent p_g is not the p_eff of Sec. 7.1: that one is defined through $\mathsf{q} = Q_V/(3M_\mathrm{Pl}^2H_0^2H)$ and this one through $g_c = Q_V/(H\rho_\mathrm{m})$, and in matter domination the two differ by exactly 3. Substituting one into a formula written for the other is an error of tens of per cent. The effective exponent is not the switch exponent: since $g_c = \bar{Q_V}/(H\bar\rho_\mathrm{m})$ carries an explicit factor of H^{-1},

C — Theorem of the model

$$p_g = p + \tfrac{3}{2} \quad \text{for } \Gamma = \Gamma_0 S_\mathrm{act} e^{\alpha f+\beta M}, \qquad p_g = p \quad \text{for } \Gamma = \xi_0 H S_\mathrm{act} e^{\alpha f+\beta M}, \tag{9.9}$$

in matter domination. This is a derivation and not a fit, which matters because a level-C result is built on it: near the onset of conversion $S_\mathrm{act} \propto a^p$ and $f \ll 1$, so $\bar{Q_V} = (1-b)\Gamma\rho_I$ scales as $a^p a^{-3}$ for a constant amplitude and as Ha^pa^{-3} for the expansion-tracking one, while $\bar\rho_\mathrm{m} \propto a^{-3}$ and $H \propto a^{-3/2}$. Dividing, $g_c \propto a^{p+3/2}$ in the first case and a^p in the second. We have also checked both against the numerical solution. With that identification,

C — Theorem of the model

$$f_g = 1 + \frac{2p_g+3}{2p_g+5}\,g_c + \mathcal{O}\!\left(g_c^2\right), \qquad f_\mathrm{RSD} = 1 - \frac{2}{2p_g+5}\,g_c + \mathcal{O}\!\left(g_c^2\right). \tag{9.10}$$

The two move in *opposite* directions: density growth is enhanced while the velocity field, and hence $f\sigma_8$, is suppressed. The reason is that $\delta Q_{V,\mathrm{com}} = 0$ makes overdense and underdense regions lose the same absolute energy density, so overdense regions lose the smaller fraction and δ_m is boosted, while ϑ is not. That the opposite signs are a consequence of *this* closure and not a generic feature of an energy drain is worth proving rather than asserting, since the abstract rests on it.

Proposition 9.7 (A density-proportional drain has no opposite-sign signature)**.** *Replace Eq. (9.3) by a transfer proportional to the local dust density, $\delta Q_{V,\mathrm{com}} = \bar{Q_V}\,\delta_\mathrm{m}$, keeping everything else in Theorem 9.5. Then*

$$\dot\delta_\mathrm{m} = -\vartheta, \qquad \dot\vartheta + 2H\vartheta = -\frac{\bar\rho_\mathrm{m}}{2M_\mathrm{Pl}^2}\delta_\mathrm{m}, \tag{9.11}$$

so $f_g = f_\mathrm{RSD}$ identically: the enhancement of δ_m disappears, the velocity equation is unchanged, and no signature of opposite signs exists at any order in g_c.

Proof. In the proof of Theorem 9.5 the residual transfer term in the summed continuity equation is $-\delta Q_V/\bar\rho_\mathrm{m}$, and the $+\gamma_c\delta_\mathrm{m}$ comes not from it but from the drained background in the definition of δ_m. With $\delta Q_{V,\mathrm{com}} = \bar{Q_V}\delta_\mathrm{m}$ the residual is $-\gamma_c\delta_\mathrm{m}$, which cancels it exactly. The Raychaudhuri equation contains no transfer perturbation at all under geodesic motion, so it is untouched. □

The cancellation is exact and independent of γ_c, which is the content: it is the comoving-uniformity of the transfer, not its magnitude, that produces the opposite signs. Our numerical solution reproduces Eq. (9.10), but the comparison has to be made against the right reference and an earlier version of this paragraph did not make it. Eq. (9.10) expands about matter domination, where the unperturbed rate is 1; by $z = 2$ dark energy has already pulled even the ΛCDM rate below 1, so comparing the two rates to unity there tests the expansion history and not the signature. Removing g_c from the growth equation while keeping the same background isolates it. At $z = 2$, for the parameters recorded with the check, the density growth rate is 0.931 against

0.8973 with the transfer term switched off, while $f_{\rm RSD}$ is 0.8833: raised and lowered respectively, as predicted. The same holds at every one of the 9 parameter and redshift combinations tried.

For the gravitational potential the relativistic Poisson equation gives $\mathrm{d}\ln\Phi/\mathrm{d}\ln a = f_{\rm RSD} - 1$, so during matter domination

C — Theorem of the model

$$\left.\frac{\mathrm{d}\ln\Phi}{\mathrm{d}\ln a}\right|_{\rm MD} = -\frac{2}{2p_g+5}\,\frac{\bar{Q_V}}{H\bar{\rho}_{\rm m}} < 0, \tag{9.12}$$

whereas in ΛCDM the potential is exactly constant then. The integrated Sachs–Wolfe contributions therefore start earlier than in ΛCDM and add to the late-time one rather than cancelling it. It is worth resisting the natural shortcut here: one might expect Φ to decay at the full rate $\bar{Q_V}/(H\bar{\rho}_{\rm m})$, since that is the rate at which the matter is drained, but that answer holds only if $\delta_{\rm m}$ is frozen. Including the back-reaction of the drain on the growth reduces it by the factor $2/(2p_g+5)$, which for the fitted rate law and $p = 6$ means $p_g = p + \frac{3}{2}$ by Eq. (9.9) and so a factor of one tenth.

9.4 No large-scale instability

Interacting dark energy models are notorious for a super-horizon instability of the curvature perturbation, which afflicts couplings with a propagating dark energy component and a constant equation of state [39].

Proposition 9.8 (Absence of the instability)**.** *Under Assumption 9.2 the system Eq. (9.5) contains no independent dark-energy degree of freedom, no pressure perturbation, and no k-dependent coefficient. Its coefficients are of order H and H^2, so $|\delta_{\rm m}|$ is bounded by a power of a on every scale.*

The instability arises when a dark-energy fluid is given a sound speed and a density perturbation of its own while being coupled; an exact vacuum has neither, and Eq. (9.3) is precisely the statement that its perturbation is determined rather than assigned. The scope of Proposition 9.8 should not be inflated. The absence of a k-dependent coefficient is the same fact as the vanishing scalar sound speed recorded in Sec. 8.4: the scalar sector is degenerate rather than hyperbolic, with no restoring force at any wavelength. What the proposition establishes is the absence of the specific super-horizon instability of coupled fluids carrying a sound speed, and nothing about the ultraviolet behaviour of this sector or about caustics.

What this section establishes. A closed and stable linear system under Assumption 9.2, a theorem that the branching ratio is unobservable in linear theory, a corrected growth equation, and a quantitative prediction for $f\sigma_8$ and the ISW amplitude. What it does not establish is nonlinear behaviour, the fate of caustics, or a full CMB calculation: the compressed distance priors used in Sec. 10 are a geometric summary, not a substitute for a Boltzmann code.

10 Observational constraints

This section confronts the model with data. The framing matters: the aim is not to show that the model fits, but to find out whether the sign prediction of Theorem 5.4 and the jerk bound of Eq. (7.2) survive, and to report the answer either way.

10.1 Data and likelihood

Table 1 lists the combination. Supernova distances constrain the shape of the late expansion history; baryon acoustic oscillations tie it to the sound horizon; cosmic chronometers measure $H(z)$ directly and without a distance ladder; and the compressed CMB likelihood supplies the

Probe	Data set	Content	Reference
Type Ia supernovae	Pantheon+	1701 light curves, full covariance	[40]
Baryon acoustic oscillations	DESI DR2	13 measurements, $0.30 \leq z \leq 2.33$	[41]
Cosmic chronometers	Moresco compilation	15 measurements of $H(z)$; statistical errors plus the IMF and "model, odd one out" systematic terms	[42]
Compressed CMB	Planck 2018 distance priors	$\mathcal{R}$, ℓ_A, ω_{b}	[43, 44]
Redshift-space distortions	SDSS DR16 (eBOSS)	4 measurements of $f\sigma_8$, $0.38 \leq z \leq 1.48$; the $f\sigma_8$ sub-block of the consensus covariance	[45]

Table 1. Data sets used. All are public releases; a script downloads each and verifies it against a recorded checksum, and none is redistributed with the code.

early-universe anchor. The absolute magnitude of the supernovae is marginalised analytically, so the baseline combination contains no distance-ladder calibration; a variant including it is reported separately.

Three choices deserve to be stated rather than buried, since each could otherwise be mistaken for an error.

First, the sound horizon is computed from the matter density *before* activation. In this model the late-time matter density is lower than the early one by the deficit Δ of Eq. (5.3), and it is the early value that fixes the acoustic scale. Using the late value would misattribute the deficit to the early universe.

Second, the compressed CMB likelihood is a geometric summary derived within a model family. Its use here is legitimate because the model's pre-recombination physics is identical to the standard one, by Assumption 6.2: the conversion rate at recombination is below 10^{-21} of the expansion rate. It is not a substitute for a Boltzmann calculation, and the late-time-only combination is reported alongside so the reader can see what the compressed likelihood is doing.

Third, our semi-analytic recipe for the acoustic scale differs from a full Boltzmann calculation at the few-tenths-of-a-percent level, which is several standard deviations of ℓ_A. Since every model considered here shares the same early physics, the recipe is calibrated once at a reference cosmology and the same constant correction is applied throughout; a test asserts that it is not re-derived per model.

10.2 Method

Posteriors are sampled with an affine-invariant ensemble sampler; chains run until they exceed fifty integrated autocorrelation times with a stable estimate of that time. Priors are uniform and identical across models wherever a parameter is shared, so that the model comparison is not driven by prior volume. Because $\epsilon = 0$ reduces every absorption model to ΛCDM exactly, the comparison is nested and ΛCDM sits inside the prior of each extension rather than beside it. Details, including the exact prior ranges, are in Appendix F.

Two degeneracies are exact and are worth recording, because they mean certain parameters are simply not measurable here. The branching ratio b does not appear in the background separately from the combination ϵ (Proposition 5.3) and cancels identically from linear growth (Theorem 9.5); it is therefore held fixed. And the memory parameters enter the jerk only through

Model	k	$\chi^2_{\rm min}$	$\Delta\chi^2$	$\ln Z$	$\ln B$
ΛCDM	3	1436.52	+0.00	-732.21 ± 0.20	—
$\Gamma = \xi H$	6	1434.88	-1.64	-733.83 ± 0.21	-1.62 ± 0.29
reinforced, $\beta = 0$	7	1434.66	-1.85	-734.03 ± 0.21	-1.82 ± 0.29
reinforced with memory	9	1434.66	-1.85	-734.54 ± 0.21	-2.32 ± 0.29
with reverse channel	10	1429.21	-7.31	-733.92 ± 0.21	-1.71 ± 0.29

Table 2. Model comparison on the baseline combination, by Bayesian evidence. The $\chi^2_{\rm min}$ column is $-2\ln\mathcal{L}$ at the best fit, and includes the Jacobian $\ln(\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}/2\pi) = 9.33$ of the analytic marginalisation over the supernova absolute magnitude. It is therefore not a χ^2 and a χ^2 per degree of freedom formed from it would be wrong by that amount; only differences between rows are meaningful, and the Jacobian cancels from them. Differences are taken with respect to ΛCDM; k is the number of sampled parameters and a negative $\ln B$ favours ΛCDM. The information criteria are deliberately absent: at $\epsilon = 0$ each extension reduces to ΛCDM identically for *any* values of its rate parameters, so those parameters are unidentified under the null and the effective parameter count there is not k. Both $2k$ and $k\ln n$ would then penalise a number that does not describe the model at the point being tested, and both would do so in the direction of the conclusion drawn below. The evidence requires no such count. The $\Delta\chi^2$ column is a maximisation and carries its own numerical error, discussed in Appendix F.5: for the extended models the multi-start agreement is poor and the best fits sit on a prior edge in the unidentified directions, so the column should be read as what the search reached — verified against a frozen brute-force run — and not as a converged goodness of fit. Nothing in the conclusions rests on it; the evidence column does.

a factor degenerate with the rate normalisation (Theorem 7.5), so any information about them must come from the shape of the expansion history rather than from its low-order derivatives. The posterior says there is none: of the nine sampled parameters of the full model, five have a 5–95 range covering more than seventy per cent of their prior, and the two memory parameters are among them at 75% and 87%. They are not measured here, and Table 3 should be read as a statement about which directions the data leave open rather than as a measurement of nine numbers.

10.3 Results

Tables 2 and 3 report the fits. Four things follow, and the first is that the pipeline reproduces the standard answer: on the baseline combination ΛCDM gives $H_0 = 68.38 \pm 0.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_{\rm m0} = 0.3014$, with $j_0 = 1.00016$ — not unity, because ΛCDM with radiation and one massive neutrino eigenstate carries the two terms of Remark 7.2, which is the point of that remark.

The convertible fraction is not measurable, for a reason. Profiling the likelihood in ϵ over $[0, 0.6]$ never raises χ^2 by more than 1.64 above its minimum, so the one-sided threshold $\Delta\chi^2 = 2.71$ is never reached and there is no upper limit to quote. This is a statement about the parametrization rather than about the data, and the mechanism is worth naming because it recurs throughout this section: ϵ is the fraction available for conversion, not the fraction converted, so the optimizer can always send the rate to zero and recover ΛCDM exactly at any ϵ. The same degeneracy appears geometrically in the posterior. The vacuum present at activation and the vacuum produced by conversion are two sources of one quantity, and the data constrain their sum, not the split. The profile shows this in its starkest form: along it the activation vacuum sits at the lower end of its prior, $\Omega_{V,\rm act} = 0$, for every $\epsilon > 0$, with conversion supplying the whole of the present vacuum; at $\epsilon = 0$ it jumps to $\Omega_{V,\rm act} = \Omega_{V0}$, supplying all of it itself. The fit is equally good at both ends because it is the same fit. An earlier version of this analysis reported a 95% upper limit on ϵ; it was the right-hand end of the grid, and it is withdrawn.

Parameter	Posterior (median, 68%)	prior filled
$\alpha + \beta$	*unconstrained*	89%
ϵ	*unconstrained*	83%
$\log_{10}(\Gamma_0/H_{100})$	$-1.60^{+0.48}_{-0.26}$	34%
$\log_{10}(\tau_M H_{100})$	*unconstrained*	87%
$\omega_{V,\mathrm{act}}$	$0.295^{+0.029}_{-0.097}$	18%
ω_b	$0.0226^{+0.0001}_{-0.0002}$	6%
ω_c	$0.1182^{+0.0007}_{-0.0009}$	1%
$\beta/(\alpha+\beta)$	$0.50^{+0.27}_{-0.25}$	75%
z_act	8^{+11}_{-6}	71%
derived		
H_0 [km s^{-1} Mpc^{-1}]	$68.49^{+0.40}_{-0.31}$	
$\Omega_{\mathrm{m}0}$	$0.2982^{+0.0053}_{-0.0067}$	
Ω_{V0}	$0.7018^{+0.0067}_{-0.0053}$	
f_0	*prior-limited*	
Δ	< 0.0109 (95%)	
q_0	$-0.553^{+0.008}_{-0.010}$	
j_0	$1.0050^{+0.0087}_{-0.0042}$	
z_acc	0.673 ± 0.012	

Table 3. Posterior for the full model on the baseline combination. Derived quantities are computed from samples, not propagated linearly. The last column is the fraction of each prior covered by the 5–95 span, by the same definition the chain-stability gate uses; a parameter above 75% of its prior is marked *unconstrained* and no number is quoted for it, because the number would be the prior's. A derived quantity is marked *prior-limited* when it is bounded only in that sense — when no profile of it crosses $\Delta\chi^2 = 2.71$ — which is why Δ carries a limit and f_0 does not. The limit shown for Δ is the marginal 95% bound for this nine-parameter model; Eq. (10.1) is the profile-likelihood limit for the $\Gamma = \xi H$ benchmark, and the two agree to about eight per cent.

Is $\Delta\chi^2 = 2.71$ the right threshold here? It is the regular-theory value for a one-sided limit on one parameter, and this section has just argued that regular theory does not apply: at $\epsilon = 0$ the rate parameters are unidentified, which is why the information criteria were dropped. Using the non-regularity where it removes an unfavourable number and ignoring it where it would loosen a favourable one would not be defensible, so the threshold is measured rather than assumed. Synthetic data are generated at the ΛCDM best fit with the real covariances, both models are maximised on each — the extension seeded with the null's answer embedded, so the statistic cannot come out negative — and the null distribution of $\chi^2_{\Lambda\mathrm{CDM}} - \chi^2_\mathrm{ext}$ is accumulated over 96 realizations. Its ninetieth percentile, which is what a 95% one-sided limit inverts, is 2.42 ± 0.41 against the regular-theory 2.71: the threshold in use is if anything *conservative*, with probability 0.08 that the true value exceeds it. The distribution has an atom at zero — in a fraction 0.05 of realizations the extension cannot improve on the null at all — which is the signature of the boundary and the reason the naive χ^2_1 intuition fails in the direction it does. This is the Davies statistic for the unconstrained comparison, which explores more of the unidentified space than a profile at fixed derived quantity does, so it bounds the correction the profiles would need. We therefore keep 2.71, and record that the limits below are marginally weaker than a calibrated threshold would give.

What the data do constrain is the deficit. The quantity with physical content is not how much substrate could convert but how much vacuum the conversion has actually produced, which is the matter deficit Δ of Eq. (5.3). Profiling it rather than ϵ gives a genuine one-sided limit,

P — Fitted to data

$$\Delta < 0.0100 \qquad (95\%), \tag{10.1}$$

with the profile rising to $\Delta\chi^2 \simeq 52.7$ across the grid, so this is a constraint and not another flat direction. About one per cent of the critical density is the most that can have been moved from matter into vacuum since activation. The constraint is imposed exactly rather than by a penalty: at each grid point every other parameter is free and ϵ is solved for so that the deficit takes the target value, which it does to a relative error of 2.15×10^{-11}, and the profile has 0 decreasing steps above its minimum. Those two diagnostics are reported because an earlier version of this analysis used a quadratic penalty too weak to hold the constraint: its grid points missed their targets by up to a factor of three and its profile was not monotone, and the limit it produced was not meaningful.

This does *not* bound the jerk excess through Eq. (7.3), in either of the two ways one might hope. That relation is a floor, so an upper limit on Δ says nothing about the excess from above; and its hypothesis fails for the activation law used here, for the reason set out in Sec. 7.1. The excess is therefore profiled directly, as its own derived quantity ε_0 — the transfer term alone, not $j_0 - 1 - 2\Omega_{r0}$, which also contains the massive-neutrino pressure that ΛCDM carries too. That gives

P — Fitted to data

$$\varepsilon_0 < 0.0273 \qquad (95\%), \tag{10.2}$$

with the same exact constraint and the same two diagnostics: targets met to 3.06×10^{-9} and 0 decreasing steps. Set against the $\sigma(j_0) \approx 0.4$ of present late-time cosmographic reconstructions, that is the honest statement of the model's current testability: the signature of Theorem 7.1 is real, has the sign the model predicts, and is more than an order of magnitude below anything existing data can resolve. The deficit limit does not tighten it, because the two quantities measure different things (Sec. 7.1).

ΛCDM **is preferred, and by less than it first appeared.** Table 2 gives the evidences. Every extension is disfavoured, and all of them by a similar amount — odds of a few to one, not a decisive margin. We draw attention to the size of these numbers because an earlier version of this comparison, using AIC and BIC with the sampled parameter count, reported a far larger margin; those criteria are inapplicable here for the reason given in the caption, and both of their errors happened to favour the conclusion we were drawing. The result is a null one either way and is reported as such: nothing in this data combination requires conversion, and the model's free parameters are bounded rather than measured.

How much of that is the prior. Enough to matter, and the reason is structural rather than a matter of taste. The extension contains ΛCDM in two distinct ways: exactly, on the face $\epsilon = 0$, and asymptotically, as the rate goes to zero at any ϵ. The second means the region of parameter space in which the two models are observationally indistinguishable is unbounded below in $\log_{10}\xi$, so how much prior mass is assigned to it is a choice and not a datum. Widening that prior adds volume where the likelihood equals ΛCDM's, and the evidence rises. Measured, with two seeds at each width:

Prior on $\log_{10}\xi$	$\ln B$	odds against
$[-3, 1]$ (baseline)	-1.93	6.92:1
$[-5, 1]$	-0.80	2.2:1
$[-7, 1]$	-0.55	1.73:1

The range is 1.39 in $\ln B$, comparable to $\ln B$ itself. The verdict does not change sign anywhere in it — ΛCDM is preferred under every prior we would defend — but its strength is

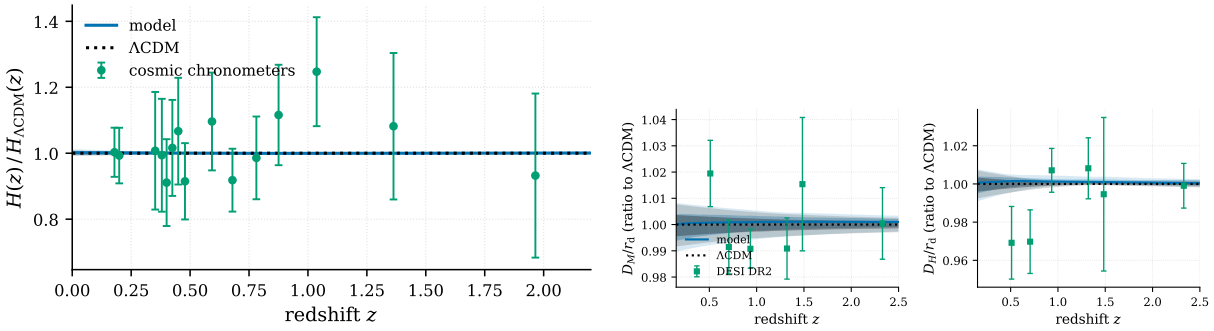


Figure 4. The two geometric probes, each as a ratio to the ΛCDM best fit so that the comparison is visible at all. *Left:* $H(z)$ against the cosmic chronometers. *Right:* the DESI DR2 distance ratios. Bands are 68 and 95 per cent of the posterior for the full model. The model tracks ΛCDM to well within the error bars over the whole range, which is the graphical form of the null result: the data neither require the extension nor exclude it.

not determined by the data. That is a property of comparisons in which one model reaches the other as a limit over an unbounded range of a logarithmic parameter, and it is the reason this section reports the preference as modest and declines to quote $\ln B$ to two decimal places.

By contrast the sampler's own noise is small: three independent seeds at the baseline prior give $\ln Z$ values whose standard deviation is 0.146 against a reported error of 0.208, a ratio of 0.70. The uncertainty in the comparison is the prior, not the sampling.

One feature of the table is worth more than the headline: adding parameters buys nothing. The nine-parameter model with memory is placed *worse* than the six-parameter one, because the parameters it adds are the ones the data do not identify, and the evidence charges them their prior volume and credits them nothing. The reverse channel, with the most parameters of all, recovers part of its penalty by fitting better — its $\Delta\chi^2$ is four times the others' — and still does not come out ahead.

The ordering among the 4 extensions should not be read as a result. The gap between the best and the worst of them is 0.71 in $\ln B$, against an error of 0.30 on a difference between two of them — the ΛCDM reference cancels, so the two log-evidence errors add in quadrature — which is 2.4σ; and against a prior-width range of 1.39 documented above, which is larger than the whole spread. On the evidence the four are indistinguishable, and the reverse channel is carried through the comparison because the theory requires it to be — it is what the sign predictions are predictions *against* — and not because the data prefer it.

One ensemble chain is quoted from and two are archived. Three ensemble chains were run. The ΛCDM one has cumulative acceptance 0.4386, runs for 157 autocorrelation times, and is the source of the ΛCDM row. The other two, for the $\Gamma = \xi H$ and reinforced models, are archived failures kept so that the diagnosis below can be repeated; the four extended rows of Table 2 come from nested sampling and not from any ensemble chain. The chains for the extended models are not, and this is a consequence of the geometry just described rather than of insufficient running: with the rate parameters unidentified along the ΛCDM boundary the ensemble spread to fill its prior in those directions, after which most proposals fell outside the prior box and were rejected before the likelihood was consulted, so the move rate *fell* as the run went on. Their posteriors are therefore taken from nested sampling, which integrates over the prior rather than trying to walk along a degenerate ridge inside it. The diagnostic that failed to catch this has been rewritten and now does; Appendix F records both.

10.4 Reading the result

Whatever the numbers, three things should be kept distinct.

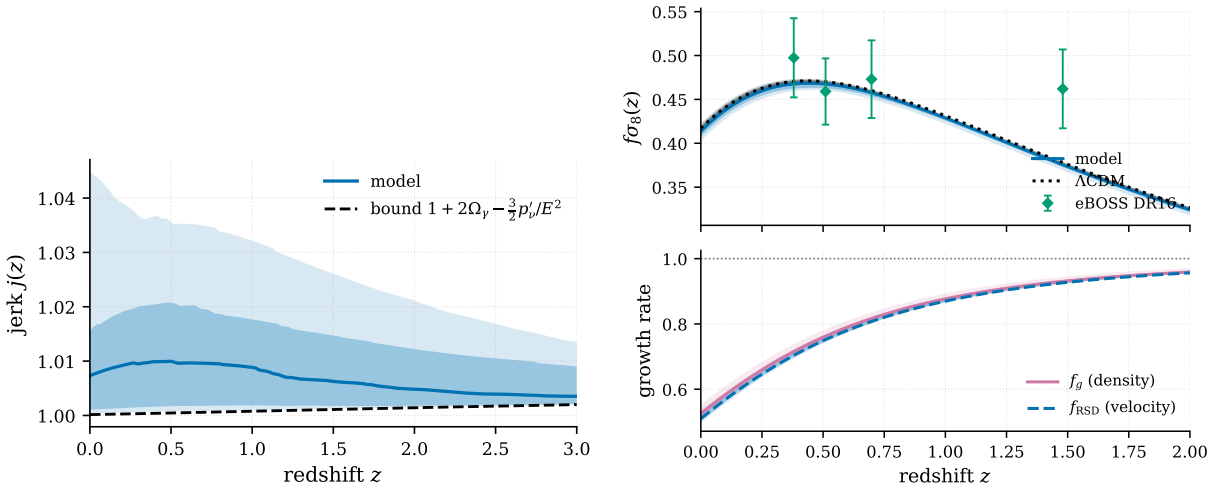


Figure 5. The two signatures the model predicts, at the amplitude the data allow. *Left:* the jerk against the bound $j \geq 1+2\Omega_{\rm r}$ of Theorem 7.1. The posterior sits above the bound everywhere, as it must, and the excess is bounded by Eq. (10.2) — more than an order of magnitude below what late-time cosmography resolves. *Right:* $f\sigma_8$ against the four SDSS DR16 measurements [45], described in Appendix F.4. Including them moves $\ln B$ from -1.62 to -1.46, a shift of $+0.16$ — smaller than the uncertainty on either, and in the same direction as the model's prediction without approaching significance. What this establishes is that the model's sharpest linear signature is not detected: the growth data neither rescue it nor bury it. Both numbers are for the $\Gamma = \xi H$ benchmark, which is the model fitted to the growth combination, and not for the full memory model. *Lower right:* the two growth rates themselves, f_g and $f_{\rm RSD}$, which is where the model's actual signature lives: they separate in opposite senses about the ΛCDM value, by the amounts of Eq. (9.10), and the data plotted above constrain the second while the first is not directly observed.

A fit that prefers $\epsilon \to 0$ would say the data are consistent with no conversion, not that the mechanism is impossible; it would bound Γ_0 and ϵ, which is a useful and reportable outcome for a paper whose main claims are structural.

A fit that prefers $\epsilon > 0$ would not by itself be evidence for the mechanism, because the model can partially mimic an evolving dark-energy equation of state within the band of Theorem 5.6. What would distinguish it is the combination of signatures: a positive jerk excess (Eq. (7.2)), a matter deficit of a consistent size, and the opposite-sign growth signature of Eq. (9.10). Any one of these alone is degenerate with other physics; together they are not.

And a fit that requires $\delta_{\rm eff} > 0$ would falsify the one-way model outright, by Theorem 5.4. Current DESI-era analyses lean weakly in that direction [28, 29], which is why the reverse channel of Sec. 5.5 is fitted alongside rather than after the fact. That the evidence for evolving dark energy is itself sensitive to supernova systematics rather than settled [46] is a reason to fit both branches and not to choose between them here.

What this section establishes. Constraints on the model's parameters from public late-time data with an early-universe anchor, obtained with code and checksums archived. What it does not establish is a full CMB analysis or a nonlinear structure-formation comparison; both are absent. The evidence is quoted together with its sensitivity to the prior it depends on, which is the honest way to report a quantity that has one.

11 Localized defects

Assumption 4.1 posits a pressureless product carrying a fraction b of the converted energy. This section asks what such an object could be. The answer offered is a topological texture, and the section is careful about what is and is not established: a finite-energy solution with an

integer charge and a preferred size is exhibited and solved, but its identification with dark matter remains a conjecture, and its abundance is not computed.

11.1 An obstruction, and how to evade it

Proposition 11.1 (Derrick scaling)**.** *In three spatial dimensions let* $E[\phi] = E_2 + E_0$ *with* $E_2 = \frac{1}{2}\int |\nabla\phi|^2$ *and* $E_0 = \int W(\phi)$*, both finite and* $W \geq 0$ *after subtracting the vacuum. Under* $\phi_\sigma(\mathbf{x}) = \phi(\sigma\mathbf{x})$ *one has* $E(\sigma) = \sigma^{-1}E_2 + \sigma^{-3}E_0$*, so a non-trivial stationary point would require* $E_2 + 3E_0 = 0$*, impossible for* $E_2 > 0$*.*

A single static canonical scalar therefore has no non-trivial static localized solution at all, which is stronger than the absence of a stable one and is what is needed here. The hypotheses are what matter: time dependence, a conserved charge, gauge fields, gravity, or additional terms in the energy each evade the conclusion. We take the last route. Let $n^A(\mathbf{x})$, $A = 1, \ldots, 4$, satisfy $n^A n^A = 1$ with a constant value at infinity, so that configurations are maps $S^3 \to S^3$, and take

A — Postulate

$$E = \int \mathrm{d}^3x \left[\frac{F_*^2}{2}\partial_i n^A \partial_i n^A + \frac{\kappa_s}{4}\mathcal{W}_{ij}^{AB}\mathcal{W}_{ij}^{AB} \right], \qquad \mathcal{W}_{ij}^{AB} = \partial_i n^A \partial_j n^B - \partial_j n^A \partial_i n^B, \tag{11.1}$$

with $F_* > 0$ of mass dimension one and $\kappa_s > 0$ dimensionless. Under the same scaling the two terms go as σ^{-1} and σ^{+1}, so Proposition 11.1 no longer applies and a preferred size exists. This is the Skyrme mechanism, and Eq. (11.1) is the SU(2) Skyrme energy with $F_\pi^2 = 4F_*^2$ and $e^2 = 1/(2\kappa_s)$; we use it as a calculable representative of a localized product, not as a new proposal.

11.2 Degree, bound, and the trial family

The topological degree

$$B = \frac{1}{12\pi^2}\int \mathrm{d}^3x\, \epsilon^{ijk}\epsilon_{ABCD} n^A \partial_i n^B \partial_j n^C \partial_k n^D \tag{11.2}$$

is an integer on such maps. A pointwise application of the arithmetic-geometric mean inequality to the singular values of $\mathrm{d}n$ gives the Bogomolny-type bound

I — Identity

$$E \;\geq\; 6\sqrt{2}\,\pi^2\, F_*\sqrt{\kappa_s}\,|B|\,. \tag{11.3}$$

For the rational-map family $n^a = 2R_s x^a/(r^2+R_s^2)$, $n^4 = (r^2-R_s^2)/(r^2+R_s^2)$, direct quadrature gives $B = 1$ and

I — Identity

$$E(R_s) = 6\pi^2\left(F_*^2 R_s + \frac{\kappa_s}{R_s}\right), \qquad R_s^* = \frac{\sqrt{\kappa_s}}{F_*}, \qquad E_{\mathrm{trial}} = 12\pi^2 F_*\sqrt{\kappa_s}. \tag{11.4}$$

11.3 The solution

The draft stopped at Eq. (11.4), which is only an upper bound on the infimum of the sector. Solving the field equations is straightforward and sharpens the statement. With the hedgehog ansatz $n^4 = \cos F(r)$, $n^a = \sin F(r)\,\hat{x}^a$ and $r = (\sqrt{\kappa_s}/F_*)x$, the Euler–Lagrange equation of Eq. (11.1) is

I — Identity

$$\left(x^2 + 4\sin^2 F\right)F'' + 2xF' + 2\sin 2F\, F'^2 - \sin 2F - \frac{2\sin 2F \sin^2 F}{x^2} = 0, \tag{11.5}$$

with $F(0) = \pi$ and $F(\infty) = 0$, regular series $F \simeq \pi - \kappa_1 x$ at the origin and tail $F \simeq C/x^2$. Solving this boundary-value problem numerically (Appendix F) gives

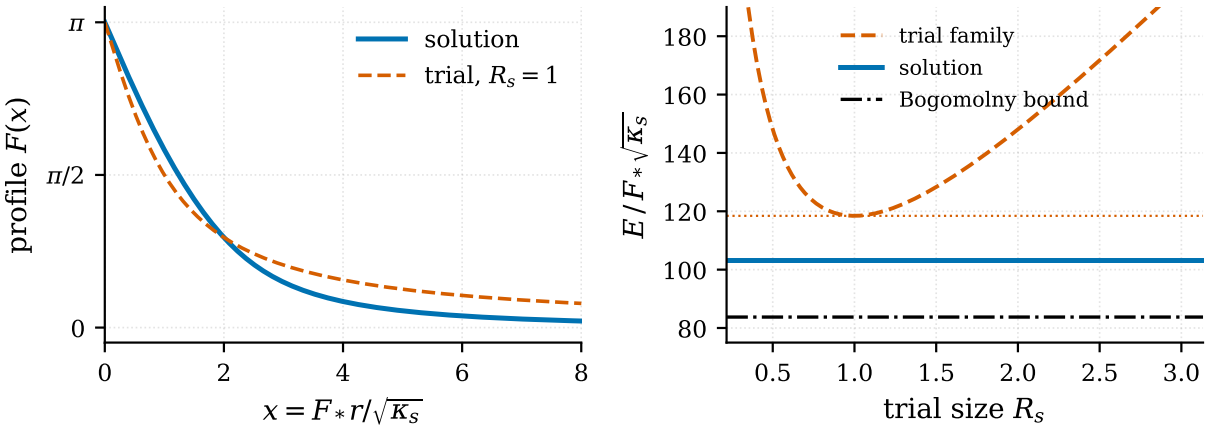


Figure 6. The solved defect. *Left:* the hedgehog profile obtained from Eq. (11.5) compared with the trial family at its preferred size; the solution has the faster x^{-2} tail. *Right:* energies in units of $F_*\sqrt{\kappa_s}$. The trial family (dashed) is minimised at 118.435, the solution (solid) sits at 103.129, and the Bogomolny bound Eq. (11.3) (dash-dotted) is 83.746.

N — Numerical

$$E_{B=1} = 103.129\, F_*\sqrt{\kappa_s} = 1.2314 \times \left(6\sqrt{2}\pi^2 F_*\sqrt{\kappa_s}\right), \tag{11.6}$$

which is 0.871 of the trial value Eq. (11.4), and which satisfies the virial identity $E_2 = E_4$ to machine precision and returns $B = 1$ to quadrature accuracy. The trial family therefore overestimates the energy by about thirteen per cent, and the solved configuration sits above the Bogomolny bound by the factor familiar from the Skyrme model.

Limits of this calculation. A minimum within the hedgehog ansatz is not a proof of stability against all deformations: that would require the full Hessian. And the energy of a solution is not a particle mass until the sector is quantised; the collective coordinates, their moments of inertia, and the Finkelstein–Rubinstein constraints all enter. What Eq. (11.6) establishes is that the localized sector of Assumption 4.1 has a concrete, calculable representative with an integer charge, a finite energy and a preferred size.

11.4 Why the equation of state is what it is

Being localized is not by itself enough to be pressureless. For an isotropic population of objects of mass m_D with distribution $\mathcal{F}(\mathbf{p})$,

$$\rho_D = \int \mathrm{d}^3p\, E_{\mathbf{p}}\mathcal{F}, \qquad p_D = \frac{1}{3}\int \mathrm{d}^3p\, \frac{\mathbf{p}^2}{E_{\mathbf{p}}}\mathcal{F}, \qquad \frac{p_D}{\rho_D} \simeq \frac{\langle v^2\rangle}{3} \ll 1 \ \text{ iff } \ |\mathbf{p}| \ll m_D. \tag{11.7}$$

So the pressureless assumption is a statement about the momentum distribution of the products, and formation by absorption could well produce fast objects. That the conversion is the same process which sets the momentum distribution is a constraint the model does not currently compute.

Dimensionality matters as much as localization. For a frozen comoving population of defects of spatial dimension d with energy proportional to their extent, $\rho \propto a^{d-3}$ and hence $w = -d/3$: point-like relics give $w = 0$, frozen strings $-1/3$, frozen walls $-2/3$. A $U(1)$ vortex in three dimensions is a string, not a particle, and a string's energy per unit length being finite does not make an infinite string's energy finite. The texture above is genuinely three-dimensional and point-like, which is why it is the representative chosen here.

11.5 Caustics: what conversion can and cannot do

The clock congruence of Sec. 8.3 is pressureless and geodesic, so it forms caustics, and it is natural to ask whether conversion regularizes them: the transfer $Q_V = (1-b)\Gamma\rho_I$ diverges where ρ_I does, which looks like a runaway sink. An earlier version of this work conjectured that it was. It is not, and the reason is worth stating because it is elementary and general.

Along a flow line with proper time τ and expansion Θ,

I — Identity

$$\rho_I(\tau) = \rho_I(\tau_0)\,\exp\left[-\int_{\tau_0}^{\tau}(\Theta+\Gamma)\,\mathrm{d}s\right]. \tag{11.8}$$

Conversion enters only through the *integral* of the rate.

Theorem 11.2 (Bounded rates cannot regularize a caustic)**.** *Let $0 \le \Gamma \le G < \infty$ on $[\tau_0, T)$ with $T-\tau_0$ finite, and let the compression be non-integrable, $\int^T \Theta\,\mathrm{d}s = -\infty$. Then $\rho_I \to \infty$.*

Proof. $\int^\tau \Gamma\,\mathrm{d}s \le G(T-\tau_0) < \infty$, so the conversion factor lies in $[e^{-G(T-\tau_0)}, 1]$ and is bounded below by a positive constant, while the compression factor diverges. □

Under Assumption 6.2 the rate obeys $\Gamma \le \Gamma_0 e^{\alpha+\beta}$, so Theorem 11.2 applies to the model as it stands. The error in the earlier conjecture was to confuse the diverging transfer per unit volume, Q_V, with the fractional loss rate Γ; only the latter appears in Eq. (11.8), and only it is bounded.

What a compression-driven rate would have to do. Take $\Theta = -c/(T-t)$ and $\Gamma = g/(T-t)$ plus a bounded part. Then $\rho_I \sim (T-t)^{-(c-g)}$, and three separate conditions must hold, not one. The precursor is finite for $g \ge c$. The vacuum increment integrates $(T-t)^{g-c-1}$ and therefore needs $g > c$ *strictly*: at $g = c$ the precursor is finite while the vacuum still diverges logarithmically. And the defect density obeys $\dot{\rho_D} = -\Theta\rho_D + b\Gamma\rho_I$. Its homogeneous part is $\rho_D(t_0)\,|J_0/J|$, which diverges as $(T-t)^{-c}$ for *any* non-zero initial defect density, whatever b is: the localized product is pressureless dust and is compressed by the same Θ as everything else. Setting $b = 0$ removes the source, not that. Regularity therefore requires both that none be produced during the collapse and that none be present when it begins. Its column density stays finite either way, which is the distinction between a finite mass and a finite density. Hence

C — Theorem of the model

$$\begin{gathered}\text{for } \Theta = -c/(T-t),\ \Gamma = g/(T-t) + \text{bounded},\\ \text{along a geodesic pressureless congruence:}\\ T^{(\mathrm{dark})}_{\mu\nu} \text{ finite at the caustic} \iff g > c \ \text{ and } \ b = 0 \ \text{ and } \ \rho_D(t_0) = 0.\end{gathered} \tag{11.9}$$

The hypotheses are displayed with the statement because without them it reads as a theorem about caustics in general, which it is not: it is a statement about the self-similar fold profile above, and $T_{\mu\nu}$ means the dark sector's own stress.

It is worth being careful about what that buys. Pointwise finiteness of the stress tensor is not a criterion for the absence of a spacetime singularity, and a generic fold caustic of a dust congruence is not a strong curvature singularity in the sense of Tipler [47, 48]: the double integral of $R_{\mu\nu}k^\mu k^\nu$ along a causal geodesic converges, and the spacetime is expected to be extendible through it. What the conditions above control is therefore the divergence of the source in the field equations, not the existence of a genuine singularity. That makes the result useful for deciding whether the effective description of Assumption 4.1 remains applicable, and not for any claim about singularity avoidance.

For $\Gamma_{\mathrm{collapse}} = \kappa\,[-\Theta]_+$ one has $g = \kappa c$, so the condition is the dimensionless statement $\kappa > 1$, independent of the strength of the collapse; for a generic Zel'dovich pancake $c = 1$.

The requirement sits awkwardly with the rest of this section, and the tension should be named rather than left for the reader. A finite relic abundance of defects needs $b > 0$ (Sec. 11.6); regularity at a caustic needs $b = 0$, and the third condition of Eq. (11.9) makes the conflict sharper

than a statement about b alone. One might hope to separate the two channels — form the defects earlier, by some other mechanism, and let the compression-driven channel feed only the vacuum — but that does not help: defects already present on the collapsing sheet are pressureless dust, are compressed by the same Θ, and carry the divergence by themselves. Pointwise regularity of $T_{\mu\nu}$ at a fold caustic is therefore incompatible with the presence of *any* localized product there, however it was made. Either the localized sector acquires pressure, in which case it is no longer the pressureless product of Assumption 4.1 and its equation of state must be recomputed, or the collapsing sheets are free of defects for a reason this paper does not supply. We record this as an unresolved tension.

The covariant price. Adding such a channel is not free. Eq. (2.6) forces the transfer four-vector to be closed, and for a geodesic clock that means the transfer must be constant on comoving slices, which a rate depending on Θ is not, precisely at a caustic. Transferring instead to a second pressureless sector is covariantly unconstrained, because two dust sectors sharing a congruence exchange energy with no integrability condition, but then the total dust obeys the unmodified continuity equation and the divergence is relocated rather than weakened. Keeping the exact vacuum requires a momentum transfer $q^\nu = -D^\nu V$, which makes the dust non-geodesic and whose force density itself diverges at the caustic. The sector that can absorb the divergence is the one that does not dilute, and that is exactly the sector whose transfer is constrained.

One further option is worth recording because it is the least damaging and is easy to miss. Give the receiver a traceless spatial anisotropic stress $\Pi^{\mu\nu}$ alongside the vacuum. Total conservation with a geodesic dust then requires only $\nabla_\mu \Pi^{\mu i} = D^i V$, whose isotropic part is unchanged, so the equation of state stays $p = -V$; and Π vanishes identically both in FLRW, by isotropy, and at linear order under Assumption 9.2, because $\delta V_{\rm com} = 0$ there. It switches on only where the vacuum fails to be comoving-uniform. This is not a repair, however, and the size of the gap can be stated exactly: a symmetric traceless spatial tensor has five independent components and the conservation condition is three equations, so a closure must supply two further relations per point. Until it does, the system is underdetermined — Π is fixed only up to a divergence-free traceless tensor — and what the option achieves is to replace an obstruction by a counted deficit. Supplying such a closure is left open.

What is not disturbed, and in what sense. The candidate channel $\kappa[-\Theta]_+$ vanishes identically on the open set $\{\Theta > 0\}$. An expanding background has $\Theta = 3H > 0$, so the background is untouched, and so is any perturbation small enough in the supremum norm that Θ stays positive everywhere: for such a perturbation every order of the expansion vanishes, not merely the linear one. Nothing in Secs. 4, 7, 9 and 10 changes.

The qualification is not decorative. The statement is pointwise, and the topology has to be L^∞: in L^p with $p < \infty$ a perturbation can keep its amplitude while shrinking its support, so it is small in norm and still drives Θ negative somewhere. The statement likewise fails for any coarse-grained description in which Θ has a distribution with support below zero. If Θ is Gaussian about $3H$ with width σ, then $\langle[-\Theta]_+\rangle$ is positive for every σ and is suppressed only as $e^{-9H^2/2\sigma^2}$: non-perturbative rather than zero, and numerically 8.3×10^{-2} of $3H$ at $\sigma = 3H$ but 7.4×10^{-26} at $\sigma = 0.3H$. So once structure formation has produced collapsing regions, an averaged treatment of this channel is not exactly zero; what is exactly zero is its contribution wherever the local expansion is positive, which is what the observational sections use.

A rate proportional to the expansion has no positive local extension. The solvable family of Sec. 6.1 uses $\Gamma = \xi_0 H$. Covariantizing it in the obvious way, $H \to \Theta/3$, gives a rate that is negative wherever $\Theta < 0$, which violates the non-negativity assumed throughout and in particular in Proposition 3.7. That family therefore has no unambiguous local extension into

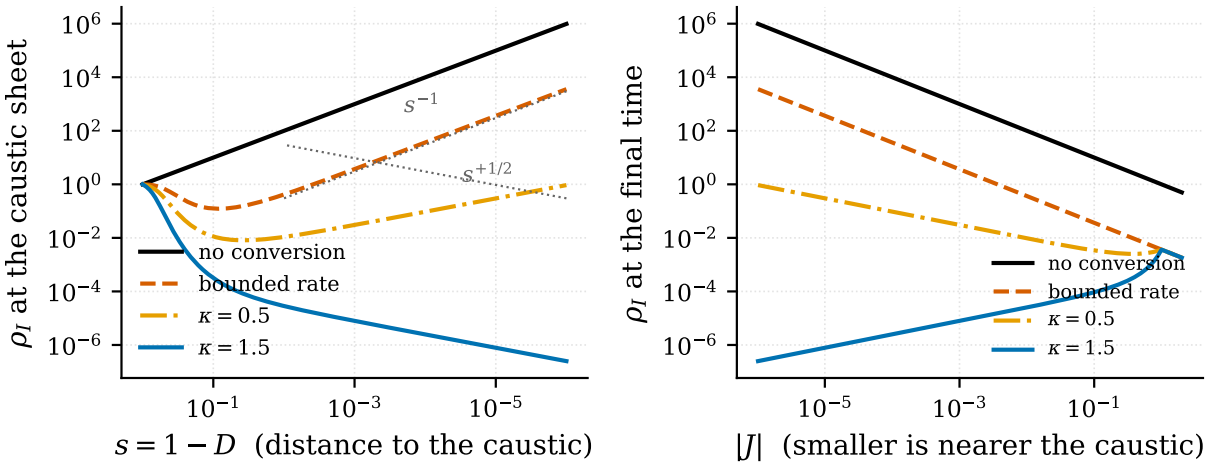


Figure 7. One-dimensional Zel'dovich pancake with conversion, followed along Lagrangian sheets. *Left:* precursor density at the sheet that forms the caustic, against the distance $s = 1 - D$ to it, with time running to the right. With no conversion, and with the bounded rate of Eq. (3.4), the density follows s^{-1}: the bounded rate lowers the amplitude but not the exponent, which is Theorem 11.2. A compression-driven channel changes the exponent to $s^{\kappa-1}$, so only $\kappa > 1$ removes the divergence. *Right:* converted fraction and Jacobian across the pancake at the final time, showing that conversion is concentrated where $|J|$ is smallest.

a collapsing region, and the kinetic law it belongs to has to be read as a statement about the homogeneous mode, as Assumption 9.2 already requires.

Numerical test, and what it does not cover. Fig. 7 reports a one-dimensional Zel'dovich pancake carrying the local sector along Lagrangian sheets. Three limitations should be stated before the numbers. The trajectories are prescribed by the Zel'dovich map, so the test isolates the effect of conversion from the gravitational back-reaction and is not a self-consistent solution. The runs approach the caustic but do not cross it: with the fiducial profile the Jacobian is $J = 1 - D\cos kq > 0$ for $D < 1$, so the caustic sits at $D = 1$ and the multi-stream region beyond it is untested. And, most importantly for the mechanism, after a crossing the Jacobian grows again, so Θ turns positive and $\kappa[-\Theta]_+$ *switches itself off* exactly where it was meant to act. What the compression-driven channel can do, it must therefore do on the approach; the integrated hazard $\kappa \log(1/|J|)$ does diverge there, which is why the exponents below are well defined, but nothing here describes the multi-stream regime.

A fourth limitation is one of measurement rather than of physics. The sheets are discrete, so the one nearest the caustic sits at $q = \Delta q/2$ and its Jacobian bottoms out at $J = s + (1 - s)\varepsilon$ with $\varepsilon = 1 - \cos(k\Delta q/2) \simeq (k\Delta q/2)^2/2$. That floor, not the time integration, is what limits how close to the caustic an exponent can be read: at the resolution used here it is $\varepsilon \simeq 2 \times 10^{-8}$, and a window at $s \sim 10^{-5}$ therefore returns an exponent a few parts in 10^4 short of its limit. The run with no conversion is best read as a check of the solver against the exact kinematic result $\rho_I = 1/J$, which it reproduces to a relative error of 5×10^{-11}; what the other runs add is the behaviour of the coupled system, where the compression channel acts together with the bounded reinforcing rate and its memory, and where the exponent is consequently not available in closed form.

The measured exponents reproduce the analytic values. With no conversion the central density follows $s^{-0.9993}$, and with the bounded rate of Eq. (3.4) it follows $s^{-0.9989}$: the same exponent, with the amplitude lowered by a finite factor of 273.9. That is Theorem 11.2 made visible — a bounded rate buys an amplitude, never an exponent. The compression-driven channel gives $\kappa - 1$, measured as -0.4992, 0.0004989, 0.5002 and 1.999 for $\kappa = 0.5$, 1, 1.5 and 3 against the predicted -0.5, 0, 0.5 and 2, and the exponent is independent of the branching ratio, the

memory time and the reinforcement strengths — asymptotically, and measurably so: the spread of the fitted exponent across those choices falls by about a decade for each decade closer to the caustic. Across all 12 runs the largest violation of the three identities the simulation conserves exactly — $\rho_I\,|J| = 1 - f$, $\rho_D\,|J| = bf$, and their sum — is 4.54×10^{-11}.

11.6 What would be needed for an abundance

By Eq. (4.3), $\rho_D a^3 \to \rho_{D,\mathrm{act}} + b\rho_{I,\mathrm{act}}$, so the comoving abundance is fixed by the branching ratio and the initial data. Converting that into a number density requires the mass m_D, which the background does not constrain, and a viable relic further requires either an asymmetry or a production rate low enough that annihilation is negligible, together with a free-streaming length short enough not to erase structure. None of this is computed here. Where such a computation has been carried through for a Skyrme-type soliton — in the electroweak sector rather than here — the constraints are severe and instructive [49]; nothing in this paper is close to that standard, and the comparison is the honest measure of the distance. The topological charge is also not a spin: although $\pi_4(S^3) = \mathbb{Z}_2$ permits a Finkelstein–Rubinstein quantisation in which the soliton is a fermion, establishing it requires showing that the 2π rotation loop is non-contractible in the configuration space and choosing a representation of its fundamental group. Sec. 13 lists both as open.

What this section establishes. A solved, finite-energy, unit-charge configuration with a preferred size, and the exact conditions under which such a population is pressureless. What it does not establish is that these objects exist, that they are stable against general deformations, that they are fermions, or that their relic abundance matches the observed dark matter.

12 Quantum absorption, memory and irreversibility

The kinetic law of Sec. 3 is irreversible. Closed quantum evolution is not. This section states the obstruction precisely, gives an open-system realization that reproduces the classical chain exactly, and is explicit about what the quantum language adds and what it does not.

12.1 The obstruction

Proposition 12.1 (No absorption under unitary flow)**.** *Let $\mathscr{H}$ be finite-dimensional, $U(t) = e^{-iHt/\hbar}$ unitary, and $\mathcal{A} \subset \mathscr{H}$ a subspace with $U(t)\mathcal{A} \subseteq \mathcal{A}$ for all $t \geq 0$. Then $U(t)\mathcal{A} = \mathcal{A}$ and $U(t)\mathcal{A}^{\perp} = \mathcal{A}^{\perp}$, so the probability of occupying $\mathcal{A}$ cannot increase by transfer from its complement.*

Proof. U is invertible and $\mathcal{A}$ finite-dimensional, so an inclusion between spaces of equal dimension is an equality. Unitarity preserves orthogonality, hence the complement is preserved too. □

Equally, two orthogonal states cannot evolve unitarily into the same state, since inner products are preserved. Absorption of distinct configurations into one description therefore requires an environment, a restriction of the observables, or a limit not covered by these hypotheses. The proposition does not forbid effective irreversibility in open systems; it delimits what may be claimed about a closed finite system without adding a mechanism.

12.2 An open realization

For a single unit with basis $\{|I\rangle, |A\rangle\}$, take $L = \sqrt{\Gamma}\,|A\rangle\langle I|$ and $\dot{\varrho} = L\varrho L^{\dagger} - \frac{1}{2}\{L^{\dagger}L, \varrho\}$. This is a completely positive, trace-preserving semigroup, and its matrix elements give $\dot{\varrho}_{II} = -\Gamma\varrho_{II}$, $\dot{\varrho}_{AA} = \Gamma\varrho_{II}$ and $\dot{\varrho}_{IA} = -\frac{\Gamma}{2}\varrho_{IA}$, so with $f = \varrho_{AA}$,

I — Identity

$$\dot{f} = \Gamma(1 - f). \tag{12.1}$$

The absorbing state is stationary for the reduced dynamics. This is not an analogy: Eq. (12.1) is the $N = 1$ case of Eq. (3.5), obtained exactly from a specified open quantum model.

Reinforcement is obtained without making quantum mechanics nonlinear. With $n_i = |A_i\rangle\langle A_i|$ and $\hat{f}_{\neg i} = (N-1)^{-1} \sum_{j \neq i} n_j$, take

A — Postulate

$$L_i = |A_i\rangle\langle I_i| \otimes \sqrt{\Gamma_0 e^{\alpha \hat{f}_{\neg i}}}. \tag{12.2}$$

The generator remains linear in ϱ. Since $\hat{f}_{\neg i}$ commutes with the operators of unit i,

I — Identity

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle n_i \rangle = \left\langle (1 - n_i)\, \Gamma_0 e^{\alpha \hat{f}_{\neg i}} \right\rangle, \tag{12.3}$$

which for diagonal states is a birth process whose rates depend on other occupations. The macroscopic nonlinearity of Eq. (3.4) is a mean-field closure of Eq. (12.3), not a modification of the underlying evolution.

Proposition 12.2 (Exact classical reduction)**.** *With the jump operators Eq. (12.2), which map each occupation basis state to a single basis state and have diagonal $L_i^\dagger L_i$, populations decouple from coherences. The populations obey exactly the birth chain with rates $(N-k)\Gamma_0 e^{\alpha k/(N-1)}$, and coherences decay at rate at least $\Gamma/2$.*

A consequence worth checking numerically, and which we do: the trajectory of $\langle n_i \rangle$ is identical for a diagonal initial state and for a coherent product state, to solver tolerance. The quantum structure supplies complete positivity and a decoherence rate; it does not supply new kinetics.

12.3 How good is the mean-field closure?

Replacing $\left\langle (1 - n_i) e^{\alpha \hat{f}_{\neg i}} \right\rangle$ by $(1 - f) e^{\alpha f}$ discards correlations. Two computations bound the cost. Integrating the full Liouvillian, which is tractable only to $N = 8$, the maximum discrepancy between the exact $\left\langle \hat{f} \right\rangle$ and the mean-field solution falls with a fitted exponent -1.190 ± 0.027. That number should not be quoted on its own: over so short a range it is still contaminated by small-N curvature. Because the jump operators preserve permutation symmetry, the same quantity can be computed exactly from the $(N+1)$-state symmetric chain, which reaches $N = 512$; there the fitted exponent is -1.011 ± 0.003 and the local slope at the largest N is -1.001 (Fig. 8). The closure is therefore $\mathcal{O}(N^{-1})$, and its error is subleading to the $\mathcal{O}\left(N^{-1/2}\right)$ fluctuations of Theorem 3.4.

12.4 Absorption does not imply increasing entropy

A tempting shortcut is to identify absorption with entropy production. It is false for the subsystem. Starting from $|I\rangle$, the solution of the two-state model is diagonal with $p_I = e^{-\Gamma t}$, so the von Neumann entropy $-p_I \log p_I - (1 - p_I) \log(1 - p_I)$ starts at zero, rises to $\log 2$ and returns to zero when absorption completes. The second law applies to the system together with its environment, and if each jump releases energy ΔE to a bath then that bath cannot be omitted while total energy conservation is being claimed.

This is the honest statement of where the cosmological model stands. Taking $Q_V \geq 0$ is a postulate of Assumption 4.1; exhibiting a microdynamics that conserves total energy, produces the right entropy, and feeds precisely that tensor remains open. Sec. 8.5 proposes a candidate and Sec. 13 lists what would have to be shown.

What this section establishes. That irreversible conversion is incompatible with closed unitary evolution, and that an explicitly specified open model reproduces the classical chain with a mean-field closure error measured to scale as $\mathcal{O}(N^{-1})$ — measured on a numerical family, not

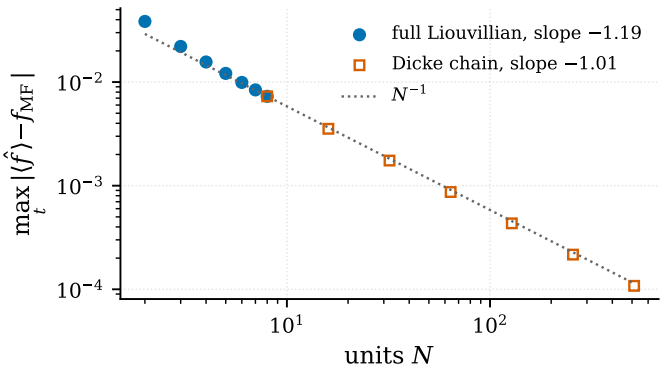


Figure 8. Mean-field closure error of the collective Lindblad dynamics: the maximum over time of the difference between the exact $\left\langle \hat{f} \right\rangle$ and the mean-field solution. Filled circles are the full Liouvillian, which is tractable only to $N = 8$ and whose fitted slope is still bent by small-N curvature; open squares are the exact permutation-symmetric chain, which reaches $N = 512$ and tracks the dotted N^{-1} reference. The closure error is subleading to the intrinsic $N^{-1/2}$ fluctuations of the jump process.

bounded by a proof. What it does not establish is that the substrate is such a system, nor that the environment required by the construction is the one the cosmological model needs.

13 Outlook: open directions

Almost everything in this section is conjecture, and the exception should be flagged rather than discovered. Sec. 13.1 to Sec. 13.7 are open directions: each is stated with what would have to be shown for it to become a result, every displayed result in them carries the conjecture level, and the framework of the preceding sections stands or falls independently of all of it. Sec. 13.8 is different in kind. It proves nothing new — it assembles results proved in Sec. 9, Sec. 11 and Sec. 4 and shows they have two roots between them — but it is an argument rather than a proposal, and the abstract and Sec. 14 rest on it. It sits here because what it identifies is what a successor would have to change, which is the business of this section, and not because it is speculative.

13.1 Where the rate law would come from

The reinforced rate Eq. (3.4) and the activation law Eq. (6.4) are effective descriptions with five parameters. Sec. 8.5 proposed that a nucleation event in a four-form sector is an absorption event. Part of that proposal is now a calculation rather than a conjecture, and it is worth being exact about which part.

The reinforcement is derived: Eqs. (8.8) and (8.9) give $\log\Gamma_\uparrow$ and fix $\alpha = 3(N\mathsf{q}/h)\Delta S$, with the range of validity of the exponential law and its truncation error, under the thin-wall and detailed-balance hypotheses stated there. What those hypotheses do not cover is the one the mechanism needs most: detailed balance is an equilibrium relation, and the nucleation here is assumed to be driven by the precursor rather than thermal. Redoing the calculation for a driven process is the open problem, and it is the same computation that would supply the prefactor and convert a rate per four-volume into a rate per correlated region. That last conversion would also decide whether the memory time τ_M is the light-crossing time of that region, which remains the only physical candidate we can point to.

What is not derived, and is now known not to follow from this bridge, is the identification of b with a fraction ending in localized objects. By Remark 8.5 the membrane leaves no stable remnant in this action, and the object of Sec. 11 is built from different fields carrying a different charge. The conjecture that survives is therefore narrower:

S — Conjecture

$$\text{the driven rate retains} \quad \frac{\mathrm{d}\log\Gamma}{\mathrm{d}h} > 0 \quad \text{with} \quad b = \frac{E_{\mathrm{wall}}}{\epsilon} \quad \text{a fraction, not a defect abundance.} \tag{13.1}$$

13.2 A local rate law off the background

Assumption 9.2 defines the inhomogeneous theory by requiring the transfer to be uniform on comoving hypersurfaces, which makes the kinetic variables diagnostic at linear order. Reading the rate law of Eq. (3.4) instead as a genuinely local law overdetermines the perturbations by one condition, Eq. (9.4), which the dust dynamics do not satisfy. Two routes out are available and neither has been followed here: allow momentum transfer, so that the donor is no longer geodesic and the extra condition is absorbed by its acceleration; or give the receiving sector its own dynamics, so that it is no longer an exact vacuum. Both change the linear theory of Sec. 9, and the second changes Assumption 4.1. Until one is carried out, the local reading of the rate law is a statement about the homogeneous mode, and the hysteresis of Theorem 3.8 is a property of that mode rather than of two regions side by side in one universe.

13.3 Caustics: what is now settled, and what is not

The clock congruence is pressureless and geodesic, so it forms caustics [35], and no linear analysis speaks to their fate. An earlier version of this paper conjectured here that because the transfer

is proportional to the precursor density, and therefore diverges at a shell crossing, the precursor would stay finite. That conjecture has since been examined and is false: conversion enters the local solution only through the integral of the rate, and the rate is bounded. Sec. 11.5 states what replaces it, and carries out the one-dimensional test proposed here.

What remains open is the self-consistent problem. The analysis of Sec. 11.5 prescribes the compression rather than solving for it, so it establishes what a regular configuration would have to satisfy and not that one exists as a solution of the coupled system. Settling that requires either a self-gravitating sheet calculation with the transfer included, or an existence argument for the self-similar profile; neither is attempted here.

13.4 Spin and the Standard Model

The topological charge of Sec. 11 is not a spin operator. Since $\pi_4(S^3) = \mathbb{Z}_2$, a Finkelstein–Rubinstein quantisation in which the soliton is a fermion is permitted,

S — Conjecture

$$\pi_4(S^3) = \mathbb{Z}_2 \quad \Longrightarrow \quad \text{the } B = 1 \text{ sector admits a fermionic quantisation,} \tag{13.2}$$

but establishing it requires showing that the 2π rotation loop is non-contractible in configuration space and choosing a representation of the fundamental group. Even granting that, a $\mathbb{Z}_2$ label is not a Standard-Model fermion: reproducing one would need a covariant spinor sector and a mass matrix with mixing. Nothing in this paper constrains either. The same applies to neutrinos: weak electromagnetic coupling is not evidence of proximity to an absorbing state, and their free-streaming affects structure growth in ways this model does not address [50].

13.5 Electromagnetism

A phase alone is not a gauge field. Writing $A_\mu = q_*^{-1}\partial_\mu\theta$ gives $F_{\mu\nu} = 0$ identically, so holonomies and phase singularities, while real, do not by themselves produce the two propagating polarizations of Maxwell. Nor is quantised flux the same as electric charge: a vortex can carry magnetic flux with no net charge, and winding does not fix the value of e. A condensate that breaks the relevant $U(1)$ gives the photon a mass $m_A^2 = Z_\Psi q_*^2 v^2/Z_A$, so a viable realization must preserve the observed $U(1)$ or exhibit a different massless combination. Finally, the degree-of-freedom count is an obstruction, not a detail: Einstein gravity about Minkowski space has two helicity-±2 modes and Maxwell adds two of helicity ±1, and no invertible local change of variables in the gravitational sector alone can create them.

13.6 Averaged stresses and turbulence

There is a structural identity worth recording, provided it is not oversold. When fast oscillations of zero mean ride on a slow background, the slow sector inherits a stress quadratic in the oscillations: in hydrodynamics this is the Reynolds stress $\langle w_i w_j\rangle$, and in general relativity it is the Isaacson effective tensor $\left\langle \partial_\mu h_{\alpha\beta}\partial_\nu h^{\alpha\beta}\right\rangle$. The two have the same form, and in both cases the mean sector acquires something no individual mode carries. What follows from this is only that the inherited tensor's anisotropy depends on the background, so it need not be an isotropic bath. What does *not* follow is a route from oscillations to negative pressure, and this can be made quantitative rather than left as a caveat. For transverse oscillations $w = a\cos(\xi\cdot x + \varphi)$ with $a\cdot\xi = 0$, the angular average gives $\langle w_i w_j\rangle = \frac{1}{2}a_i a_j$, hence

$$p_{\mathrm{R}} = \tfrac{1}{3}\operatorname{Tr}\langle w\otimes w\rangle = \tfrac{1}{6}\,|a|^2\,, \qquad \varepsilon_{\mathrm{kin}} = \tfrac{1}{2}\left\langle |w|^2\right\rangle = \tfrac{1}{4}\,|a|^2\,, \qquad w_{\mathrm{eff}} = \frac{p_{\mathrm{R}}}{\varepsilon_{\mathrm{kin}}} = \frac{2}{3}, \tag{13.3}$$

exactly. Both statements are in fact unconditional, and the qualification we first attached to them — that the modes have vanishing cross terms — is not needed for either.

The value is an identity. Writing $R_{ij} = \langle \rho\, \delta v_i \delta v_j \rangle$ for a density-weighted average, $p_{\mathrm{R}} = \operatorname{Tr} R/3$ and $\varepsilon_{\mathrm{kin}} = \frac{1}{2}\left\langle \rho\, |\delta v|^2 \right\rangle = \operatorname{Tr} R/2$ by definition, so $w_{\mathrm{eff}} = 2/3$ for *any* velocity field and any non-negative weight: the two quantities are the same trace with different factors, and transversality never enters. The positivity needs exactly one hypothesis. For any direction n, $n^i R_{ij} n^j = \langle \rho\, (n \cdot \delta v)^2 \rangle \geq 0$ whenever $\rho \geq 0$, so R is positive semidefinite and its trace cannot be negative whatever the correlations, the anisotropy, or the number of scales. Replacing ρ by a signed weight does produce negative eigenvalues, which locates the content of the result: it rests on the non-negativity of the mass density and on nothing else. Measured instead against a total energy density including rest mass one gets $w_{\mathrm{eff}} = \langle v^2 \rangle / 3c^2 \geq 0$, the warm-dark-matter result. Both normalizations are strictly positive. An averaged quadratic stress of this kind therefore cannot approach $p = -\rho$ by any choice of amplitudes, correlations or number of modes; reaching it needs a different object, not a better average.

One distinction is worth making explicit, because the intuition that motivates the attempt comes from a different quantity. The pressure depression that sustains a vortex core is not p_{R}. For incompressible flow the pressure is determined only up to an additive function of time, $p \to p + C(t)$, which changes no motion, so a core depression is a difference within a solution and not an absolute value. What a cosmological reading needs is the full stress tensor, where that freedom is fixed by the coupling to gravity — and the fixed value is the one bounded below by zero above. A different calculation — Isaacson's effective stress tensor for high-frequency gravitational waves, which averages a quadratic in the wave's own curvature rather than a kinetic stress — gives $p = \rho/3$ instead. The two are not in tension; they are not the same computation, and quoting them side by side without saying so invites the reader to think one of them is wrong. Concentration phenomena in the continuum limit are suggestive of where a description of this kind would break down, but a singularity of a continuum equation says only that the solution leaves a regularity class; it does not say what replaces it, and it does not convert a singular region into a stable absorbing state.

13.7 Emergent quantum mechanics

The Madelung and Klein–Gordon identities of Appendix C are exact rewritings of equations that were assumed, not derived. Writing $n = |\psi|^2$ after starting from the Schrödinger equation adopts the quantum structure. A theory of emergent quantum mechanics would have to produce the tensor product of subsystems, superposition, entanglement and the Born rule, and would have to explain why a two-particle state lives on a six-dimensional configuration space rather than on physical space. Likewise, the semiclassical equation $G_{\mu\nu} = M_{\mathrm{Pl}}^{-2} \left\langle \hat{T}_{\mu\nu} \right\rangle_{\mathrm{ren}}$ couples a quantum expectation to a classical metric; it does not quantise the metric.

13.8 The obstructions are one obstruction

This paper accumulates negative results, and they read as a list. They are not a list but two families, and sorting them is more useful than either the list or a claim that they are all one thing. This subsection deals with the first family: three results, proved in three different sections, that are corollaries of a single hypothesis — that the recipient is a *silent exact vacuum*, meaning $T_V^{\mu\nu} = -V g^{\mu\nu}$ with no momentum transfer in the frame u^μ — which is also the hypothesis that makes the model work at all. The derivation below is three lines and claims no depth; what it claims is that three things found separately are one thing, which is what tells a successor what to change.

The second family is not of this kind and is not lifted by anything said here. The obstruction at a caustic (Sec. 11.5) is carried by the localized product, which is pressureless dust compressed by the same expansion as everything else and therefore divergent on a collapsing sheet however it was made, and by the conflict between the branching ratio a relic abundance needs and the one regularity needs. Those turn on the *product*'s stress, not the recipient's, and the modification

they call for — a localized sector with pressure — is stated where they are. Two roots, two modifications, and neither substitutes for the other.

What the hypothesis buys. By Proposition 4.2, conversion between components cannot create pressure: the total is the sum of the components' own. Acceleration therefore has to come from an equation of state, and $p = -V$ supplies it, $\rho + 3p = -2V < 0$. Nothing in the kinetics substitutes for that. The rate, the reinforcement and the memory decide how much vacuum exists by today, not that it accelerates.

What the same hypothesis forbids. The stress $-Vg^{\mu\nu}$ has divergence $-\nabla^\nu V$, a gradient. With no momentum transfer the source $Q_V u^\mu$ must therefore be closed, and three consequences follow in succession, none of them independent choices:

1. $Q_V u^\mu$ closed makes u^μ hypersurface-orthogonal, so $Q_V\, \omega_{\mu\nu} = 0$: no vorticity wherever the transfer is on (Proposition 9.1). Any picture driving the vacuum with vortical structure is excluded at this step, before any dynamics.

2. Locally $Q_V u^\mu = \nabla_\mu \phi$ for some scalar, so Q_V and V are functions of one common clock. A rate depending on the local compression independently of that clock is then inconsistent, which is the dichotomy of Sec. 11.5 and the reason the compression-driven channel has no covariant realization.

3. V a function of the clock gives $\delta V_{\rm com} = 0$ on comoving hypersurfaces. The geodesic realization of Assumption 9.2 is therefore not a choice among four at all: with the donor pressureless it is forced outright (Proposition 9.4), acceleration included.

The consequence for a successor. What all three obstructions require is the same single capacity: the recipient must be able to exert a transverse force. Every step above turns on the source being purely longitudinal, and nothing else about the recipient is used. That is a more precise demand than "give the vacuum dynamics", and it admits a graded set of answers rather than one.

The cheapest is already in this paper. Giving the receiver a traceless spatial anisotropic stress alongside the vacuum (Sec. 11.5) supplies the transverse force while leaving the equation of state at $p = -V$, and it vanishes identically in FLRW and at linear order under Assumption 9.2, so nothing fitted in Sec. 10 moves. Its cost is a counted deficit rather than a vague one: five independent components against three conservation equations, so a closure must supply two further relations per point, and none is on offer here.

The expensive answer is a recipient with its own pressure and rest frame. It lifts all three obstructions with no ambiguity, and the price is that an exact vacuum is silent and a recipient with pressure is not: it has a sound speed, it perturbs, it has a frame, and Sec. 9 would have to be built again rather than inherited — Proposition 9.4 is what breaks first, since its hypothesis is exactly what fails. That is a different theory, not an extension of this one.

The ordering is the useful output. A successor is not obliged to pay the full price, and the intermediate rung is a well-posed problem — find a closure for $\Pi^{\mu\nu}$ — rather than a programme. This paper records the obstructions instead of routing around them because neither rung can be climbed without leaving the model it is about.

It is worth being clear that this is not an argument that such a theory fails. It is a statement of what it must contain, derived from what the present one cannot do.

13.9 What would falsify the programme

Being clear about this matters more than any of the above. A specified version of this model should be abandoned if it violates its own positivity conditions, if it cannot maintain a conserved

total tensor, if it develops an instability in the regime where it is being used, or if it cannot describe the data with its parameters fixed in advance. Three sharp tests already exist: the sign prediction of Theorem 5.4, the jerk bound of Eq. (7.2), and the opposite-sign growth signature of Eq. (9.10). Having many parameters, or requiring tuning, is not by itself a contradiction or a falsification; but it does weaken explanatory power, and it obliges the statistical comparison of Sec. 10 to account for the added freedom.

14 Conclusions

A single kinetic idea has been followed through to its consequences. Regions of a substrate convert irreversibly; the conversion reinforces itself; the reinforcement responds to a lagged memory of what has already converted; and the products are a pressureless component and a growing vacuum. The question was what that implies once written as equations.

At the level of the kinetics, the population-level memory law is exactly the average of a per-unit relaxation law, so it is equivalent to a local assumption rather than an extra non-local one, and the pair (f, M) is Markov even though f alone is not (Proposition 3.2). The finite-population process has a controlled deterministic limit (Theorem 3.4); the $N^{-1/2}$ law for its fluctuations is exact where the rate does not depend on the state, and is supported by direct simulation where it does. Its distinguishing property is path dependence: along a *prescribed* expansion history, two histories reaching the same converted fraction with different memory states convert at different epochs and therefore accumulate different vacuum energies (Theorem 3.8). That hypothesis is not cosmetic (Remark 3.9): it makes the property one of the kinetics rather than of the coupled system, and Sec. 13.2 says what would be needed to promote it. An interaction written as a function of the densities and the expansion rate, including one whose sign switches with redshift, closes on those densities, so they fix its entire future and it has no such freedom. The difference is the dimension of the state space, not the shape of the coupling.

Coupled to general relativity under an explicit closure, the model has exact identities relating the converted fraction to the surviving densities, global positivity and bounded expansion, and a de Sitter attractor reached without the Hubble rate ever increasing (Theorems 4.4 and 4.5). Positivity holds structurally, because every outflow is proportional to the density of the sector it leaves, which is the property that the recipient-proportional couplings among those catalogued as pathological lack (Theorem 5.1 and Proposition 5.2). The model maps onto a restricted corner of the standard parametrizations and never onto the other family (Proposition 5.3).

Three consequences of the one-way model are falsifiable and cannot be tuned away within it. The effective dark matter can only lose energy, never gain it (Theorem 5.4); adding the reverse channel of Sec. 5.5 removes that prediction, and is a different model, which is why it is carried through the comparison rather than mentioned and dropped. The jerk parameter satisfies $j \geq 1 + 2\Omega_{\mathrm{r}}$, which has the opposite sign to both running vacuum and the currently favoured evolving dark-energy quadrant (Theorems 7.1 and 7.3). And linear growth is enhanced while the velocity field that redshift-space distortions measure is suppressed (Eq. (9.10)). The memory parameters, invisible in the jerk, reappear in the snap, although mixed there with the activation profile (Theorem 7.5).

Confronted with Pantheon+, DESI DR2, cosmic chronometers and the Planck distance priors, the model is not preferred, and we report that as the result it is. ΛCDM wins on the Bayesian evidence against every extension fitted here (Table 2). The convertible fraction ϵ cannot be measured at all — the rate may always vanish, so no value of ϵ is excluded — and the quantity that does carry information, the matter deficit, is bounded by Eq. (10.1). The jerk excess is bounded separately, by profiling it directly: the deficit does not constrain it, the relation between the two being a floor whose hypothesis this model's activation law fails to satisfy (Sec. 7.1). Profiled directly it is bounded too, and by Eq. (10.2) it lies more than an order of magnitude below what late-time cosmography can resolve. That speaks to the jerk alone. The no-gain

result is a statement about the sign of a quantity no present measurement isolates. The growth signature is confronted with redshift-space data, and the confrontation settles nothing: adding the four SDSS DR16 $f\sigma_8$ measurements moves $\ln B$ from -1.62 to -1.46 for the $\Gamma = \xi H$ benchmark, a change of $+0.16$ against an uncertainty of 0.29 on either. The shift is in the direction the model predicts and is not close to significant. Of the three sign predictions above, then, one is measured and out of reach, one is tested and undetected, and one is not isolated by any present measurement. They remain refutable in principle, which is the most that can honestly be claimed for them today.

Along the way two defects of the earlier version of this work were found and repaired rather than quietly dropped. The rate law with a positive floor has no viable early universe (Theorem 6.1), and the minimal repair is an activation switch whose exponent must exceed the donor's dilution rate: $p > 1$ for the constant-rate law fitted here, $p > 3$ for the expansion-tracking benchmark. And the growth equation previously written for the precursor contrast is wrong when both products are present; the correct variable is the total dust contrast, in which the branching ratio cancels identically (Theorem 9.5).

What has *not* been shown should be stated with equal clarity. There is no microscopic derivation of the rate law; Sec. 8.5 proposes a mechanism and Sec. 13 lists the calculation that would settle it. The clock congruence forms caustics, and this is the paper's sharpest limitation rather than a remote one: the congruence is the dark-sector dust congruence, so shell crossing is present throughout the redshift range the data cover, and where it has occurred the clock is not a coordinate and Sec. 9 has no content at all. What licenses the use made of it here is a separation of scales and not the absence of caustics, and no controlled estimate of the resulting error is supplied. The earlier conjecture that defect formation regulates them is settled in the negative for the model as published: a rate bounded on the approach cannot change the exponent of the divergence, only its amplitude, and this model's rate is bounded (Theorem 11.2). A compression-driven rate can change the exponent, but only above a threshold and only with no branching, which contradicts what a relic abundance requires; and every covariant realization we could construct runs into the dichotomy set out in Sec. 11.5, which we leave unresolved. What replaces the conjecture is therefore an obstruction, which is a weaker claim about the model and a stronger statement about what would have to be true for it to work. The localized configuration of Sec. 11 is solved but not quantised, and its identification with dark matter is a conjecture, not a result. Quantum mechanics is used, not derived. Electromagnetism and the Standard Model are untouched, and the degree-of-freedom counting in Sec. 13.5 shows they will not emerge from a change of variables.

Those obstructions are less independent than a list of them suggests, though they are not all one either: they sort into two families with two roots. Three of them — the absence of vorticity where the transfer is on, the rate forced to be a function of the clock rather than of the local compression, and the geodesic realization of the transfer — descend from the single hypothesis that also makes the model work: an exact vacuum, $T_V^{\mu\nu} = -V g^{\mu\nu}$, receiving energy with no momentum transfer. That hypothesis supplies the only source of acceleration available, because conversion between components cannot manufacture pressure (Proposition 4.2); and the same stress, whose divergence is a gradient, is what forbids vorticity wherever the transfer is active, what forces the rate to be a function of the clock rather than of the local compression, and what leaves the geodesic closure of Sec. 9 as the only consistent one rather than a choice among four. What a successor needs in order to want any of the three is one capacity, not a whole sector: the recipient must be able to exert a transverse force. The cheapest way to supply it is already recorded here, a traceless anisotropic stress that leaves $p = -V$ and vanishes wherever the model has been fitted, at the cost of being underdetermined by two functions without a closure; the expensive way is a recipient with its own pressure and rest frame, which would mean building the perturbation sector again rather than inheriting it. The remaining obstructions have the other root and are untouched by all of this: at a caustic the divergence is carried by the localized

product, which is dust, so what they ask for is a localized sector with pressure and not a different recipient (Sec. 13.8). Two roots and two modifications, neither substituting for the other, is a sharper statement than a list of separate failures, and we record it as such.

The claim this paper is prepared to defend is therefore narrow and, we think, worth making: *there exists an effective model of self-reinforced absorption with memory whose cosmological consequences are exact under stated hypotheses, whose positivity is structural rather than tuned, and which makes three sign predictions that current and near-future data can refute.* The identification of its ingredients with the physical universe, and any unification of gravity with quantum theory, remain what they were before this paper: open.

Acknowledgments

This work used public data releases from the Pantheon+ collaboration, the Dark Energy Spectroscopic Instrument, the cosmic-chronometer compilation of Moresco and collaborators, and the Planck collaboration. It made use of NumPy, SciPy, Matplotlib, SymPy, `emcee` and `getdist`.

Use of artificial intelligence tools

The conceptual proposal, the physical hypotheses and the scientific decisions in this work are the author's. Large language models (Anthropic Claude, through the Claude Code interface) were used as writing and computational assistants: drafting and editing English prose, proposing and checking steps in derivations, writing and refactoring the numerical code, and producing internal critical reviews of the manuscript. All analytical results were re-derived and checked by the author; every numerical value quoted in the text and every figure is produced by the archived code from public data whose checksums are recorded in the repository; and every bibliographic reference was resolved against the arXiv, INSPIRE or Crossref record by an automated check whose log is archived with the code. No AI system is an author, and the author takes full responsibility for the content of this paper.

Data and code availability

The analysis code, data checksums, chain summaries and the results file from which every number and table in this paper is generated are archived at `10.5281/zenodo.22726266`. No data are redistributed: a script downloads each public release and verifies it against a recorded checksum.

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A Conventions, dimensions and domain of validity

A.1 Symbols

The draft version of this work reused several symbols for unrelated objects. The table below is the manuscript's single source of truth. Where a clash was inherited it is listed explicitly with its resolution, so that a reader comparing the two versions is not misled.

Symbol	Name	Meaning	Dim.
Geometry and gravity			
$g_{\mu\nu}$	metric	signature $(-+++)$	0
$M_{\rm Pl}$	Planck mass	reduced, $M_{\rm Pl}^2 = (8\pi G)^{-1}$	1
$T_{\mu\nu}$	stress tensor	total; sector tensors carry a label	4
u^μ	four-velocity	of the precursor/clock congruence	0
T	clock field	constrained scalar of Eq. (8.3)	-1
λ	multiplier	enforces the clock constraint	4
$\mathcal{H}$	four-form	field strength of the three-form potential	2
Densities			
ρ_I	precursor	pressureless, converted away	4
ρ_D	defects	pressureless, produced	4
V	vacuum	$p_V = -V$ exactly	4
$\rho_{\rm m}$	total dust	$\rho_I + \rho_D + \rho_{\rm b}$	4
$\rho_{\rm dm}^{\rm eff}$	effective dark matter	$\rho_I + \rho_D$	4
ω_X	reduced density	$\rho_X/(3M_{\rm Pl}^2 H_{100}^2)$	0
Kinetics			
f	converted fraction	$\dot f = (1-f)\Gamma$	0
M	memory	$\tau_M \dot M = f - M$	0
Γ	rate	$\Gamma_0 S_{\rm act}(a) e^{\alpha f + \beta M}$	1
b	branching ratio	share of converted energy to defects	0
ϵ	convertible fraction	$(1-b)\rho_{I,{\rm act}}/\omega_c$	0
ε	jerk excess	$Q_V/(2M_{\rm Pl}^2 H^3)$; *not* ϵ	0
E_p	precursor energy quantum	per patch, in Sec. 8.5 only	5
$\epsilon_{\mu\nu\rho\sigma}$	Levi-Civita tensor	Sec. 8.1 only	0
Q_V	transfer	$(1-b)\Gamma\rho_I$	5
Δ	matter deficit	Eq. (5.3)	0
$a_{\rm act}$, p	activation	scale and sharpness of $S_{\rm act}$	0
$p_{\rm eff}$	effective slope	of q in Sec. 7.1	0
p_g	effective slope	of g_c; see Eq. (9.9). Not $p_{\rm eff}$: the two differ by 3 in matter domination	0
$p_{\rm R}$	effective pressure	of the Reynolds stress in Sec. 13.6	—
ξ_0	tracking amplitude	of the rate $\Gamma = \xi_0 H S_{\rm act} e^{\alpha f + \beta M}$	0
$\mathcal{B}$	memory lag	$\beta(f-M)/(\tau_M H)$	0
Expansion			
H, E	expansion rate	$E = H/H_{100}$, $E(a{=}1) = h$	1, 0
N	time variable	$\ln a$, used throughout	0
q, j, s	cosmography	deceleration, jerk, snap	0

Clashes resolved. The conversion rate keeps Γ, and Christoffel symbols are written with their indices, $\Gamma^\rho{}_{\mu\nu}$, appearing only in Sec. 2. The memory variable keeps M and the Planck mass is always written $M_{\rm Pl}$. The branching ratio keeps b while baryons carry a roman subscript. The clock keeps T while the stress tensor always carries indices. Where the draft used A for three different objects, $\ddot a/a$ is written out, the four-acceleration is A^μ and the gauge field is A_μ.

A.2 The claims ladder in full

Every displayed key result in this paper is wrapped in an environment carrying its evidential level, which is printed with the result, and the table below is extracted from the source by the build rather than maintained by hand. A build check refuses any boxed result that is unclassified,

and any conjecture appearing outside Sec. 13. The check is a check of bookkeeping and not of physics: it verifies that a level is present and, for conjectures, that it appears in the right section. Whether a result deserves the level it carries is our judgement, and the numbered theorems and propositions that are not wrapped in this environment carry their hypotheses in their statements instead.

Result	Level	Meaning
Eq. (2.4)	I	Identity or standard result
Eq. (2.5)	I	Identity or standard result
Eq. (2.6)	I	Identity or standard result
Eq. (2.8)	I	Identity or standard result
Eq. (2.9)	I	Identity or standard result
Eq. (3.1)	A	Postulate of the model
Eq. (3.2)	A	Postulate of the model
Eq. (3.3)	I	Identity or standard result
Eq. (3.4)	A	Postulate of the model
Eq. (3.5)	C	Theorem of the model
Eq. (3.6)	C	Theorem of the model
Eq. (3.7)	C	Theorem of the model
Eq. (3.8)	I	Identity or standard result
Eq. (3.9)	I	Identity or standard result
Eq. (4.1)	A	Postulate of the model
Eq. (4.2)	C	Theorem of the model
Eq. (4.3)	I	Identity or standard result
Eq. (4.4)	I	Identity or standard result
Eq. (4.5)	C	Theorem of the model
Eq. (4.7)	C	Theorem of the model
Eq. (4.8)	I	Identity or standard result
Eq. (5.1)	I	Identity or standard result
Eq. (5.2)	C	Theorem of the model
Eq. (5.3)	I	Identity or standard result
Eq. (5.4)	C	Theorem of the model
Eq. (6.1)	I	Identity or standard result
Eq. (6.3)	C	Theorem of the model
Eq. (6.4)	A	Postulate of the model
Eq. (6.5)	I	Identity or standard result
Eq. (6.6)	C	Theorem of the model
Eq. (7.1)	I	Identity or standard result
Eq. (7.2)	C	Theorem of the model
Eq. (7.3)	C	Theorem of the model
Eq. (7.4)	C	Theorem of the model
Eq. (7.5)	C	Theorem of the model
Eq. (7.6)	C	Theorem of the model
Eq. (8.2)	I	Identity or standard result
Eq. (8.3)	A	Postulate of the model
Eq. (8.4)	I	Identity or standard result
Eq. (8.5)	I	Identity or standard result
Eq. (8.6)	X	Result of an auxiliary model
Eq. (8.8)	X	Result of an auxiliary model
Eq. (8.9)	X	Result of an auxiliary model
Eq. (9.1)	I	Identity or standard result
Eq. (9.2)	I	Identity or standard result
Eq. (9.3)	C	Theorem of the model
Eq. (9.4)	I	Identity or standard result
Eq. (9.5)	C	Theorem of the model
Eq. (9.6)	C	Theorem of the model
Eq. (9.7)	C	Theorem of the model
Eq. (9.8)	C	Theorem of the model
Eq. (9.9)	C	Theorem of the model
Eq. (9.10)	C	Theorem of the model
Eq. (9.12)	C	Theorem of the model

continued on the next page

Result	Level	Meaning
Eq. (10.1)	P	Phenomenological (fitted to data)
Eq. (10.2)	P	Phenomenological (fitted to data)
Eq. (11.1)	A	Postulate of the model
Eq. (11.3)	I	Identity or standard result
Eq. (11.4)	I	Identity or standard result
Eq. (11.5)	I	Identity or standard result
Eq. (11.6)	N	Numerical solution of the stated equations
Eq. (11.8)	I	Identity or standard result
Eq. (11.9)	C	Theorem of the model
Eq. (12.1)	I	Identity or standard result
Eq. (12.2)	A	Postulate of the model
Eq. (12.3)	I	Identity or standard result
Eq. (13.1)	S	Conjecture
Eq. (13.2)	S	Conjecture
Eq. (B.2)	I	Identity or standard result
Eq. (B.3)	I	Identity or standard result
Eq. (C.1)	I	Identity or standard result
Eq. (C.2)	I	Identity or standard result
Eq. (D.2)	I	Identity or standard result
Eq. (D.4)	I	Identity or standard result

A.3 What each result requires

Results in this paper hold on different domains, and they do not transfer automatically between them.

- The kinetic results of Sec. 3 concern a finite population and its deterministic limit. They assume a bounded, Lipschitz rate and no collective jumps; they say nothing about a spatially extended process.
- The cosmological theorems of Secs. 4 and 5 assume exact homogeneity and isotropy, and hold to the future of activation. They are statements about ordinary differential equations, not about the Einstein equations with inhomogeneities, caustics or collapse.
- The perturbative results of Sec. 9 are linear and assume the geodesic realization. They do not address nonlinear structure formation, and the compressed CMB likelihood used in Sec. 10 is a geometric summary rather than a Boltzmann calculation.
- The defect results of Sec. 11 are classical and static, in flat space, within the hedgehog ansatz.
- The quantum results of Sec. 12 concern a finite-dimensional open system with a specified generator.

A rate is a probability per unit time, not a probability: the chance of a conversion in an interval $\mathrm{d}t$ is $\Gamma\,\mathrm{d}t + o(\mathrm{d}t)$. A nucleation rate per unit four-volume must be multiplied by a volume before it can be compared with Γ. Finally, the branch $f_i = 1$ is degenerate and is excluded throughout by taking $f_i < 1$, so that the normalised identities apply.

B Deferred proofs

B.1 Moments of the jump process

For a test function $g(f, M)$ the extended generator of the process of Eq. (3.1) together with Eq. (3.2) is

$$\mathcal{L}_N g = N(1-f)\Gamma(f,M)\left[g(f+N^{-1},M) - g(f,M)\right] + \frac{f-M}{\tau_M}\,\partial_M g. \tag{B.1}$$

Taking $g = f$ and $g = f^2$ gives the first two infinitesimal moments,

I — Identity

$$\frac{\mathbb{E}[\mathrm{d}f_N \mid \mathfrak{F}_t]}{\mathrm{d}t} = (1 - f_N)\Gamma, \qquad \frac{\mathbb{E}[(\mathrm{d}f_N)^2 \mid \mathfrak{F}_t]}{\mathrm{d}t} = \frac{(1 - f_N)\Gamma}{N}. \tag{B.2}$$

Consequently $\mathrm{d}\mathbb{E}[f_N]/\mathrm{d}t = \mathbb{E}[(1 - f_N)\Gamma(f_N, M)]$, which is *not* $(1 - \mathbb{E}[f_N])\Gamma(\mathbb{E}[f_N], \mathbb{E}[M])$: the average of a nonlinear function is not the function of the average, and the difference is what Theorem 3.4 controls.

Two consequences constrain any diffusion approximation. In the interior the noise amplitude implied by Eq. (B.2) is $\sqrt{(1 - f)\Gamma/N}$, which is not the same as an amplitude chosen for convenience such as $\sqrt{f}(1 - f)$. And near $f = 0$ a Gaussian approximation admits negative excursions that the jump process forbids, so the boundaries must either be simulated exactly or justified by a separate argument. We simulate them exactly.

B.2 Absorption time of the finite chain

For a memoryless rate depending on k/N with $\Gamma \geq g > 0$, the mean time to reach $K = N$ from k_0 is

I — Identity

$$\mathbb{E}[T_N] = \sum_{k=k_0}^{N-1} \frac{1}{(N - k)\Gamma(k/N)} \leq \frac{1}{g} \sum_{j=1}^{N-k_0} \frac{1}{j}, \tag{B.3}$$

since the holding time in state k is exponential with mean $1/\lambda_k$ and each state is visited once. The restriction to a memoryless rate is material: with memory the chain is not a pure birth process in k alone, and the sum above does not apply.

The finite chain therefore reaches its absorbing state in finite time almost surely, with a mean growing like $\log N$, while the deterministic equation approaches $f = 1$ only asymptotically. The two statements are compatible, and Eq. (3.8) shows why: the continuum solution has $1 - f > 0$ at every finite time whenever the integrated rate is finite.

B.3 Positivity, in the weak form used

Theorem 5.1 was proved by the subtangent criterion, which requires only that the vector field be defined almost everywhere. Since the rate enters every equation multiplied by a density, the integral form

$$\rho_I(t) = \rho_I(t_i) \exp\left[-\int_{t_i}^{t} (3H + \Gamma)\,\mathrm{d}s\right] \tag{B.4}$$

holds for merely locally integrable Γ, and the remaining sectors follow by the integrating-factor representation with non-negative sources. This is why the theorem survives the replacement of the rate floor by the activation law of Assumption 6.2, which makes Γ vanish in the past.

B.4 The vacuum bound

The upper bound in Eq. (4.5) follows from $a \geq 1$ together with Eq. (4.3):

$$\begin{aligned} V(t) - V_i &= (1 - b) \int_{t_i}^{t} \Gamma(s)\rho_I(s)\,\mathrm{d}s \\ &\leq (1 - b)\rho_{I,i} \int_{t_i}^{t} \Gamma(s) e^{-\int_{t_i}^{s} \Gamma(r)\mathrm{d}r}\,\mathrm{d}s = (1 - b)\rho_{I,i}\left[1 - e^{-\int_{t_i}^{t} \Gamma\,\mathrm{d}s}\right]. \end{aligned} \tag{B.5}$$

The total density cannot vanish in finite time either: from $\dot{\rho_{\mathrm{tot}}} \geq -4H_i\rho_{\mathrm{tot}}$ one gets $\rho_{\mathrm{tot}}(t) \geq \rho_{\mathrm{tot}}(t_i)e^{-4H_i(t-t_i)} > 0$, so the positive Friedmann root stays away from zero and local existence extends to the whole future.

C Quantum-hydrodynamic identities

The identities below are exact rewritings of equations that are *assumed*, not derived. They are collected here because they are used in Secs. 12 and 13 and because the distinction between a rewriting and a derivation is easy to lose.

C.1 Madelung form

Starting from the Schrödinger equation $i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + W\psi$ and writing $\psi = \sqrt{n}\, e^{iS/\hbar}$ on a domain where $\psi \neq 0$, the imaginary and real parts separate into

I — Identity

$$\partial_t n + \nabla\cdot\left(n\frac{\nabla S}{m}\right) = 0, \qquad \partial_t S + \frac{|\nabla S|^2}{2m} + W + U_Q = 0, \qquad U_Q = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{n}}{\sqrt{n}}. \tag{C.1}$$

With $\mathbf{v} = \nabla S/m$ and local irrotationality this gives $m(\partial_t + \mathbf{v}\cdot\nabla)\mathbf{v} = -\nabla(W + U_Q)$. There is no viscous term, and the underlying evolution conserves the norm and is reversible for a Hermitian Hamiltonian: a hydrodynamic *form* is not a dissipative dynamics.

The same equations follow from the action $S_{\mathrm{Mad}} = \int \mathrm{d}t\, \mathrm{d}^3x\, [-n(\partial_t S + |\nabla S|^2/2m + W) - \frac{\hbar^2}{8m}|\nabla n|^2/n]$, since $\delta E_Q/\delta n = U_Q$ for $E_Q = \frac{\hbar^2}{8m}\int |\nabla n|^2/n$. This exhibits the gradient energy any microscopic theory would have to reproduce, and makes plain that symmetry alone does not fix the coefficient $\hbar^2/8m$.

C.2 Circulation is extra information

Single-valuedness of ψ requires, around a closed loop not crossing a node,

I — Identity

$$\oint \nabla S\cdot \mathrm{d}\boldsymbol{\ell} = 2\pi\hbar\, n_{\mathrm{w}}, \qquad n_{\mathrm{w}} \in \mathbb{Z}. \tag{C.2}$$

The local equations do not imply this. On a two-dimensional annulus, $n = n_0$, $S = \ell\theta - Et$ and $W(r) = E - \ell^2/(2mr^2)$ satisfy Eq. (C.1) with $U_Q = 0$ for *any* real ℓ, while single-valuedness forces $\ell/\hbar \in \mathbb{Z}$. Any programme deriving quantum mechanics from a fluid must therefore supply this topological input separately. On a time-dependent spatial metric, conservation of the norm must also be written with the measure $\sqrt{h}\,\mathrm{d}^3x$; replacing the Laplacian alone is not enough.

C.3 Klein–Gordon and its non-relativistic limit

For $\Psi = Ae^{iS/\hbar}$ with $A > 0$ the same separation of $(\Box - m^2c^2/\hbar^2)\Psi = 0$ gives $\nabla_\mu(A^2\nabla^\mu S) = 0$ and $\nabla_\mu S\nabla^\mu S + m^2c^2 - \hbar^2\Box A/A = 0$. The conserved Klein–Gordon current is not a positive probability density for every solution, so the relativistic interpretation needs quantum field theory. Writing $\Psi = e^{-imc^2t/\hbar}\psi$ in Minkowski space gives $2im\hbar^{-1}\partial_t\psi + \nabla^2\psi - c^{-2}\partial_t^2\psi = 0$, which reduces to the free Schrödinger equation when the second time derivative is negligible. That is a low-energy limit of a relativistic *quantum* equation, not a derivation of quantisation from relativity.

C.4 Why this does not amount to emergence

A two-particle state $\psi(\mathbf{x}_1, \mathbf{x}_2)$ lives on a six-dimensional configuration space, which a local scalar fluid on physical space does not represent. Identifying $n = |\psi|^2$ after starting from the Schrödinger equation adopts the quantum structure rather than deducing it. The same applies to the semiclassical equation $G_{\mu\nu} = M_{\mathrm{Pl}}^{-2}\left\langle \hat{T}_{\mu\nu}\right\rangle_{\mathrm{ren}}$: it couples an expectation value to a classical metric and does not quantise it.

D Relativistic hydrodynamics and causality

The transport statements used in passing in the main text are collected here, with the signs fixed, because a sign error in the shear term reverses the physics.

D.1 Projections and constitutive relations

With $\Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$, $D = u^\mu \nabla_\mu$ and $\Theta = \nabla_\mu u^\mu$, conservation of a perfect fluid projects into $D\rho + (\rho + p)\Theta = 0$ and $(\rho + p)Du^\alpha + \Delta^{\alpha\mu}\nabla_\mu p = 0$. Einstein's equations supply conservation of the *total* tensor; a particular fluid equation requires in addition a decomposition of matter and constitutive relations. Viscosity is not determined by $G_{\mu\nu}$.

In the Landau frame, with $\pi^{\mu\nu}$ spatial, symmetric and traceless and

$$\sigma_{\mu\nu} = \Delta_\mu{}^\alpha \Delta_\nu{}^\beta \left(\nabla_{(\alpha} u_{\beta)} - \tfrac{1}{3} \Delta_{\alpha\beta} \Theta \right), \tag{D.1}$$

the first-order constitutive relations are

I — Identity

$$\Pi = -\zeta\Theta, \qquad \pi^{\mu\nu} = -2\eta\sigma^{\mu\nu}, \qquad \eta, \zeta \geq 0. \tag{D.2}$$

The minus signs are not conventional: with the opposite sign a shear flow would sharpen gradients rather than diffuse them. The check is entropy production. For a fluid without conserved charges in local equilibrium, $\rho + p = \mathcal{T} s$ and $\mathrm{d}\rho = \mathcal{T}\,\mathrm{d}s$, the energy projection gives

$$\mathcal{T}\nabla_\mu(s u^\mu) = -\Pi\Theta - \pi^{\mu\nu}\sigma_{\mu\nu} = \zeta\Theta^2 + 2\eta\sigma^{\mu\nu}\sigma_{\mu\nu} \ \geq 0, \tag{D.3}$$

which fixes the signs and confirms positivity of dissipation.

D.2 Non-relativistic limit and dimensionless groups

Restoring c, taking small velocities and $p \ll \rho_\mathrm{m} c^2$, the conservation laws reduce to the compressible Navier–Stokes system, and for constant density and divergence-free flow to $\partial_t \mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{v} = -\nabla p/\rho_\mathrm{m} + \nu\nabla^2\mathbf{v} + \mathbf{f}/\rho_\mathrm{m}$ with $\nu = \eta/\rho_\mathrm{m}$. Non-dimensionalising gives the Reynolds number $\mathrm{Re} = UL/\nu$; in the relativistic case the appropriate combination uses the enthalpy, $(\rho + p)UL/\eta$.

For the conversion problem two further groups matter: a Damköhler number $\mathrm{Da} = \Gamma L/U$ comparing conversion to advection, and a Deborah number $\mathrm{De} = \tau_M U/L$ comparing the memory time to the flow time. A local hydrodynamic description also requires a small Knudsen number. None of these is a new law: a large Reynolds number does not produce quantisation, spin, or dark matter.

D.3 Causality

First-order theories in the Eckart or Landau frames are acausal, but it is not true that every first-order relativistic hydrodynamics is: with an appropriate frame choice, causal and hyperbolic first-order formulations exist [51]. In a Israel–Stewart-type closure with relaxation times, a transverse perturbation obeys

I — Identity

$$\tau_\pi \partial_t^2 v_y + \partial_t v_y - \frac{\eta}{h}\partial_x^2 v_y = 0, \tag{D.4}$$

whose principal speed is $v_T^2 = \eta/(h\tau_\pi)$, so $v_T \leq 1$ requires $\tau_\pi \geq \eta/h$; as $\tau_\pi \to 0$ one recovers a diffusion equation with its instantaneous propagator. This is a statement about one channel, not a proof of causality for the full system.

The same caution applies to spatially extended versions of the conversion law. A Cattaneo closure, $\partial_t f + \nabla\cdot\mathbf{J} = F(f)$ with $\tau_f \partial_t \mathbf{J} + \mathbf{J} = -D_f \nabla f$, gives a finite propagation speed $\sqrt{D_f/\tau_f}$, but a hyperbolic equation of this kind does not inherit the maximum principle of its parabolic

counterpart. We have not shown that such an extension preserves $0 \le f \le 1$ for all data. The interval theorems of Secs. 3 and 4 refer to the finite chain and to the homogeneous system, not to an arbitrary spatial extension.

E Skyrme integrals

E.1 Radial integrals

For $k > 3/2$, substituting $u = r^2/R_s^2$ in $I_k(R_s) = \int_0^\infty r^2\,\mathrm{d}r/(r^2+R_s^2)^k$ gives

$$I_k(R_s) = \frac{R_s^{3-2k}}{2}\,\frac{\Gamma_{\mathrm{E}}(3/2)\,\Gamma_{\mathrm{E}}(k-3/2)}{\Gamma_{\mathrm{E}}(k)}, \tag{E.1}$$

where Γ_{E} is Euler's gamma function, not the conversion rate. With $\Gamma_{\mathrm{E}}(1/2) = \sqrt{\pi}$ and the recurrence this yields $I_2 = \pi/(4R_s)$, $I_3 = \pi/(16R_s^3)$ and $I_4 = \pi/(32R_s^5)$, which are the three values used in Sec. 11.

E.2 Induced metric of the trial family

With $D_r = r^2 + R_s^2$, differentiating the trial configuration gives $\partial_i n^a = 2R_s\delta_{ia}/D_r - 4R_s x^a x_i/D_r^2$ and $\partial_i n^4 = 4R_s^2 x_i/D_r^2$. Summing over A, the terms proportional to $x_i x_j$ cancel and

$$G_{ij} \coloneqq \partial_i n^A \partial_j n^A = \frac{4R_s^2}{D_r^2}\,\delta_{ij}. \tag{E.2}$$

Since $\sum_{ijAB}(\mathcal{W}_{ij}^{AB})^2 = 2[(\mathrm{Tr}\,G)^2 - \mathrm{Tr}(G^2)]$, an isotropic $G = s^2 \mathbb{1}_3$ gives $12s^4$, which fixes the normalisation used in Eq. (11.4). The degree density of the same family is $\mathcal{B}(r) = 4R_s^3/[\pi^2(r^2+R_s^2)^3]$, whose integral against $4\pi r^2\mathrm{d}r$ is unity.

E.3 The bound

Let $\sigma_1, \sigma_2, \sigma_3$ be the singular values of $\mathrm{d}n$ at a point. The two-derivative density is $\frac{1}{2}F_*^2 \sum_i \sigma_i^2$, the four-derivative density is $\kappa_s \sum_{i<j} \sigma_i^2\sigma_j^2$, and the degree density is $\sigma_1\sigma_2\sigma_3/(2\pi^2)$. The arithmetic-geometric mean inequality applied pointwise to these three combinations gives $\frac{1}{2}F_*^2 \sum \sigma_i^2 + \kappa_s \sum_{i<j}\sigma_i^2\sigma_j^2 \ge \sqrt{2}\,F_*\sqrt{\kappa_s}\cdot 3\,\sigma_1\sigma_2\sigma_3$, and integrating yields Eq. (11.3). Equality would require all three singular values equal everywhere, which is not attainable for a configuration with a finite-energy tail; the solved profile exceeds the bound by the factor 1.2314.

F Numerical methods and reproducibility

Every number in this paper that comes out of a fit, a simulation or a solver is produced by the archived code from public data and is not typed by hand: the analysis scripts write results files from which the manuscript's macros and tables are generated, and a build check fails if the two disagree, if a macro is used that no results file defines, or if a decimal literal appears in the prose outside a short whitelist. Round figures used for orientation — an order of magnitude, a quoted error bar from the literature, a fraction written in words — are not covered by that machinery, and are not results of ours.

F.1 Background integration

Working in $N = \ln a$ and in reduced densities $\omega_X = \rho_X/(3M_{\mathrm{Pl}}^2 H_{100}^2)$, so that $E = H/H_{100}$ and $E(a{=}1) = h$, the survival identities Eq. (4.3) eliminate ρ_I and ρ_D analytically. Only the

three-vector (f, M, ω_V) is integrated, with

$$\frac{\mathrm{d}f}{\mathrm{d}N} = (1-f)\frac{\gamma}{E}, \qquad \frac{\mathrm{d}M}{\mathrm{d}N} = \frac{f-M}{\hat{\tau}E}, \qquad \frac{\mathrm{d}\omega_V}{\mathrm{d}N} = \epsilon\,\omega_c\,a^{-3}\frac{\mathrm{d}f}{\mathrm{d}N}, \tag{F.1}$$

and the Friedmann constraint evaluated in closed form at each step, $E^2 = [\omega_\mathrm{b} + \omega_c(1-\epsilon f)]a^{-3} + \omega_\gamma a^{-4} + \omega_\nu(a) + \omega_V$. Integration uses an explicit eighth-order Runge–Kutta method with relative tolerance 10^{-9}; the solution is interpolated with cubic splines in N, which is the variable it is smooth in. One background solve costs about 11 ms.

The neutrino sector carries one massive eigenstate of mass 0.06 eV and two massless ones with $N_\mathrm{eff} = 3.044$, using an interpolation that is exact in both the relativistic and non-relativistic limits. Its pressure is obtained from the exact consequence of separate conservation, $p = -\rho - \frac{1}{3}\mathrm{d}\rho/\mathrm{d}N$, so the cosmographic parameters remain analytic rather than finite-differenced.

F.2 What is checked, and against what

The exact identities of Sec. 4 are used as the Friedmann closure, so they cannot also serve as tests of the solution. The code therefore provides a second mode in which all five densities are integrated independently, and the identities become genuine checks. The test suite asserts the following, at every parameter point it exercises.

Check	Against	Residual
Friedmann closure	the integrated solution	$\sim 10^{-16}$
Survival identity Eq. (4.3)	independent five-density integration	$< 10^{-10}$
Jerk	two independent routes (see below)	$< 10^{-12}$
Snap	closed form versus differentiated j	$< 5 \times 10^{-3}$
Total continuity	analytic $\mathrm{d}E^2/\mathrm{d}N$	$< 10^{-4}$
Solvable family Eq. (6.1)	its closed form	$< 10^{-9}$
ΛCDM limit ($\epsilon = 0$)	an independent cosmology library	$< 2 \times 10^{-5}$

The jerk check is worth describing because it is not a tautology. One route uses the total-pressure identity $j = 1 - \dot{p}/(2M_\mathrm{Pl}^2 H^3)$; the other computes $j = 1 + 3E'/E + (E'^2 + EE'')/E^2$ from analytic derivatives of the Friedmann constraint, in which the transfer terms cancel between the dust and vacuum sectors. The two agree to machine precision, which tests both the algebra of Theorem 7.1 and its implementation.

The activation exponent is checked directly: with $p = 6$ the conversion rate at recombination is below 10^{-21} of the expansion rate, whereas $p = 2$ leaves a rate nine orders of magnitude too large for the expansion-tracking form of Eq. (6.4).

F.3 Growth

Eq. (9.7) is integrated from $z = 100$ with the growing mode, using splined background coefficients so that g_c' is analytic rather than finite-differenced inside the solver. Massive neutrinos are counted with the clustering matter. They are not sourced by the transfer, so including them in the denominator of g_c dilutes it slightly; at the redshifts probed they are half a per cent of the matter budget and the effect on g_c is of the same order, far below the precision of the growth data. Treating them as fully clustering also overstates their contribution on small scales, which only a Boltzmann treatment would resolve.

F.4 Likelihoods

Supernova distances use the Pantheon+ compilation with the full statistical plus systematic covariance; the absolute magnitude is marginalised analytically under a flat prior, which is verified

against a brute-force one-dimensional marginalisation. Baryon acoustic oscillations use the DESI DR2 measurements with their covariance, and the sound horizon at the drag epoch is evaluated from a fitting formula in the *pre-activation* matter density, which is the quantity the early universe actually determines. Cosmic chronometers use the Moresco compilation with a covariance built from the published systematic budget: the statistical errors plus the initial-mass-function and "model, odd one out" terms. That is the reduced budget, not the full one. The conservative variant, which adds the stellar-library and full model terms, is implemented and tested but is not what the fits use; it doubles the effective error on a coherent offset in $H(z)$, from 5.7% to 11.6%. Reporting the fits under it is left undone, and the choice is recorded here rather than described as "full covariance", which is what an earlier version of this paragraph and of the data table both said. The compressed CMB likelihood uses the shift parameter, the acoustic scale and the baryon density.

Redshift-space distortions enter only the "+growth" combination and are described here because an evidence is quoted from them. The data are the four $f\sigma_8$ entries of the SDSS DR16 consensus at $z = 0.38$, 0.51, 0.698 and 1.48 [45], taken with the $f\sigma_8$ rows *and* columns of the consensus covariance: at each effective redshift the growth measurement correlates with the two geometric ones at $|\rho| \sim 0.4$, and those are dropped because they overlap in volume with the DESI sample and would otherwise be counted twice. The model is $f\sigma_8(z) = f_g(z)\sigma_8(z)$ from the growth system of Sec. 9, and no scale cut is applied: the consensus values are already compressed to a single number per redshift.

One choice there is consequential and is not marginalised over. $\sigma_{8,0}$ is held at the Planck value 0.8111 rather than treated as a free nuisance parameter. The reason is that the question being asked is whether the parameters the geometry prefers are compatible with the observed growth, and a free $\sigma_{8,0}$ absorbs exactly the effect under test: the model's signature is a change in the *shape* of $f_g(z)$ together with a change in amplitude, and an unconstrained amplitude removes half of it. The consequence is that the evidence quoted for the "+growth" combination is conditional on that value, and a reader who prefers to marginalise should read it as an upper bound on how much growth data can help. We did not run the marginalised variant.

One caveat is disclosed rather than buried. The semi-analytic recipe for the sound horizon at recombination differs from a full Boltzmann calculation by about 0.2%, which is several standard deviations of the acoustic scale. Since all models considered here share identical pre-recombination physics, the recipe is calibrated once at the reference cosmology and the same constant correction is applied throughout.

How much that constant leaves behind is measured rather than assumed, because a constant can only absorb a constant. Isolating the fitting formula for the recombination redshift — which gives $z_* = 1091.71$ against Planck's 1089.92 — and recomputing the priors with the Planck value shifts ℓ_A by 0.3009 on average across the posterior range, which the calibration absorbs, and that shift *varies* by 0.1319 across the same range, which it does not. Against $\sigma(\ell_A) = 0.09$ that residual is 1.465σ. Over the whole prior box it is 11.54σ, so the constant calibration is adequate only because the posterior is narrow, and not in general. Two consequences follow and both are stated rather than buried: parameter values quoted from the compressed likelihood carry that residual, and it largely cancels from comparisons between the models fitted here, which share ω_b and ω_c at recombination and whose posteriors in them overlap. We have not refitted with an alternative z_* to confirm the cancellation.

A test asserts that the calibration is not re-derived per model. Users who prefer to avoid the compressed likelihood can run the late-time-only combination, which is reported alongside.

F.5 Maximisation

The χ^2_min column of Table 2 is a maximisation, so it carries a numerical error of its own, and that error has to be small compared with the differences read off the column. Three protocols were used and the first two failed in ways worth recording, because both produced numbers that

looked ordinary.

A fixed evaluation budget from a handful of random starts gave every model the same number of likelihood calls regardless of dimension, so the three parameters of ΛCDM and the nine of the memory model were searched equally hard. Forty full simplex runs from uniform prior draws did find the maxima but took 8.4 hours for five models, because a uniform draw usually lands where the background integration is stiff and the simplex then spends its whole budget there. The protocol in use screens a Latin hypercube of 3,000 points once, refines only the best few — which are by construction in the cheap region — and polishes the winner with alternating simplex and direction-set steps until it stops moving, with a budget of 300 evaluations per parameter. The alternation is not decoration: the simplex contracts across the curved degeneracy between the rate normalisation and the activation redshift instead of following it, and on the six-parameter model the direction-set steps recovered 0.7699 in χ^2 that the simplex had left behind.

Two things keep the cheaper protocol honest. Nesting is exact, so a model cannot fit worse than a model it contains; each search is seeded with every submodel's best fit embedded, and a gate fails the build if the column is not monotone in the nesting. That is not hypothetical: the nine-parameter search once returned a value 1.55 *worse* than the seven-parameter model it contains, which is impossible, and the number was on its way into the table. And the slow run's answers are frozen in the repository as a ceiling. A chi-square is attained at an actual point in parameter space, so a later run landing above the frozen value has failed to find something already known to exist, whatever its internal diagnostics say. ΛCDM reproduces the frozen value to 10^{-9}, which is what certifies that the likelihood itself has not changed. A second gate fails the build on any larger gap, and it is also not hypothetical: when it was first written — the comparison had been described in a docstring for some time without being performed — it found the reinforced model sitting 0.137 above a value this repository had already recorded, while reporting its search as converged.

What the protocol does not achieve should be said plainly. For the extended models the agreement among independent refinements is poor — for the six-parameter model one restart in nine reached the best value — and the maxima sit on a prior edge in the unidentified directions, which is what one expects when those directions are flat. The column is therefore the best value the search reached, checked against a frozen independent answer, and not a converged goodness of fit; the conclusions of Sec. 10 rest on the evidence column, which is computed by integration rather than by search.

Profiles. A profile in a derived quantity imposes the target exactly — at each grid point every parameter but ϵ is free and ϵ is solved for by a bracketed root find, so the achieved value matches the target to a relative 3.06×10^{-9} — and is computed in two sweeps. The forward sweep warm-starts each target from the previous one, which helps in one direction only: a target whose search fails then seeds the next from a failure, and the profile acquires local minima that belong to the optimiser and not to the likelihood. A profile is the lower envelope of what is attainable, so a backward sweep from the far end can only improve it, and it improved 13 of the thirteen grid points of the jerk profile and 11 of the deficit profile's — that almost every point improved is the measure of how much the forward-only sweep was propagating. Two diagnostics are reported with every profile: the worst relative miss of the target, and the number of decreasing steps *above* the minimum, which must be zero because a profile cannot fall on its rising side. Both were added after an earlier version used a quadratic penalty too weak to hold the constraint, whose grid points missed their targets by up to a factor of three.

The grid matters as much as the search. The jerk profile was first run from 10^{-4}, while the unconstrained maximum sits at a jerk excess of 1.4×10^{-8}; the grid then never sampled the region containing its own minimum, and every $\Delta\chi^2$ in it was measured from a floor 0.93 too high, which makes the limit too weak. A gate now records where the maximum sits in the profiled quantity and refuses a grid that does not bracket it.

F.6 Sampling

Posteriors are sampled with an affine-invariant ensemble sampler using differential-evolution moves, initialised in a small ball around a multi-start maximum-a-posteriori estimate. Sampling stops when the chain exceeds fifty integrated autocorrelation times and that estimate has itself stabilised to two per cent. Chains are stored with the burn-in and thinning actually applied, and the number of workers is chosen from available memory as well as from the core count.

Why the extended models are sampled differently. An ensemble move is a difference between two walkers and knows nothing about the edges of a prior box. When a posterior is wide compared with its prior in some direction — which is what happens here, because the rate parameters are unidentified along the ΛCDM boundary — most proposals land outside the box and are rejected before the likelihood is evaluated. The rejection rate then *grows* as the ensemble spreads, so the chain decelerates and running it longer makes it worse. Measured on the first attempt: cumulative acceptance 0.0073, move rate falling from 0.0160 over the first tenth of the run to 0.0064 over the last, and a fraction 0.562 of replayed proposals outside the prior at the default step scale — against acceptance 0.4386 and no invalid proposals at all for ΛCDM on the same machinery. The chains that produced these numbers are kept, so the diagnosis can be repeated. Two changes follow. The sampler now works in a logit coordinate, so no proposal can be invalid; this is a change of variables and not of prior, with the Jacobian added to the target and a test that recovers a flat prior under a constant likelihood. And the posteriors quoted for the extended models come from nested sampling, which integrates over the prior rather than walking along a degenerate ridge inside it, and which supplies the evidence used in Table 2 directly. The convergence gate was rewritten at the same time, because the version in place had no power to fail: it watched the drift of a percentile pinned against a prior edge, which is insensitive to this failure by construction and passes the frozen chains even now. It additionally checks the move rate over the last tenth of the chain against the first, the chain length in autocorrelation times, and whether any parameter has simply filled its prior.

F.7 Stochastic sector

The finite-population process is simulated exactly, not approximated. Because the memory variable relaxes monotonically between jumps, the intensity is monotone there, and two exact samplers are available: thinning with a tight bound, and direct inversion of the integrated intensity, which for the exponential rate law is expressible through the exponential integral. The two are checked against each other with a two-sample test.

These simulations verify the deterministic limit of Theorem 3.4 and its fluctuation scaling. With the rate law of Eq. (3.4), the maximum discrepancy between the ensemble mean and the deterministic solution falls from 1.42×10^{-2} at $N = 100$ to 4.81×10^{-4} at $N = 10^4$, while $\sqrt{N}\,\mathrm{std}(f_N)$ takes the values 0.8877, 0.9066 and 0.9331 at $N = 10^2$, 10^3 and 10^4. That is numerical evidence for the $N^{-1/2}$ law under the reinforced rate, and not a demonstration of it: the exact binomial result of Remark 3.5 requires a state-independent rate, which Eq. (3.4) is not, and the three values drift upwards by about five per cent over the range rather than sitting on a constant. Whether that drift is a slowly vanishing correction or a different exponent is not settled by three decades.

The hysteresis experiment of Theorem 3.8 is run in the same framework: two histories are prepared to reach the same converted fraction $f = 0.26283$ with memory states differing by 0.04164, and are then evolved under identical conditions. Their asymptotic vacuum densities differ by 0.00641, a relative difference of 0.0841. In the memoryless limit the two histories coincide to solver tolerance.

F.8 Defect profile

Eq. (11.5) is solved as a boundary-value problem on $x \in [10^{-4}, 40]$ with a collocation method, using the trial profile as the initial guess. The solution returns degree 1.000000 and satisfies the virial identity $E_2 = E_4$ with relative residual 1.70×10^{-10}.

F.9 Symbolic verification

Six analytic claims are verified independently with a computer algebra system rather than by hand: the four-form stress tensor and its contraction identities; the constrained-clock stress tensor and field equation; the FLRW curvature tensors; the jerk identity; the reduction of the perturbation pair to the second-order growth equation; and the closed-form solutions of the solvable family.

F.10 Availability

The code, the data checksums, the chains' summary statistics and the results file are archived at `10.5281/zenodo.22726266`. Data are not redistributed: a script downloads each public release and verifies it against a recorded checksum. Random seeds are fixed and recorded. The build is reproducible with a single command, and the quality gates described above run as part of it.